

Illustrated review copy

The chapter text is a snapshot of the working manuscript. Animations are represented by selected, time-stamped frames, arranged in chronological order at their location in the text. These panels show visual states, not continuous motion or every transition. They are review aids, not the intended presentation of the animated website. Manuscript wording and equations have been preserved.

Symmetry

Strike a pool ball with a cue, and the balls move in an expected way. Move the table over a few feet, and the balls move in recognizably the same way. Wait a few minutes, and the balls move in the same way. Turn the pool table a few degrees, and the balls still move the same way. Put the pool table on a train at constant velocity, and, again, the balls move in the same way. These are the manifest "symmetries" of the world we live in -- position and time translation, rotation, and velocity "boosts."

Symmetry transforms don't change how pool balls behave

Animation 1 / frame 1 of 8 / 1.00 s

The same experiment

Position translation

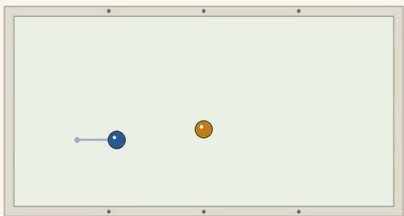
Time translation

Rotation

Velocity boost

Reference

clock 00.52 s

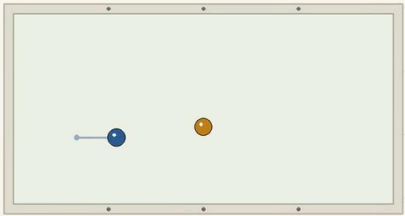


Elapsed after shot 0.52 s

separation 1.38

Moved elsewhere

clock 00.52 s



Elapsed after shot 0.52 s

separation 1.38

$$\mathbf{x}'(t) = \mathbf{x}(t) + \mathbf{a}$$

Same collision. Same paths relative to the table.

The same experiment

SYMMETRY

Position translation

Time translation

Rotation

Velocity boost

Reference

clock 02.26 s

Elapsed after shot 2.26 s

separation 1.17

Moved elsewhere

clock 02.26 s

Elapsed after shot 2.26 s

separation 1.17

$$\mathbf{x}'(t) = \mathbf{x}(t) + \mathbf{a}$$

Same collision. Same paths relative to the table.

The same experiment

SYMMETRY

Position translation

Time translation

Rotation

Velocity boost

Reference

clock 00.52 s

Elapsed after shot 0.52 s

separation 1.38

Started later

clock 10.52 s

Elapsed after shot 0.52 s

separation 1.38

$$\mathbf{x}'(t + b) = \mathbf{x}(t)$$

Same collision. Same paths relative to the table.

The same experiment

SYMMETRY

Position translation
Time translation
Rotation
Velocity boost

Reference clock 02.26 s

Elapsed after shot 2.26 s separation 1.17

Started later clock 12.26 s

Elapsed after shot 2.26 s separation 1.17

$\mathbf{x}'(t + b) = \mathbf{x}(t)$

Same collision. Same paths relative to the table.

The same experiment

SYMMETRY

Position translation
Time translation
Rotation
Velocity boost

Reference clock 00.52 s

Elapsed after shot 0.52 s separation 1.38

Turned clock 00.52 s

Elapsed after shot 0.52 s separation 1.38

$\mathbf{x}'(t) = R \mathbf{x}(t)$

Same collision. Same paths relative to the table.

The same experiment

SYMMETRY

Position translation Time translation **Rotation** Velocity boost

Reference

clock 02.26 s

Elapsed after shot 2.26 s separation 1.17

Turned

clock 02.26 s

Elapsed after shot 2.26 s separation 1.17

$$\mathbf{x}'(t) = R \mathbf{x}(t)$$

Same collision. Same paths relative to the table.

The same experiment

SYMMETRY

Position translation Time translation Rotation **Velocity boost**

Reference

clock 00.52 s

Elapsed after shot 0.52 s separation 1.38

Moving steadily

clock 00.52 s

Elapsed after shot 0.52 s separation 1.38

$$\mathbf{x}'(t) = \mathbf{x}(t) + \mathbf{u}t$$

Same collision. Same paths relative to the table.



Our pool game illustrates what we mean by symmetries of physical behavior, but why should we start our story here? We will argue that symmetry constrains both the laws that govern physical evolution and the classification of the objects that undergo that evolution. The term "symmetry" in this context may not at first glance seem like the same concept as, say, a triangle's symmetry, but it precisely is, as we will see.

We will start with the familiar world of these discrete symmetries and build up the definitions we need from there. Then we will turn to "continuous symmetries," specifically rotations and translations, where we will encounter a new branch of math, Lie algebra, that joins symmetry ideas with calculus. We will discover that waves arise naturally when we represent translational symmetry, and this will lead us to a discussion of Fourier analysis. Finally, we will situate these ideas in the context of quantum mechanics.

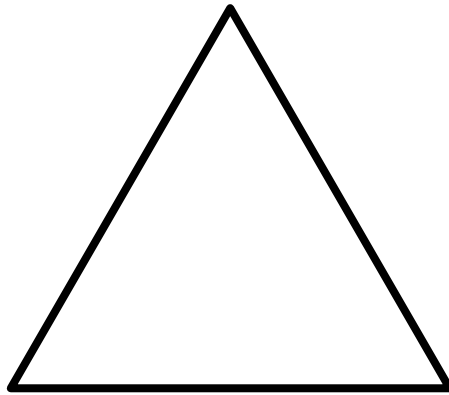
Contents

- Discrete symmetries
 - Representations
 - Invariants
- Continuous symmetries
 - Infinitesimal Generators
 - Commutators
 - Translations and Function Representation
- The Fourier Structure of Waves
 - The Heisenberg Symmetry Group
 - Position / Wave Number Uncertainty
 - Wave Propagation and Interference
- From Wave Mechanics to Quantum Mechanics

Let's start with the humble triangle.

Discrete symmetries

Consider a triangle:



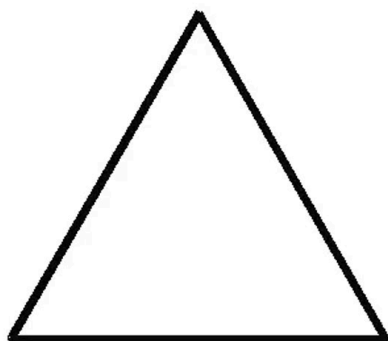
An equilateral triangle

We can see its obvious symmetry. To categorize its symmetry we can write down all the actions that leave it unchanged:

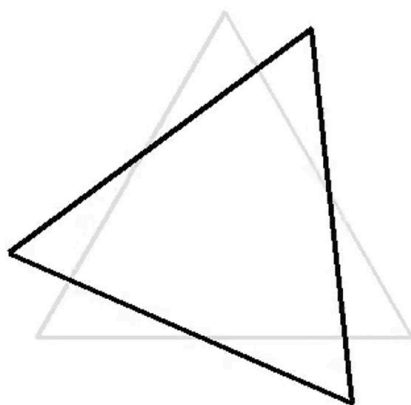
1. Do nothing
2. Rotate 120°
3. Rotate 240°
4. Flip along an axis
5. Flip then rotate 120°
6. Flip then rotate 240°

A triangle's symmetries

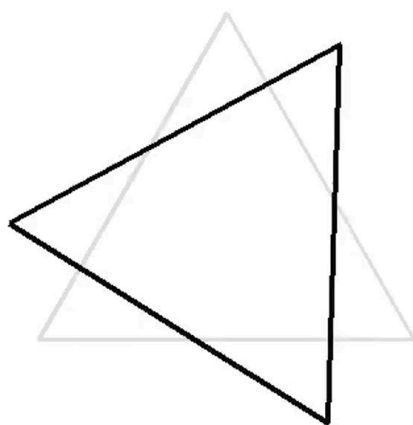
Animation 2 / frame 1 of 6 / 0.40 s



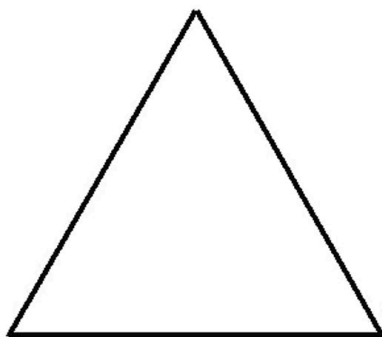
Animation 2 / frame 2 of 6 / 4.00 s



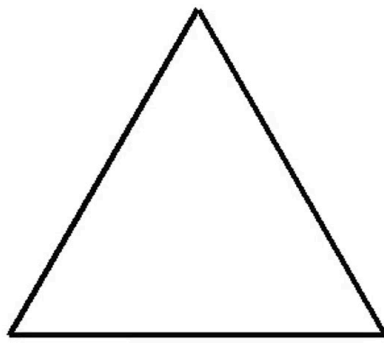
{e}



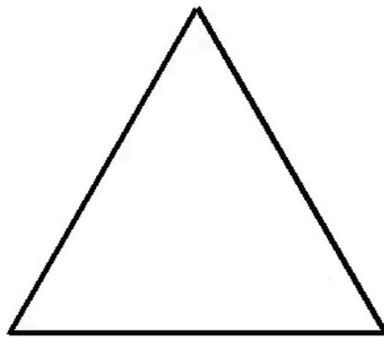
$\{e, r\}$



$\{e, r, r^2\}$

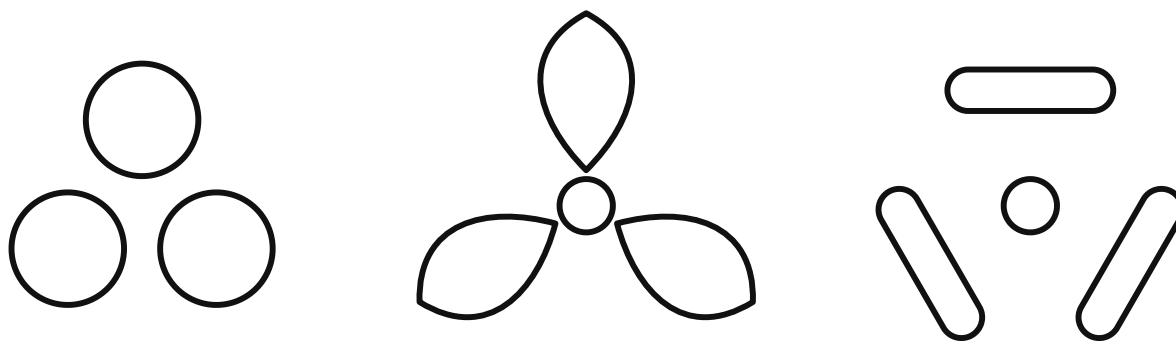


$$\{e, r, r^2, s, sr\}$$



$$\{e, r, r^2, s, sr, sr^2\} = D_3$$

We say the triangle "belongs to the D_3 symmetry **group**." Any number of objects possess D_3 symmetry.



D_3 is more general than triangles

We tend to think of a "triangle's symmetry" not the "symmetry of two rotations and flips about 3 axes," but this latter way of thinking in terms of symmetry groups is more general and is the definition of symmetry used in physics, where we ask not "what symmetry does this or that object have?" but "what mathematical objects realize the actions of a symmetry group?"

Representations

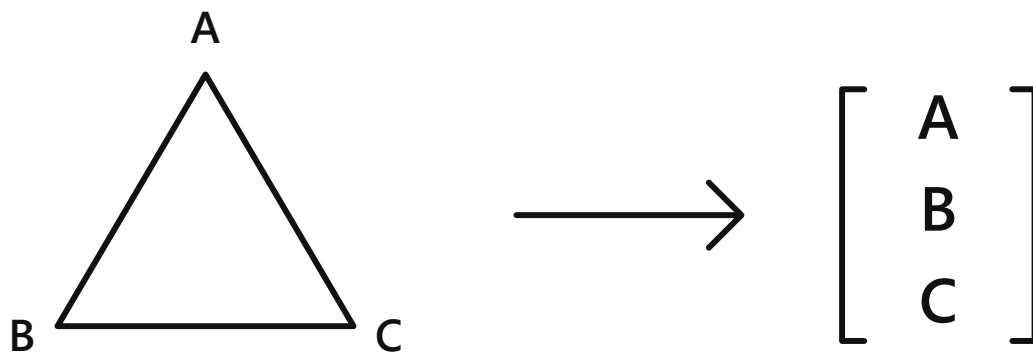
We can plainly "see" the symmetry of the triangle, but what if we want to write it down mathematically? Specifically, what if we want to write group actions in terms of how they transform labels on an object? For example, suppose we label our triangle's vertices A, B, C and ask, if we rotate twice, flip once, then rotate again, where is vertex A sent? Recognizing that the triangle's state has three ordered components, we might guess that we could represent the triangle as a vector with 3 components. We could then represent 120° rotations as **transformations** of the labels that permute the vertices in accordance with the symmetry group actions.

There is nothing special about the vertices here, we could just as easily have chosen the midpoints of the edges or any other triplet of points with the triangle's symmetries, and the same matrix that permutes the vertices would permute those vectors. That is, the group actions are represented as **linear** transformations. A **representation** of a symmetry group is a vector space and set of linear transformations that compose in the same way as the group actions:

$$D(g_1 g_2) = D(g_1) D(g_2).$$

where g_1 and g_2 are group actions, such as a rotation and a flip, and $D(g_n)$ is the matrix representing the g_n action, and D itself is the map from symmetry actions to matrices in the representation. This says that the matrix for the composed action g_2 followed by g_1 is the product of the matrices for the separate actions.

To construct a 3-dimensional representation of D_3 , we map the 3 vertices to components of a vector.



We can represent rotations and flips, respectively, with the following matrices:

Rotation:

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Flip:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

For example, a single rotation would be represented as:

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix}$$

The matrices that provide a set of transformations from which all other operations can be constructed are the symmetry's **generators**. For example, in the 3-dimensional representation of D_3 , all operations can be constructed from one rotation matrix and 3 flip matrices, one for each axis.

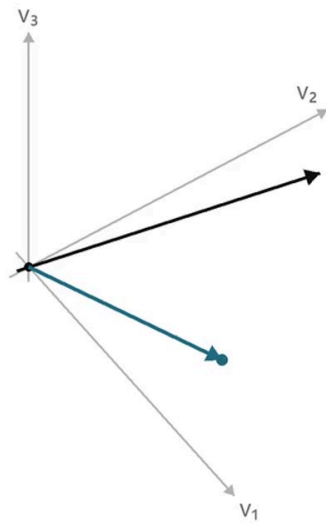
Visually, permutations that correspond to rotations of the triangle are 120-degree rotations about a diagonal axis in this vector space, while those that correspond to flips are flips about planes that contain the diagonal axis.

D_3 in 3-dimensional representation

Animation 3 / frame 1 of 4 / 0.13 s

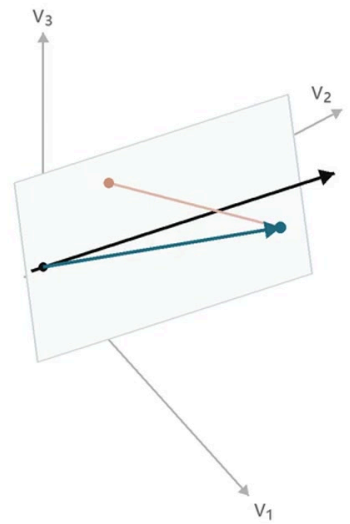
rotations

→ equal components



flips

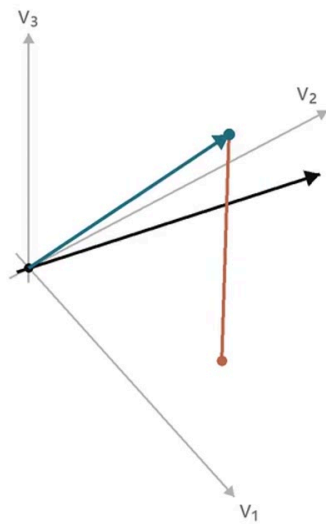
→ equal components



Animation 3 / frame 2 of 4 / 2.15 s

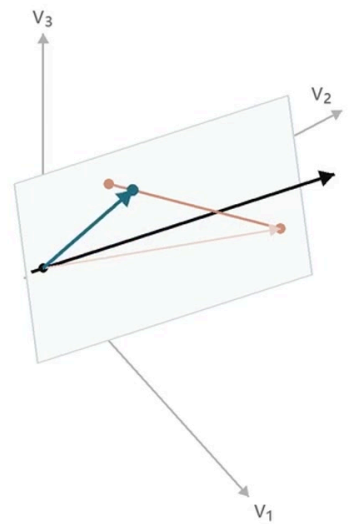
rotations

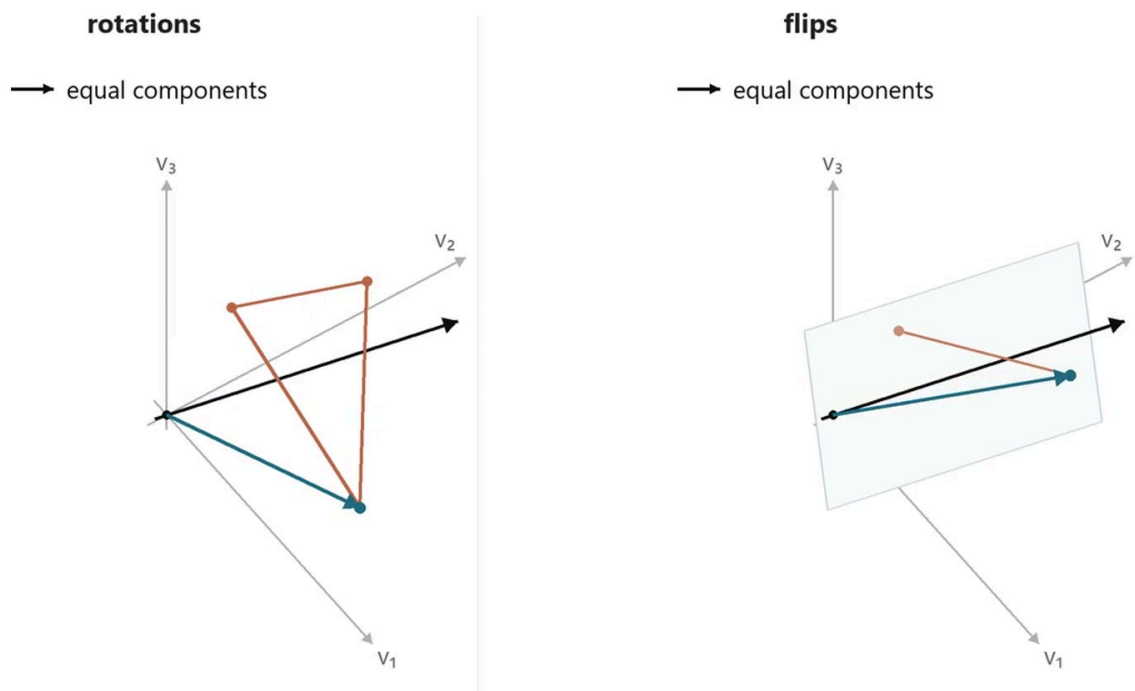
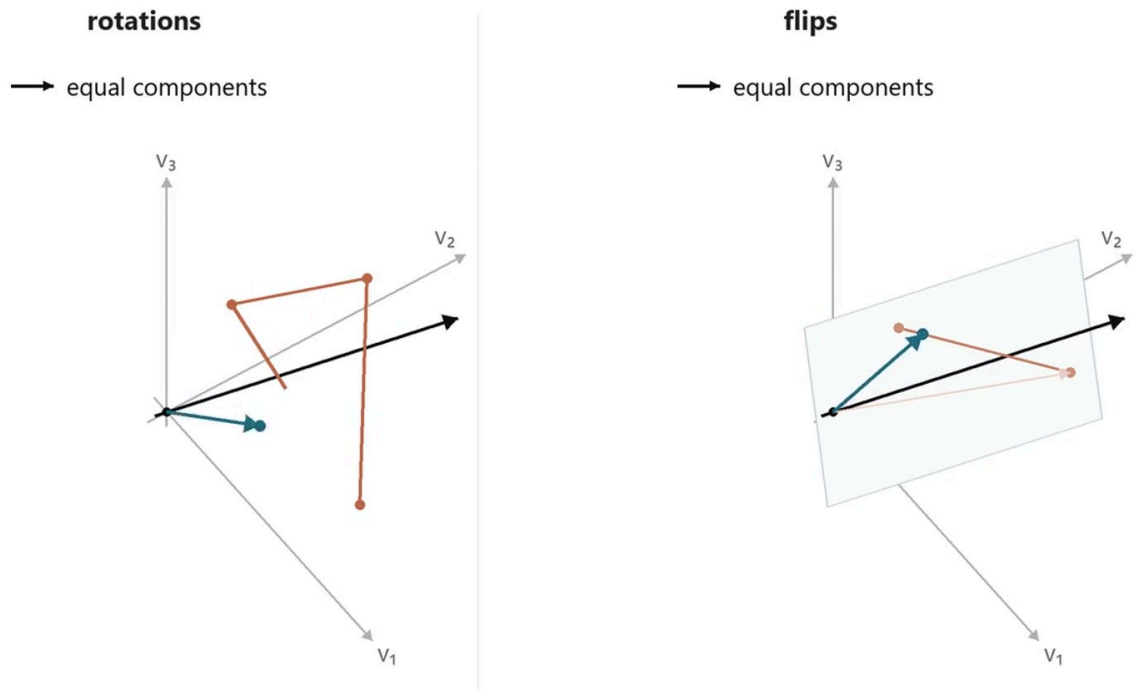
→ equal components



flips

→ equal components





Now, we can notice something about these visualizations. The transformation of any vector lies in a plane, and all such planes are parallel to one another. Thus, if we subtract the average of a vector's components from each component, that is, if we move the point where the plane intersects the axis of rotation to the origin, we preserve the permutation structure of the transformations. We thus see that the D_3 symmetry is just as well represented as 120-degree rotations in the subspace of a 2-dimensional plane.

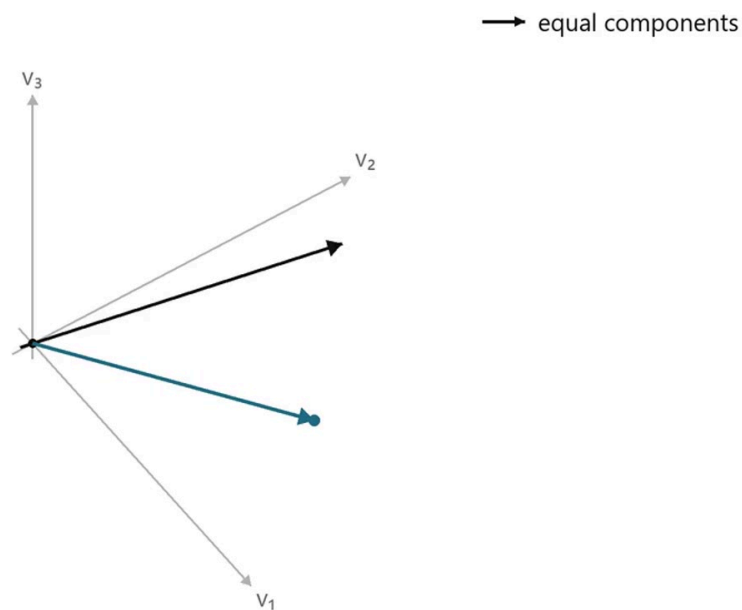
For our earlier vector, the average of the components is 3. We can separate it into a common part and a zero-sum part.

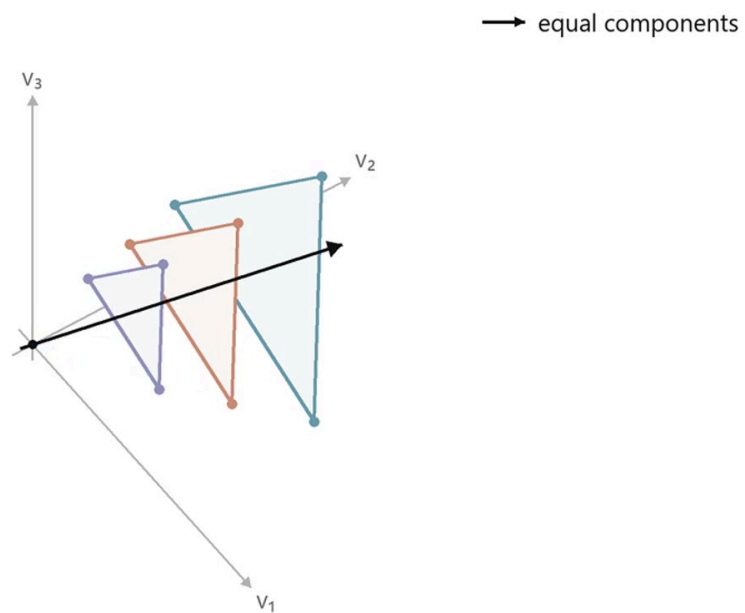
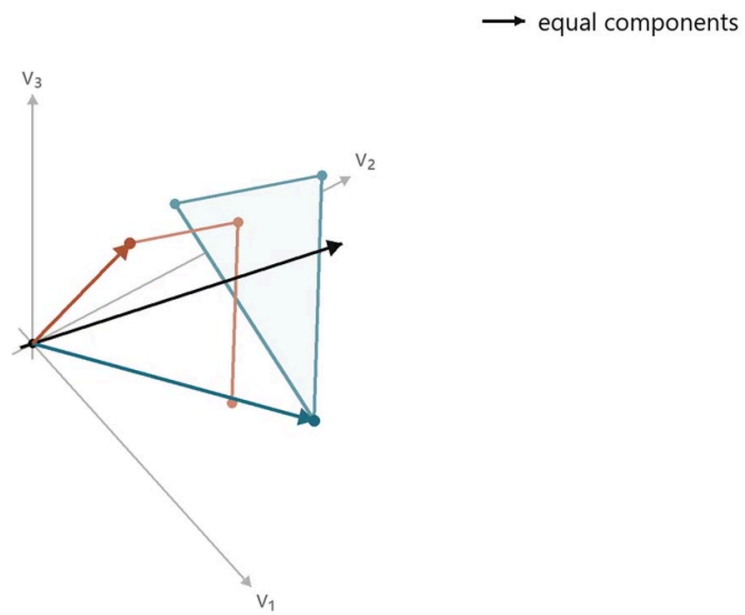
$$\begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}.$$

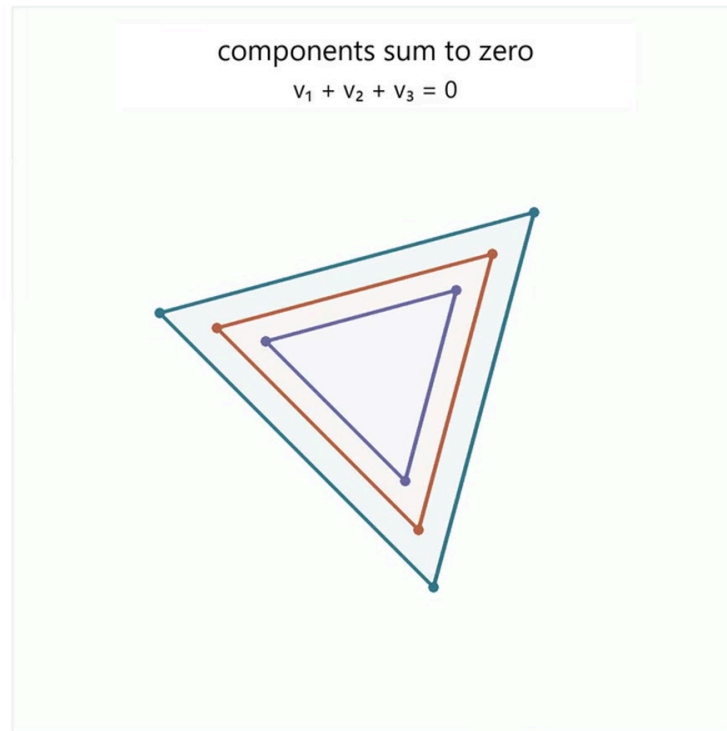
The common part lies on the rotation axis and is unchanged by every permutation. The zero-sum part lies in a plane through the origin. Although written with three components, it needs only two independent numbers, since the third must make their sum zero.

Reducing 3-dimensional representation to 2-dimensional

Animation 4 / frame 1 of 4 / 0.47 s







We also notice that vectors that lie on the axis of rotation itself are left unchanged by the transformation. A way to think of this is to allow the triangle's vertices to store some information, like a number or any numerical quantity. If the value they store is the same for all vertices, the symmetry actions have no effect, whereas if they are different, the actions permute those values in a way that can be represented in a 2-dimensional vector space. The 3-dimensional representation space we began with is thus decomposable into 2 subspaces, one 1-dimensional, the other 2-dimensional. These cannot be decomposed further. That is, there is no lower-dimensional space such that an allowable transformation of any state remains in that space. The 1-dimensional and 2-dimensional representations are called **irreducible representations** or **irreps** for short. This is admittedly heavy math. Why do we bother? In the story we have to tell of quantum physics, where constituents of matter must abide the dynamical symmetry of nature, each constituent, each type of **particle** such as electron or photon, corresponds to an irreducible representation of nature's symmetry. A particular state of the particle is encoded in an element in the corresponding irrep.

Invariants

Once we have chosen a representation for a symmetry, we might well ask, how do we know our transformations preserve the symmetry? If we look at a triangle and rotate by 100 degrees we can "see" that doesn't preserve the symmetry. In a representation, we need some set of mathematical expressions that say "this transformation left the triangle the same." We call these the **invariants** of the transformation.

In our 2-dimensional irrep of D_3 , we only need to check how a transformation acts on a single vector. The question "does the triangle overlay itself" becomes "is an arbitrary vector rotated by 120 degrees or flipped along a given axis." And this has a precise answer. Given the coordinates of a vector a transformation cannot change these invariants:

$$r^2 = x^2 + y^2 \quad (\text{the squared length of the vector}),$$

$$u = x^3 - 3xy^2 = r^3 \cos(3\theta) \quad (\text{its orientation relative to reference}).$$

These exact invariants are not important. They are obscure without seeing the derivation, but what is important is that we can write an expression in terms of coordinates that must not change under the symmetry transformation.

Why should we care about invariants? As we will see later, a system's characteristic physical behavior, the "sameness" of the pool ball's behavior, so to speak, can be found by assigning a number to each possible "history," each way the pool balls might have moved, and using a procedure called variational calculus to find the path that extremizes that number, meaning it makes it "stationary," such as at a minimum or maximum. When using the spacetime structure provided by relativity, which we will learn about in the next chapter, this number is the same for every observer in an **inertial reference frame**, that is, a perspective distinguished by a symmetry transformation. (This raises a point we glossed over earlier. In our introductory pool table example, we discussed reorienting the table. But we can just as well reorient the observer. We can move the table a few feet to the right or the observer a few feet to the left. When discussing invariants, we often think in terms of what "observers agree on.")

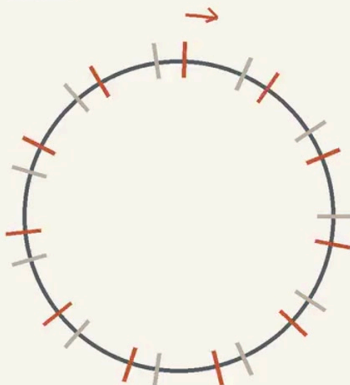
Continuous symmetries

What rotations return a circle to itself? All of them! Similarly, all translations return a line to itself. The symmetries of space and time are continuous and can therefore be represented as transformations that depend on continuously varying parameters.

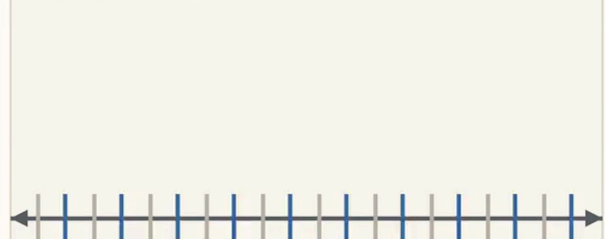
Continuous symmetries

Animation 5 / frame 1 of 4 / 0.10 s

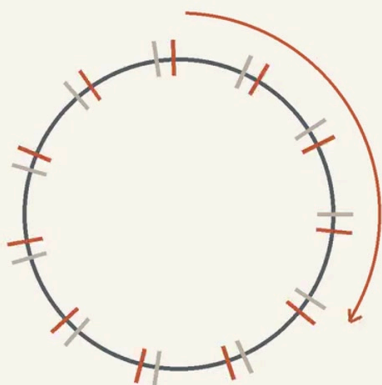
SO(2): rotation



(R,+): translation



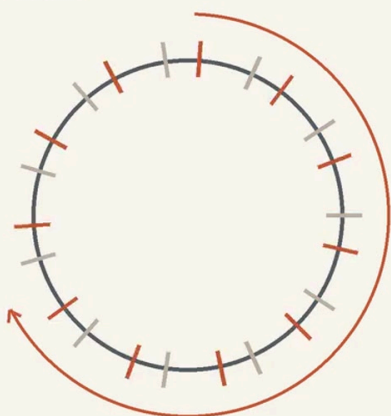
$SO(2)$: rotation



$(\mathbb{R}, +)$: translation

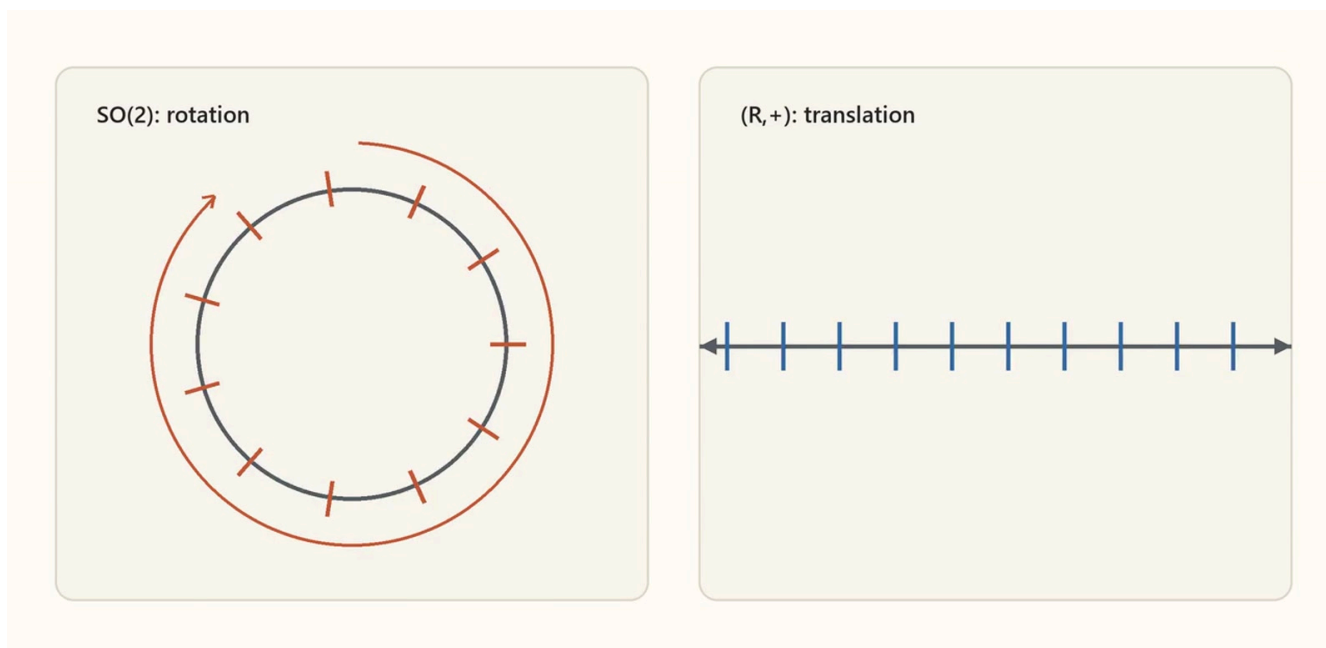


$SO(2)$: rotation



$(\mathbb{R}, +)$: translation

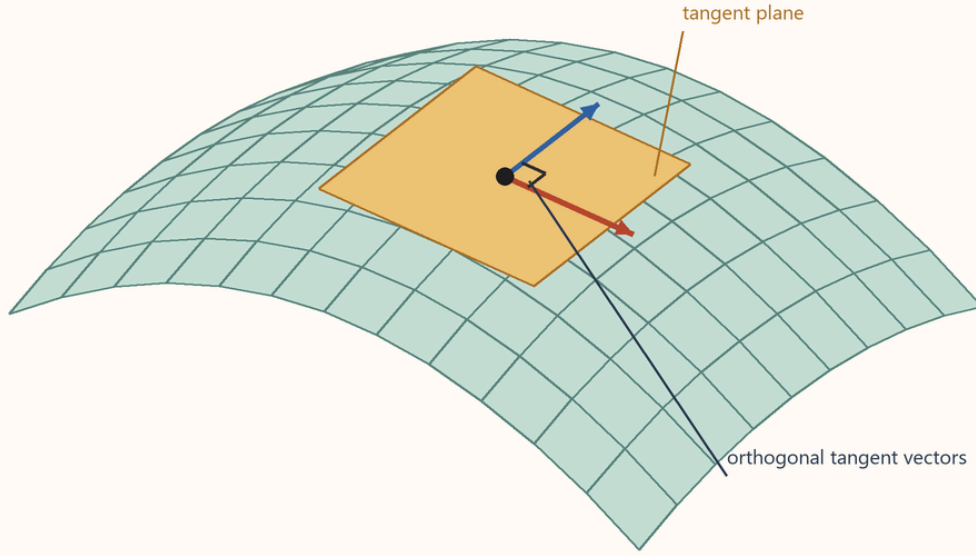




Infinitesimal Generators

As we saw, in the D_3 symmetry group, we can build any action from a combination of the elemental actions of rotations and flips about an axis. Now let us ask the question, what is the generator of a continuous transformation? Let's say we want to rotate a circle by 10° . We could compose 10 1° rotations. But what if we want to rotate by 1° ? We can see where this is going. The generators must be infinitesimal rotations. Noting that if we zoom in enough, any curved surface appears flat, we see that a continuous symmetry's infinitesimal generators are the vectors in the tangent plane to the symmetry's action.

Curved surface, tangent plane, tangent vectors



Infinitesimal generators are vectors in the plane tangent to the space of symmetry transformations

As every high school calculus student learns, a derivative of a function is the tangent slope to that function. In the case of a circle, in the 2-dimensional representation, we have a single variable that parameterizes the operator matrix:

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

We can take the derivative of the matrix valued function of θ . This is:

$$\frac{dR}{d\theta} = \begin{pmatrix} -\sin \theta & -\cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix}.$$

To obtain a single matrix representing this infinitesimal nudge, we evaluate the derivative at $\theta = 0$, where $R(0) = I$ is the identity transformation. In that case:

$$\left. \frac{dR}{d\theta} \right|_{\theta=0} = \begin{pmatrix} -\sin 0 & -\cos 0 \\ \cos 0 & -\sin 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

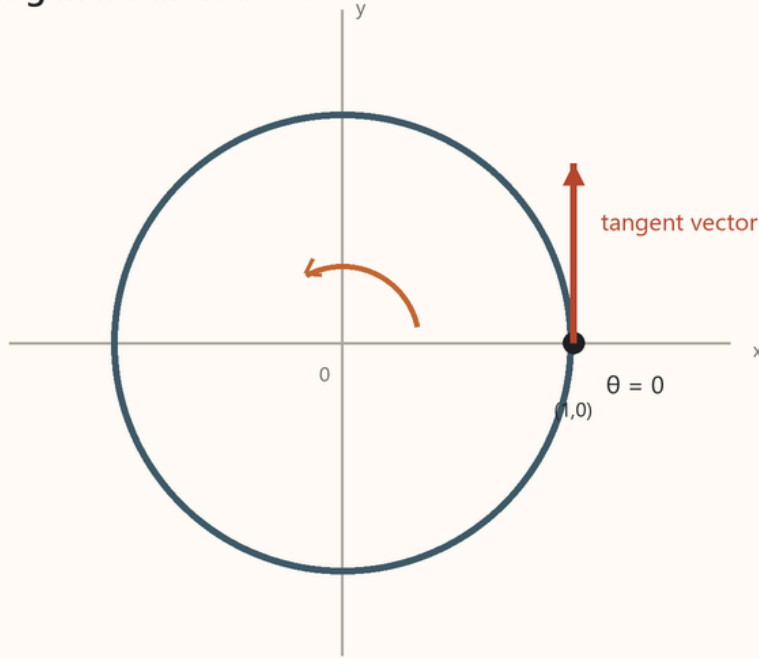
This is the infinitesimal generator matrix:

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Now apply this tangential nudge to a state vector

$$\mathbf{v}(\theta) = \begin{pmatrix} x(\theta) \\ y(\theta) \end{pmatrix}.$$

Tangent generator at $\theta = 0$



The generator matrix J acts on the point $(1, 0)$ to produce the tangent vector $(0, 1)$ shown here.

J acts on any point on the circle to produce the tangent vector at that point. Putting this into calculus notation:

$$\frac{d\mathbf{v}}{d\theta} = J\mathbf{v}.$$

We now have a differential equation for the transformation that uses the generator. We found this by starting with a known transformation and deducing its generator. But we can just as well go in the opposite direction. Given this differential equation, we can find the transformation:

$$\mathbf{v}(\theta) = e^{\theta J} \mathbf{v}(0).$$

The exponential of a matrix is defined in terms of the Taylor expansion for an exponential:

$$e^{\theta J} = I + \theta J + \frac{\theta^2}{2!} J^2 + \frac{\theta^3}{3!} J^3 + \dots$$

Now we notice something. Since:

$$J^2 = -I,$$

the even powers of J become powers of I , while the odd powers become powers of J . Therefore

$$e^{\theta J} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) I + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right) J.$$

These are the Taylor series for sine and cosine:

$$e^{\theta J} = \cos \theta I + \sin \theta J.$$

Substituting in J gives

$$e^{\theta J} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = R(\theta).$$

This is an **exponential map**. It takes the infinitesimal generator J and returns the finite symmetry transformation $R(\theta)$. Exponential maps give what one would think is the familiar purely algebraic quality of exponentiation a rich geometric interpretation.

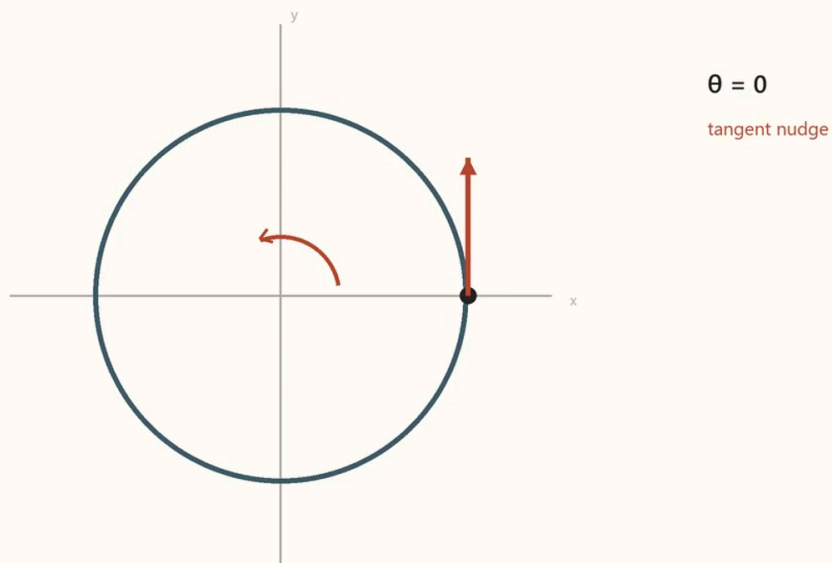
Symmetry Flows

We can think of a symmetry transform as a "uniform looking" vector field, as illustrated below:

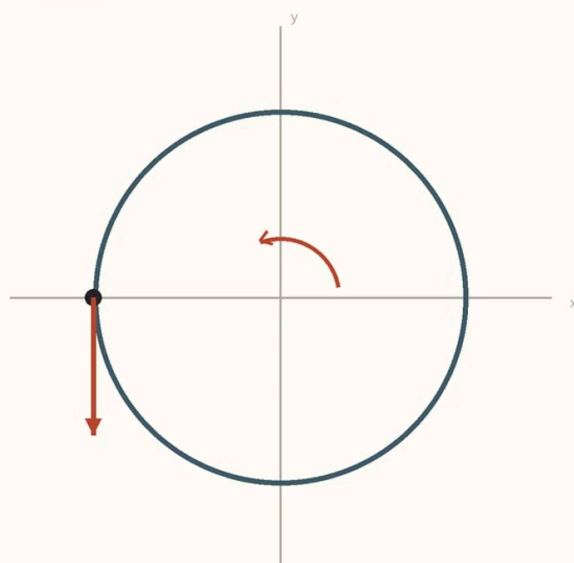
Symmetry flows provide a rich picture of symmetry transformations

Animation 6 / frame 1 of 6 / 0.17 s

A tangent direction at one state



A tangent direction at one state

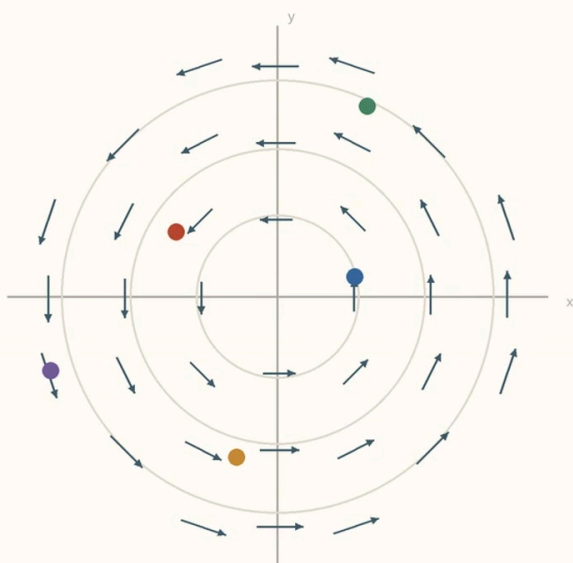


$$\theta = \pi$$

tangent nudge

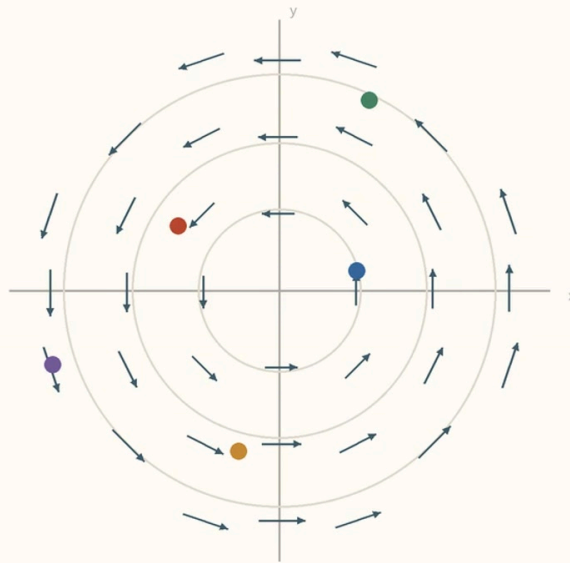
The tangent rule at every state

$$K(x,y)=(-y,x)$$



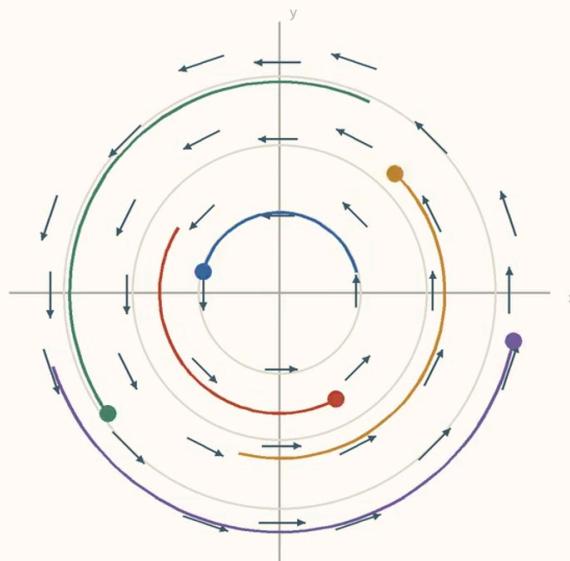
The tangent rule at every state

$$K(x,y)=(-y,x)$$



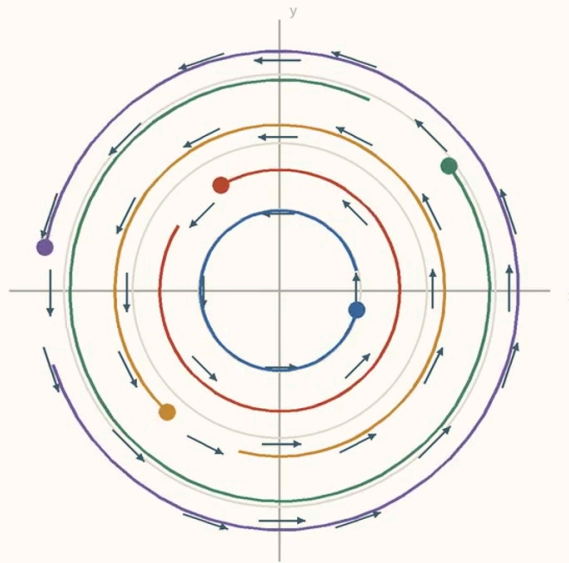
The tangent rule at every state

$$K(x,y)=(-y,x)$$



The tangent rule at every state

$$K(x,y)=(-y,x)$$



Here we ask not what happens when the transformation is applied to a single set of starting conditions, but what the transformation does to all starting conditions. This view lends itself to seeing the structure in the evolution of a collection of nearby states.

Invariants and Metrics

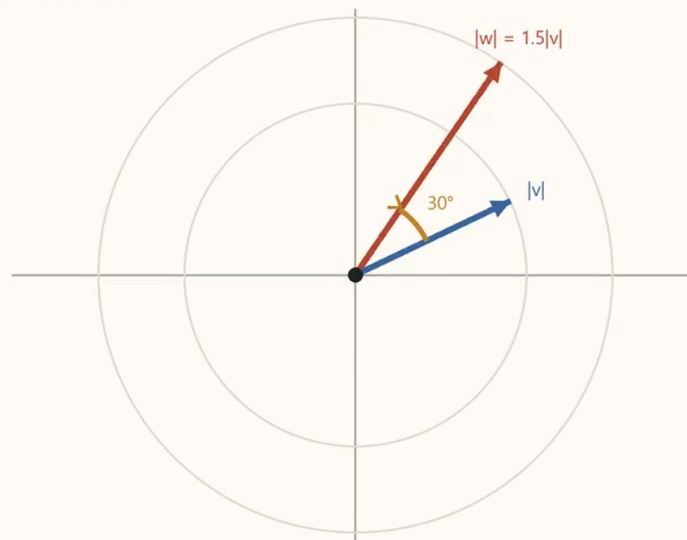
What is the invariant of rotation in this representation? It is just the length of vectors and the angles between them.

Metric structure

Animation 7 / frame 1 of 4 / 0.12 s

Rotation changes coordinates, not lengths or angle

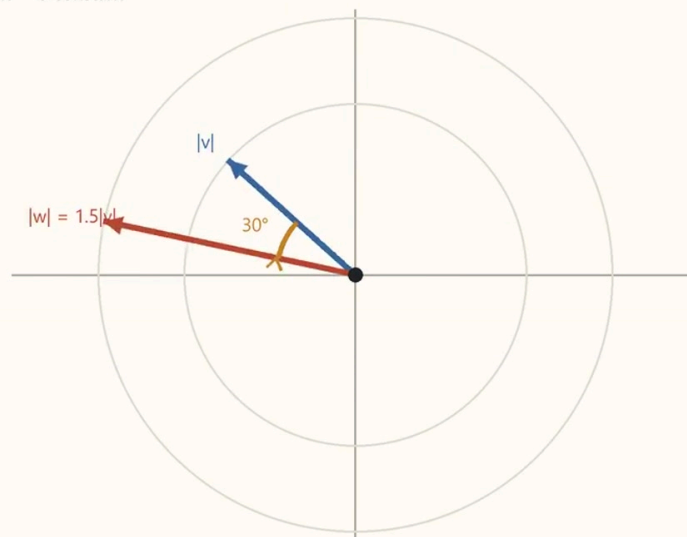
$|v|$ constant $|w|$ constant θ constant



Animation 7 / frame 2 of 4 / 1.98 s

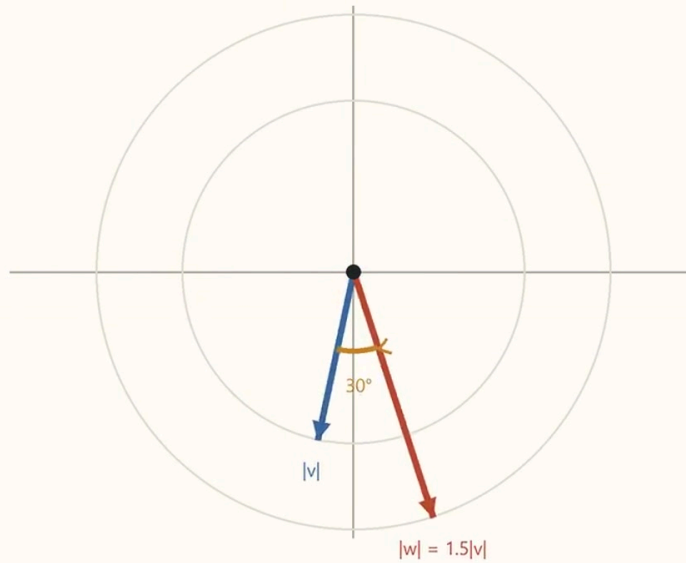
Rotation changes coordinates, not lengths or angle

$|v|$ constant $|w|$ constant θ constant



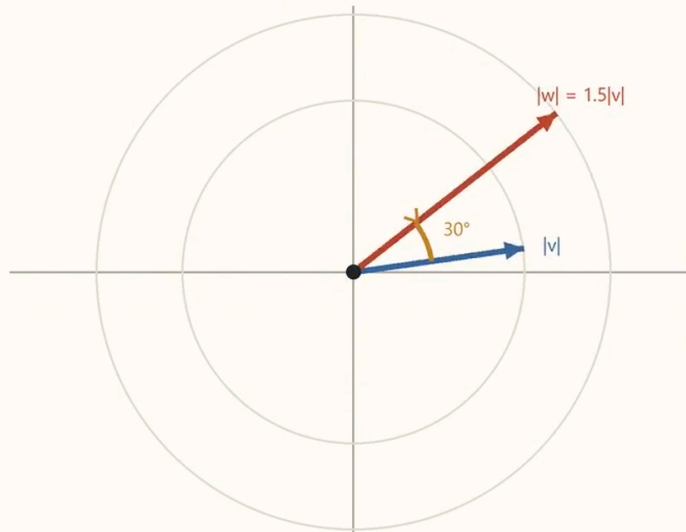
Rotation changes coordinates, not lengths or angle

$|v|$ constant $|w|$ constant θ constant



Rotation changes coordinates, not lengths or angle

$|v|$ constant $|w|$ constant θ constant



These relationships are expressed by a single invariant, the dot product.

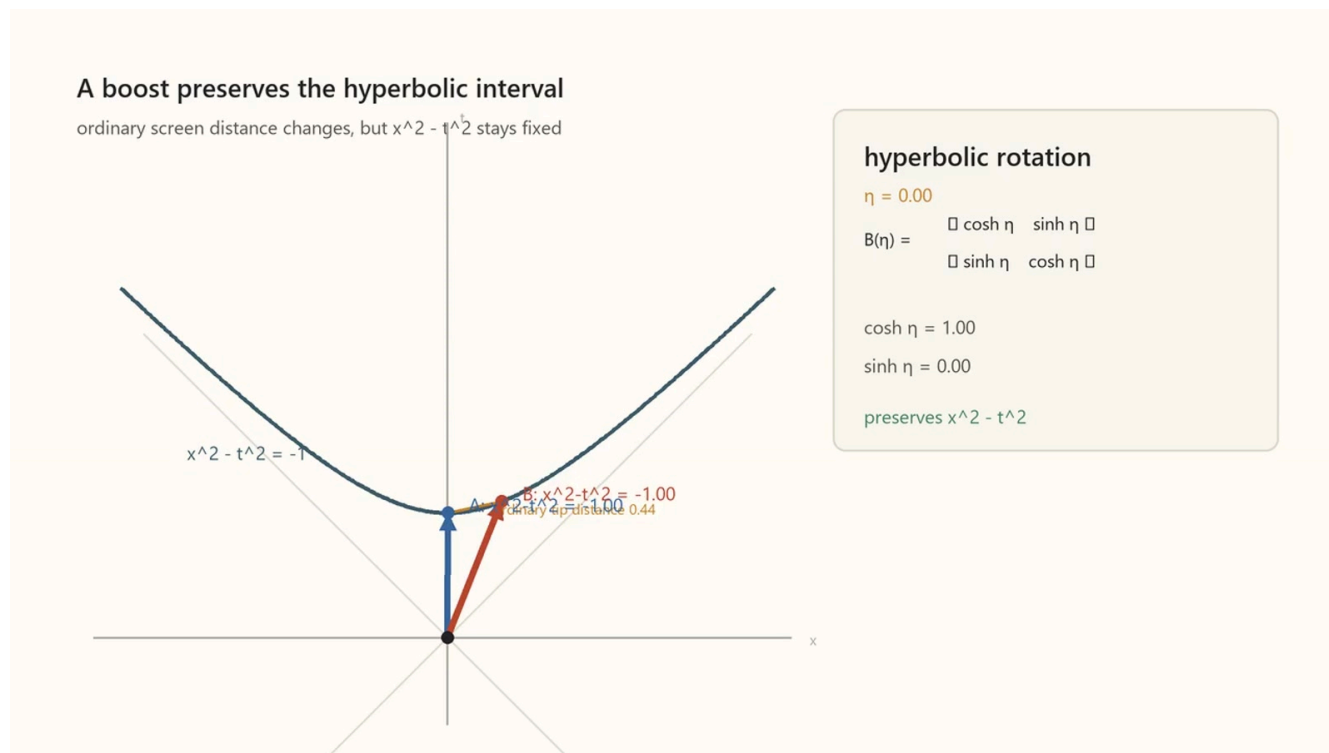
$$\mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y.$$

An invariant of this sort, that fixes some notion of placing reliable rulers on the space, is a **metric**. For normal rotations and translations, these relationships are intuitive, but a symmetry may preserve a less intuitive metric. If we define invariant length not as $x^2 + y^2$ but as $t^2 - x^2$, we have a "hyperbolic rotation." Separation between points stretches out to infinity as the rotation approaches its

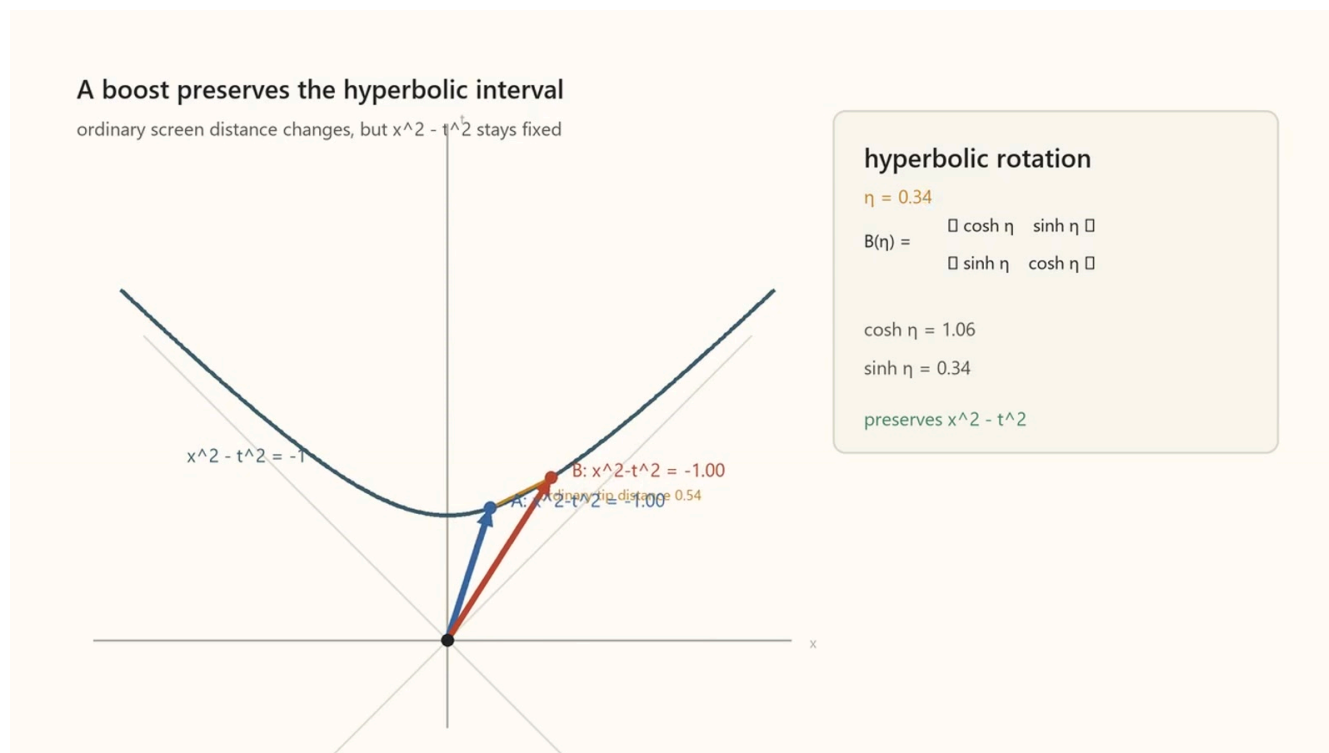
asymptotes, leaving the invariant interval unchanged even though the separation appears to grow using our everyday notion of distance. Such a hyperbolic metric, as we will see, encodes the asymptotic limit of light speed in the relativistic geometry **spacetime**.

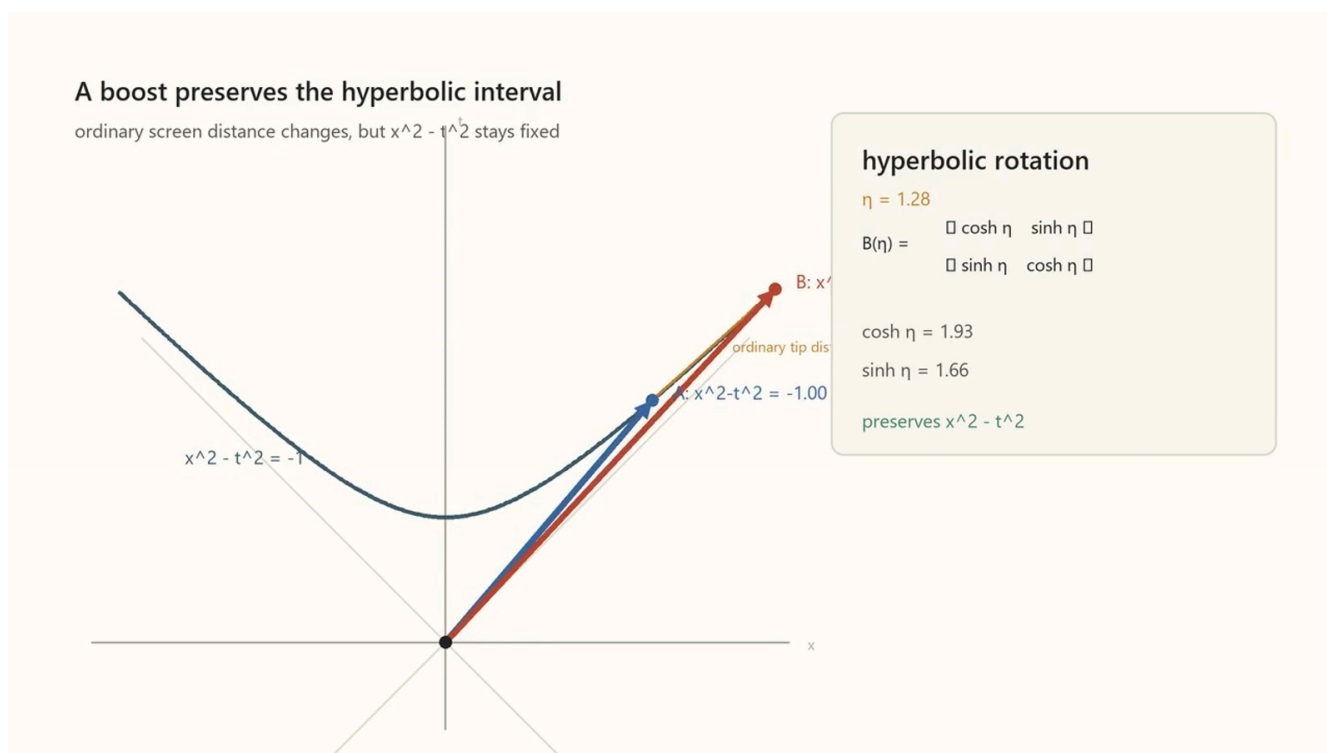
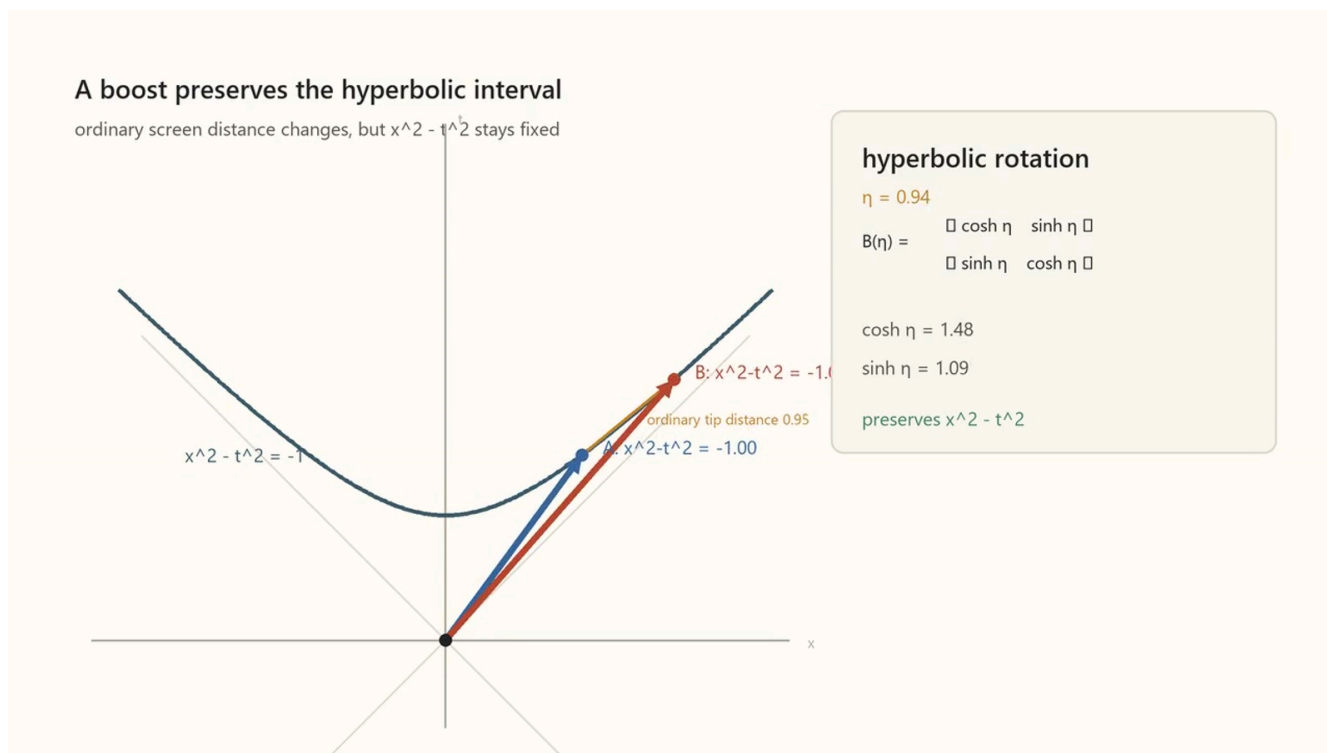
Hyperbolic rotations

Animation 8 / frame 1 of 4 / 0.14 s



Animation 8 / frame 2 of 4 / 2.31 s





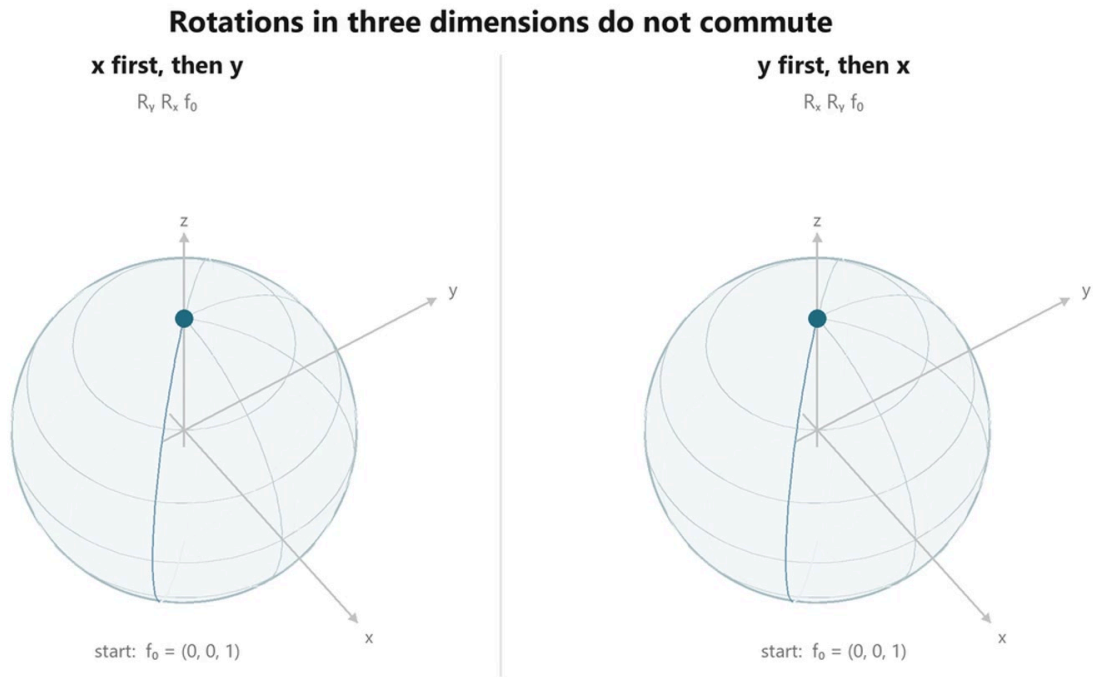
Commutators

In the case of a single continuous symmetry transformation, the generator combined with a parameter, like angle in the case of rotation, specifies an arbitrary transformation. However, when there are multiple independent transformations that can be composed to form a multi-dimensional group, the order of the composition may also need to be taken into account because it can produce different

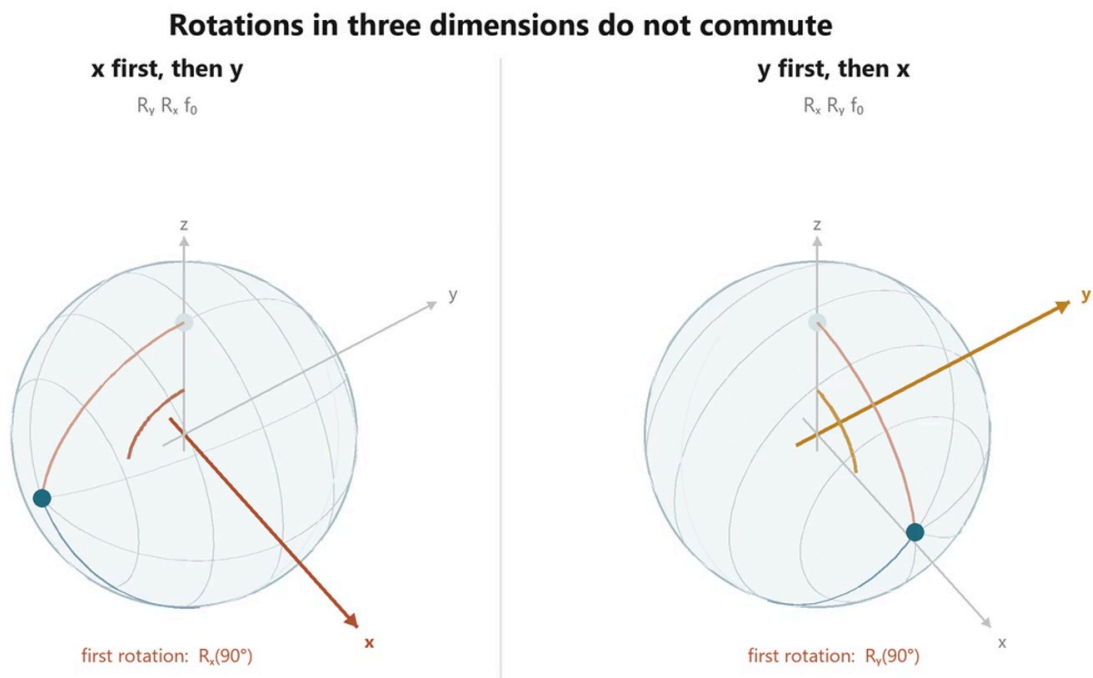
resultant states. For example, in the 3-dimensional rotation group, called $SO(3)$, rotating about the x -axis then the y -axis leaves a sphere in a different state than applying the same actions in the opposite order.

Non-commuting operations of SO_3

Animation 9 / frame 1 of 4 / 0.16 s



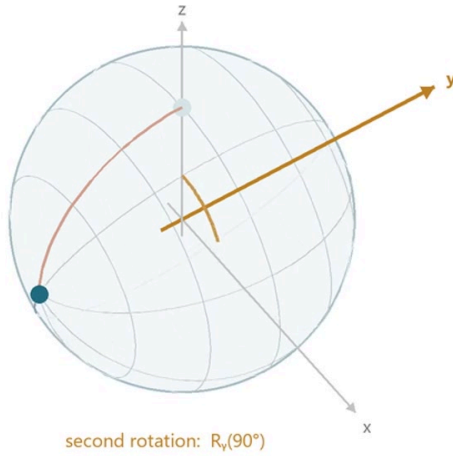
Animation 9 / frame 2 of 4 / 2.64 s



Rotations in three dimensions do not commute

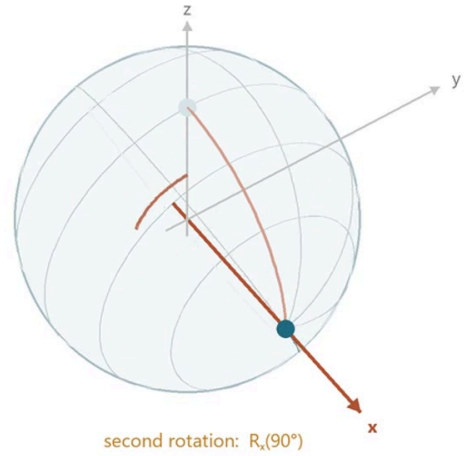
x first, then y

$$R_y R_x f_0$$



y first, then x

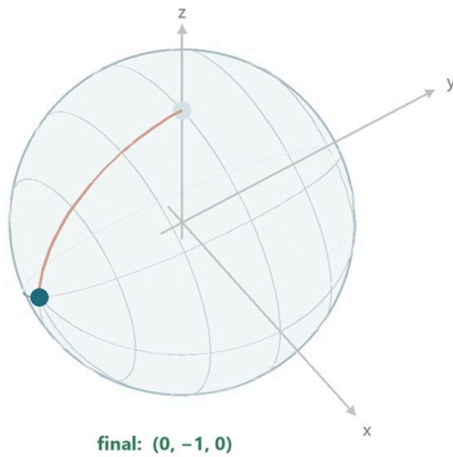
$$R_x R_y f_0$$



Rotations in three dimensions do not commute

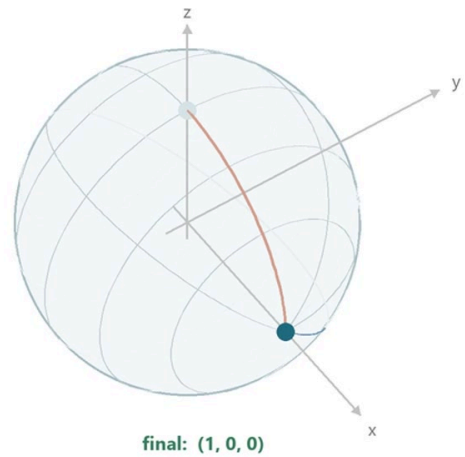
x first, then y

$$R_y R_x f_0$$



y first, then x

$$R_x R_y f_0$$



$$R_y R_x f_0 \neq R_x R_y f_0$$

This order-dependence is expressed by the **commutator**:

$$[X, Y] = XY - YX.$$

For example, for $\mathfrak{so}(3)$, the commutators are:

$$[J_x, J_y] = J_z, \quad [J_y, J_z] = J_x, \quad [J_z, J_x] = J_y.$$

In $SO(3)$, then, we know that rotating about $\mathbf{x}$ rotates the $\mathbf{y}$ axis into $\mathbf{z}$. We would write such a composite action as a product of exponential maps:

$$e^{aJ_x}e^{bJ_y}.$$

How then do we write this as a single exponential in terms of J_x and J_y ? For ordinary numbers, and for commuting operators:

$$e^Ae^B = e^{A+B}.$$

But this identity does *not* hold for noncommuting generators.

$$e^{aJ_x+bJ_y}$$

generates a rotation about a single axis in the $\mathbf{x}$ - $\mathbf{y}$ plane, which is not the same as performing the two rotations sequentially.

How can we correct the exponent so that the product can again be written as a single exponential? Keeping terms through second order in the small parameters a and b :

$$e^{aJ_x}e^{bJ_y} = \exp\left(aJ_x + bJ_y + \frac{1}{2}ab[J_x, J_y] + \dots\right) = \exp\left(aJ_x + bJ_y + \frac{1}{2}abJ_z + \dots\right).$$

The commutator supplies J_z .

A sphere gives one useful visualization of noncommutativity. On a flat plane, translations in two perpendicular directions commute. On a sphere of radius R , we can instead move along the surface by rotating about two perpendicular axes. Near the north pole, these motions act locally like translations in $\mathbf{x}$ and $\mathbf{y}$. Their generators are:

$$T_x = \frac{J_y}{R}, \quad T_y = -\frac{J_x}{R}, \quad [T_x, T_y] = \frac{J_z}{R^2}.$$

Here J_z generates rotation about the pole. The factor $1/R^2$ is the sphere's curvature. For fixed small travel distances, the residual rotation from moving, then undoing the movements in the same order, shrinks as the sphere flattens.

Lie Algebra

Taken together, the dimension of the symmetry group and the commutators specify a **Lie algebra** that contains the structure of the symmetry group, up to global topological features -- such as a symmetry group wrapping around onto itself such as a plane wrapped around a cylinder -- that are invisible to local structure. This algebra consists of the group generators, linear combinations of them, and their commutator. Every representation must preserve this commutator structure. Therefore, if we know the Lie algebra, we can use it to assess if a representation is valid.

Invariants and Casimir operators

Lie algebra provides a procedure for finding invariant operators. If we can construct an operator from the generators that commutes with all the algebra's generators, we then know that the operator is an invariant under the symmetry. As an illustration, in 3-dimensional rotation, with generators denoted L_i for angular momentum, we can see that the square of the rotation generators is invariant:

$$L_i = i\hbar J_i, \quad [L_x, L_y] = i\hbar L_z \quad (\text{cyclically}), \quad L^2 = L_x^2 + L_y^2 + L_z^2.$$

$$\begin{aligned}
&= [L_y^2, L_x] + [L_z^2, L_x] \\
&= L_y[L_y, L_x] + [L_y, L_x]L_y + L_z[L_z, L_x] + [L_z, L_x]L_z \\
&= i\hbar(-L_yL_z - L_zL_y + L_zL_y + L_yL_z) \\
&= 0.
\end{aligned}$$

By the same calculation,

$$[L^2, L_y] = [L^2, L_z] = 0.$$

This is common sense. The magnitude of angular momentum is independent of the direction of the axis.

An invariant constructed this way is a Casimir operator. Casimir eigenvalues supply labels for the irreducible representations in which particle states live and thereby help classify particle types, such as electrons and photons. For example, as we will see in detail later, the invariant mass of a particle is determined by the eigenvalue of a Casimir operator built from the generators of time and space translations. Casimir operators can be built from both spacetime symmetries and internal symmetries that rotate phases or mix components of the wave function.

Translations and Function Representation

Translation is, to the eye, the simplest possible symmetry. It would be natural to think the ideal representation space for a translation is simply a one dimensional vector space, where points move via "sliding a number line." But there is a problem. If an operation is to move x by some amount a

$$x \mapsto x + a,$$

then 0 is moved to a , but the origin must remain fixed in a linear vector space. This is a special case of the rule for linearity. A linear operation T must preserve linear combinations:

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v).$$

For translation of points on the number line, the candidate operator is

$$T_a(x) = x + a.$$

But this does not preserve addition. For two points x_1 and x_2 ,

$$T_a(x_1 + x_2) = x_1 + x_2 + a,$$

while

$$T_a(x_1) + T_a(x_2) = (x_1 + a) + (x_2 + a) = x_1 + x_2 + 2a.$$

These are not equal unless $a = 0$. Translation does act sensibly on a number line, but it is not a linear operation when acting on this space.

Shifting a number line is not a linear operation

Animation 10 / frame 1 of 4 / 0.13 s

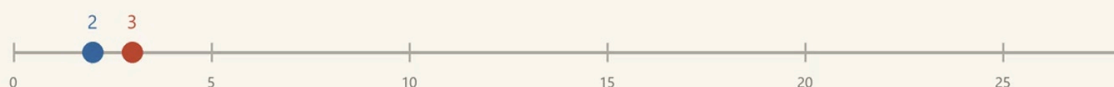
Point translation is not linear

Use $x_1=2$, $x_2=3$, and $a=10$

add first, then translate
 $2 + 3 = 5$



translate first, then add



Animation 10 / frame 2 of 4 / 2.15 s

Point translation is not linear

Use $x_1=2$, $x_2=3$, and $a=10$

add first, then translate
 $2 + 3 = 5$

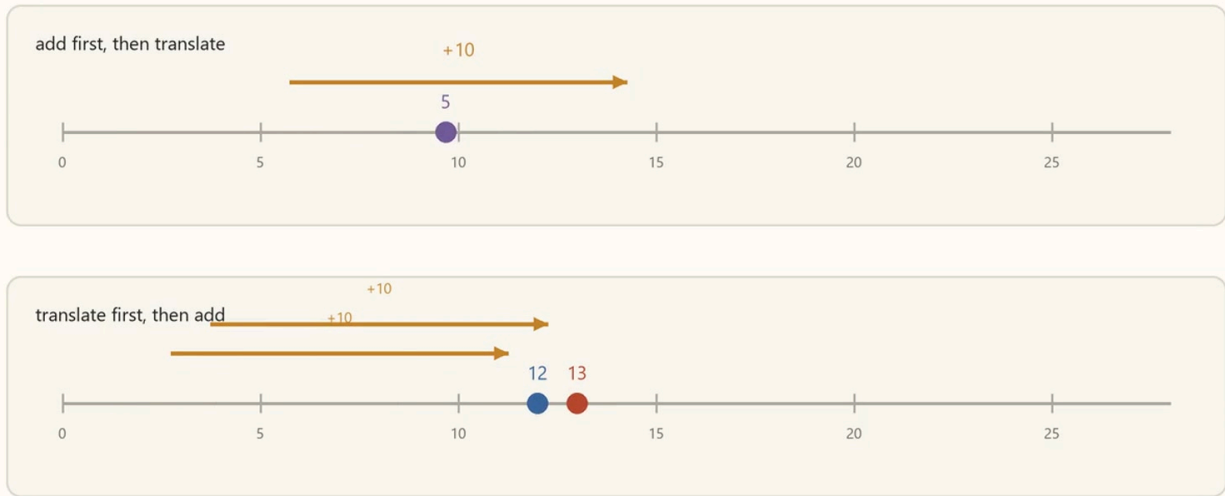


translate first, then add



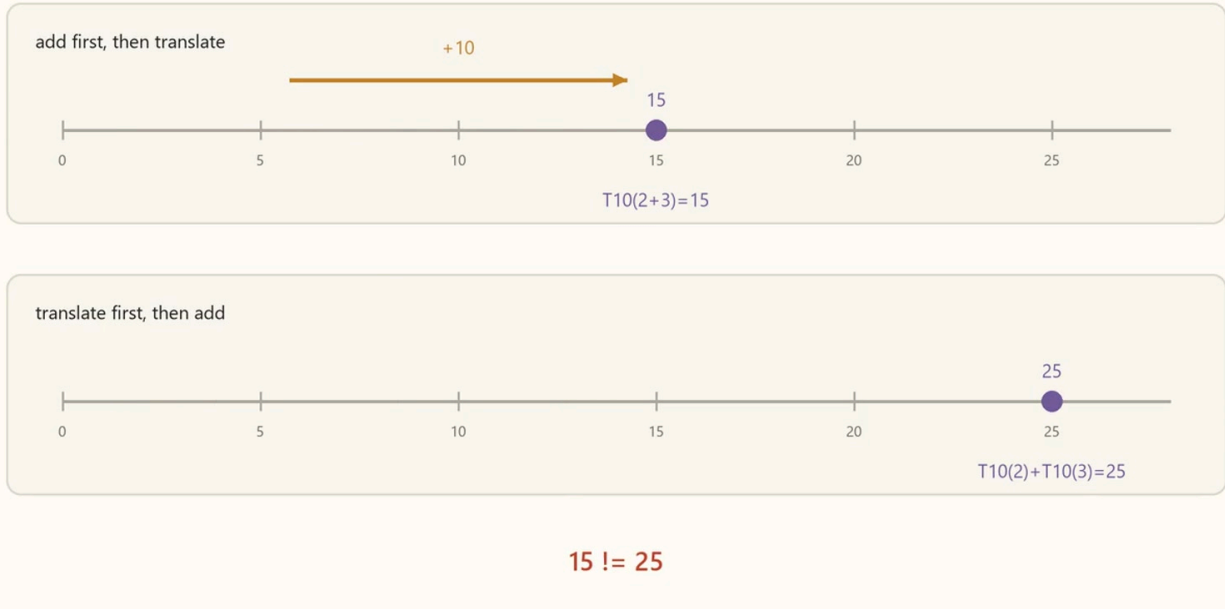
Point translation is not linear

Use $x_1=2$, $x_2=3$, and $a=10$



Point translation is not linear

Use $x_1=2$, $x_2=3$, and $a=10$



On the other hand, the space of functions on a number line does form a linear representation. Instead of translating the point x , we translate a function by shifting its argument:

$$(T_a f)(x) = f(x - a).$$

Now the state is the whole function f , and T_a is an **operator** on the vector space of functions. An operator is a generalization of a matrix when applied to continuous functions. A continuous function is as an infinite-dimensional vector, in which its domain values are "axes," or component labels, and its

range values are the component values. We can transform one function to another with an infinite dimensional matrix, but in practice, we can condense this into a well-known operation, such as the derivative operation.

Linearity works because function addition is pointwise:

$$T_a(f + g)(x) = (f + g)(x - a) = f(x - a) + g(x - a) = (T_af)(x) + (T_ag)(x).$$

So

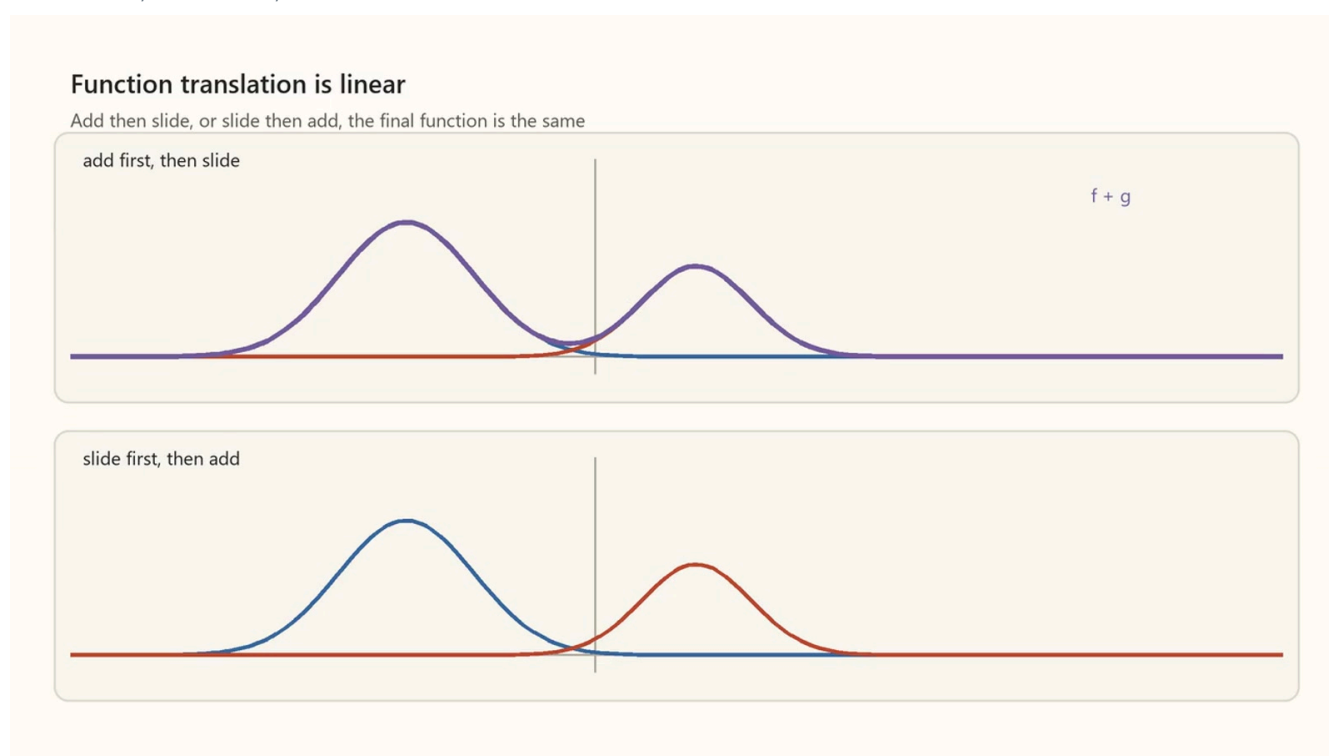
$$T_a(f + g) = T_af + T_ag.$$

The zero function is also fixed:

$$T_a 0 = 0.$$

A function representation is linear

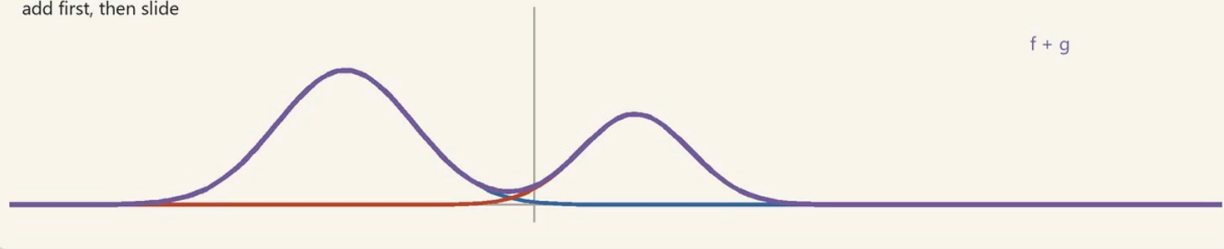
Animation 11 / frame 1 of 4 / 0.13 s



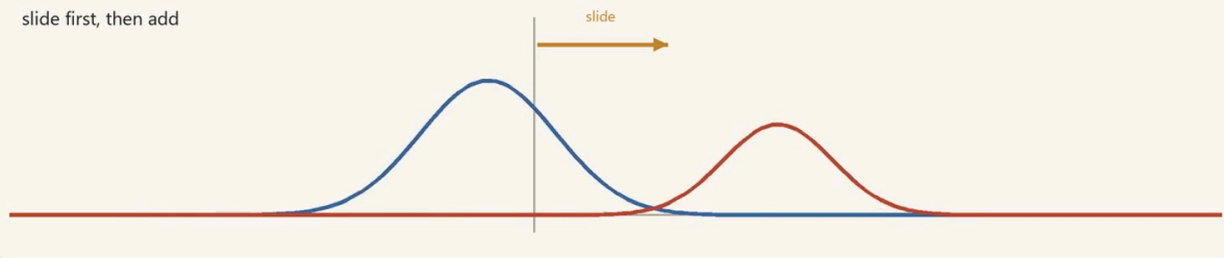
Function translation is linear

Add then slide, or slide then add, the final function is the same

add first, then slide



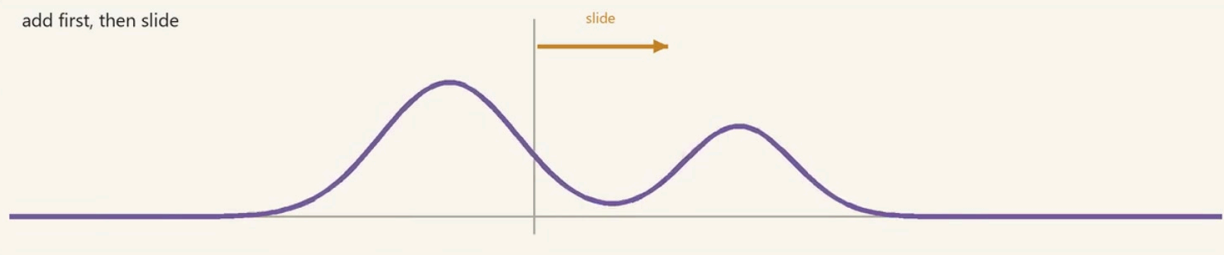
slide first, then add



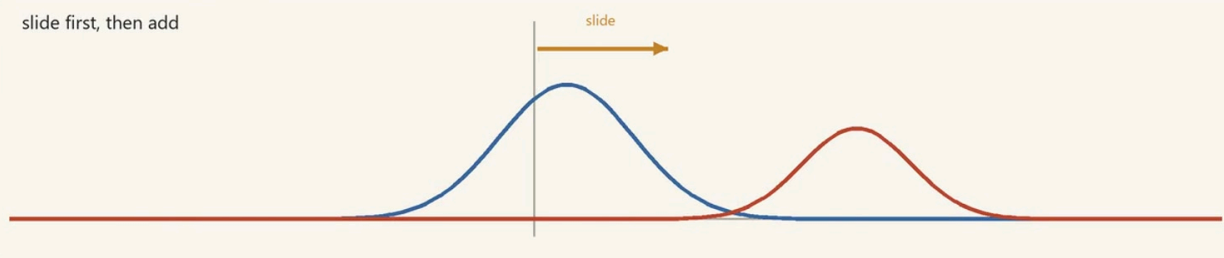
Function translation is linear

Add then slide, or slide then add, the final function is the same

add first, then slide

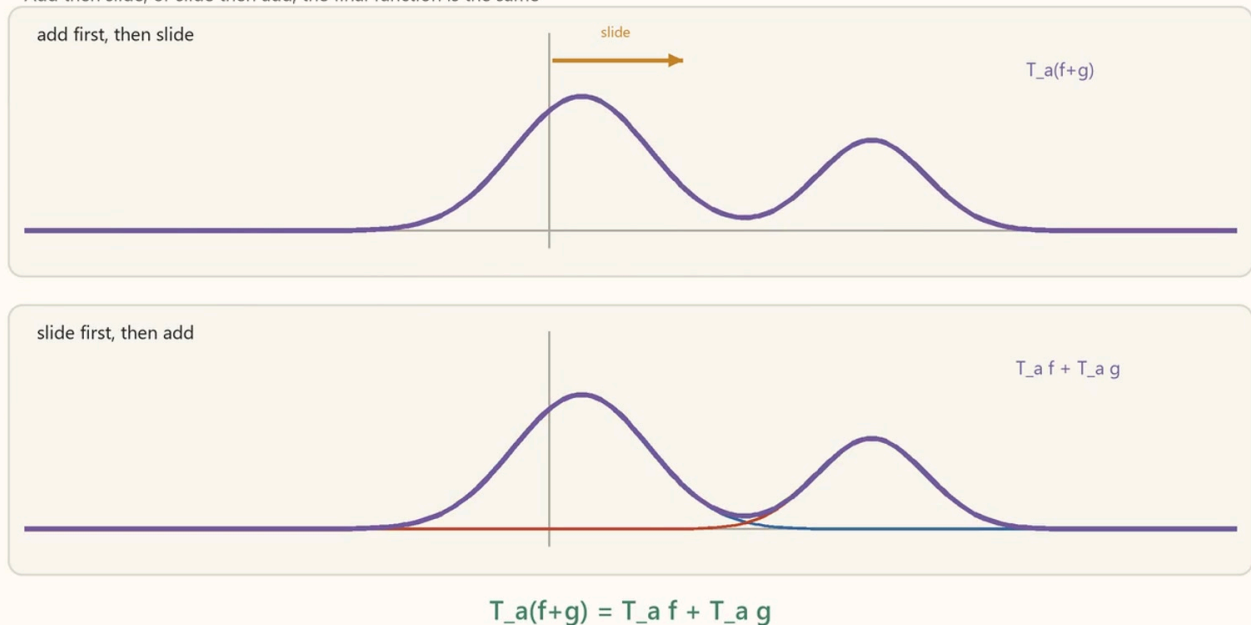


slide first, then add



Function translation is linear

Add then slide, or slide then add, the final function is the same

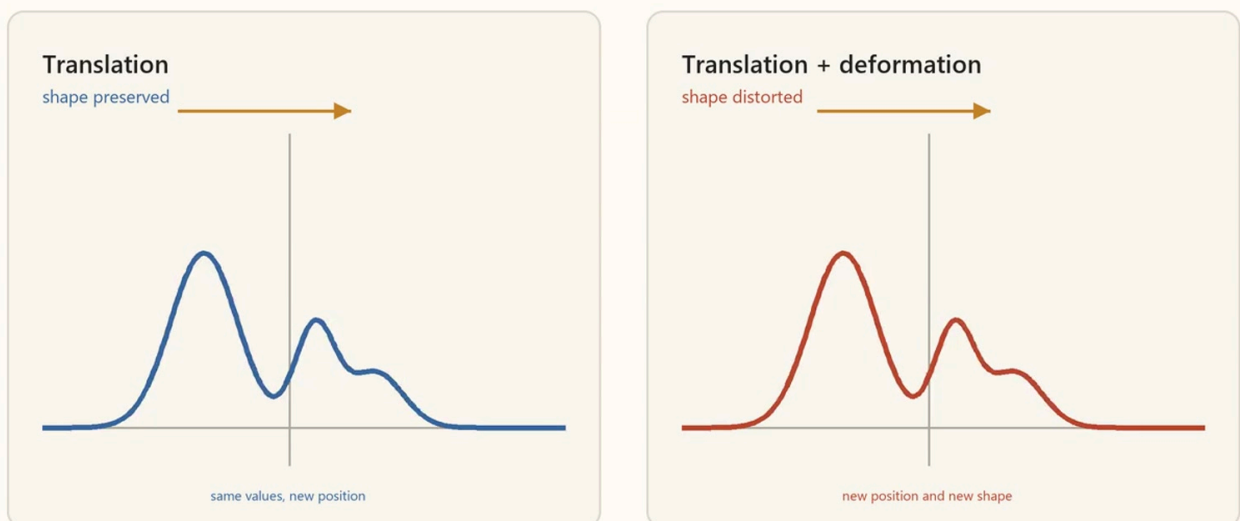


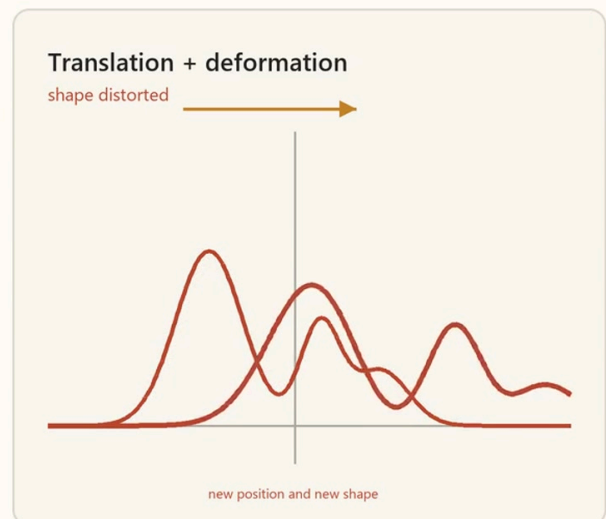
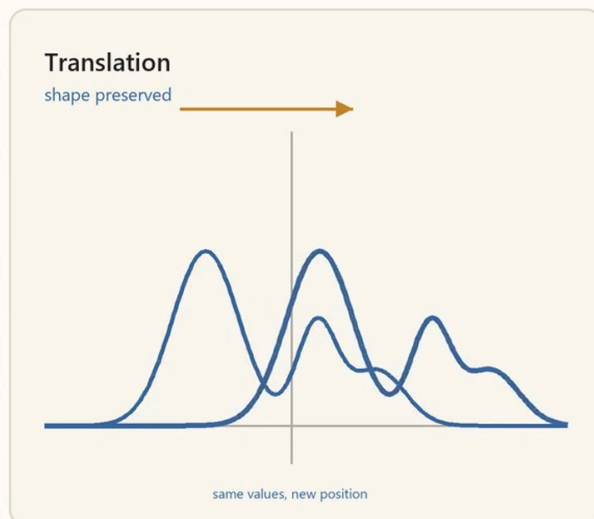
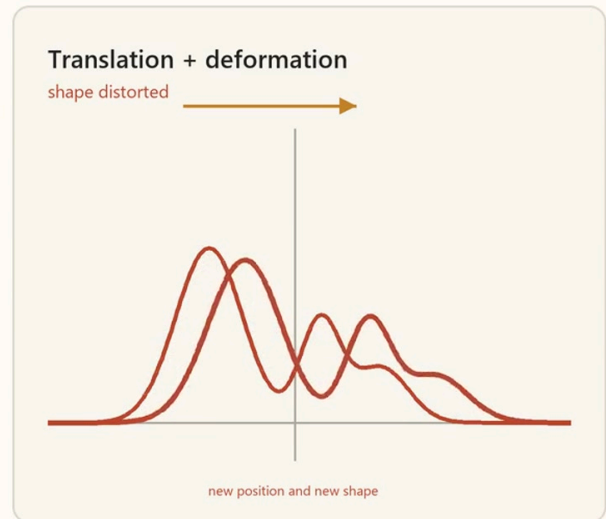
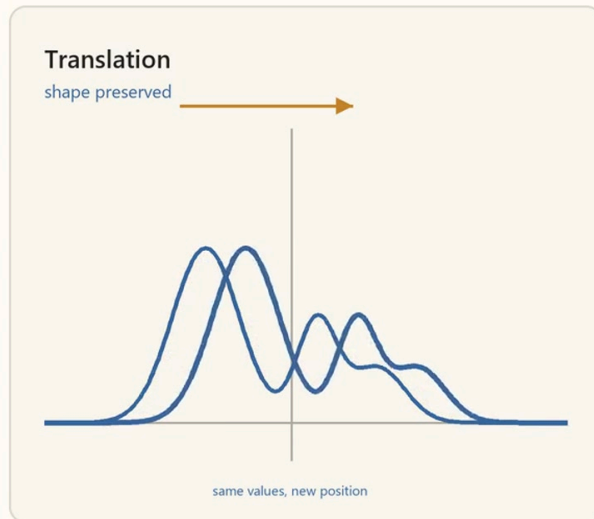
This makes sense. Translational symmetry needs an object to translate, just as D_3 symmetry needs a triangly thing to translate. We can think of the function as a shape. If a system possesses translational symmetry, that shape is preserved.

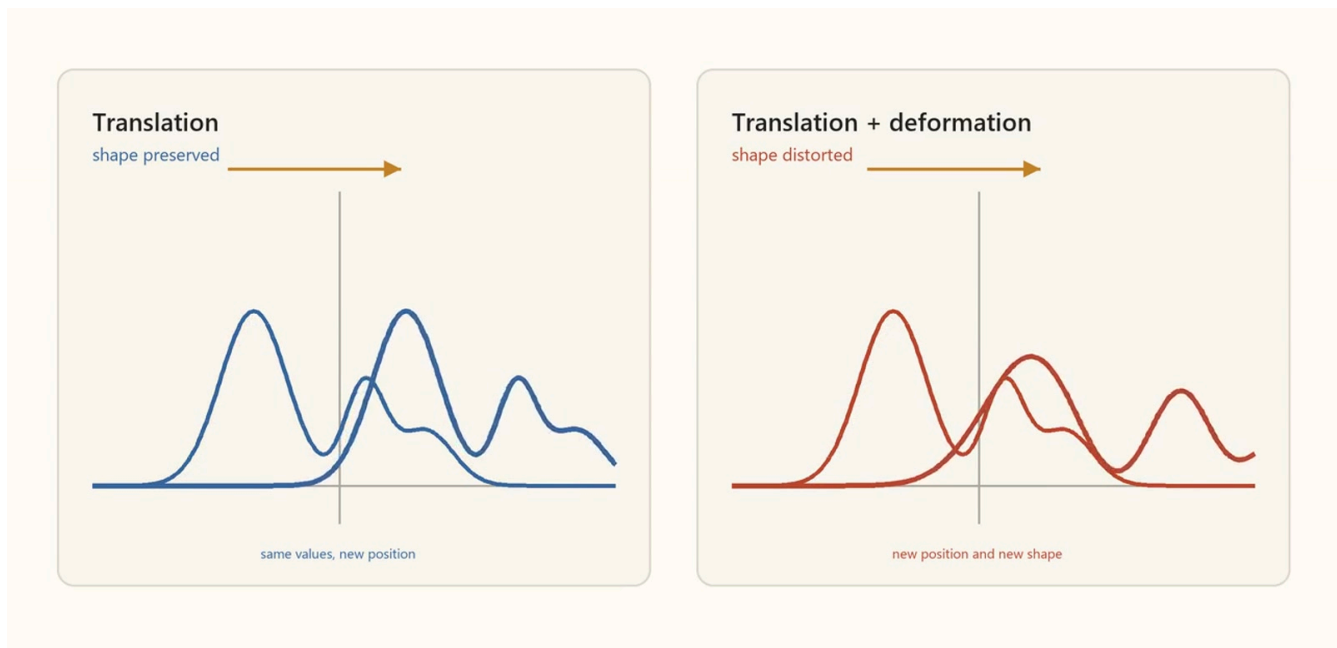
This describes how the function transforms. Whether the physical system it describes behaves the same way after translation is a separate question.

Translation of functions preserves their shape

Animation 12 / frame 1 of 4 / 0.11 s







Indeed, this representation is completely analogous to our 3 dimensional representation of D_3 . There a vertex contained a component value, here an input value does. Indeed, the D_3 representation is itself a function representation, with a domain that contains only three members and is cyclic.

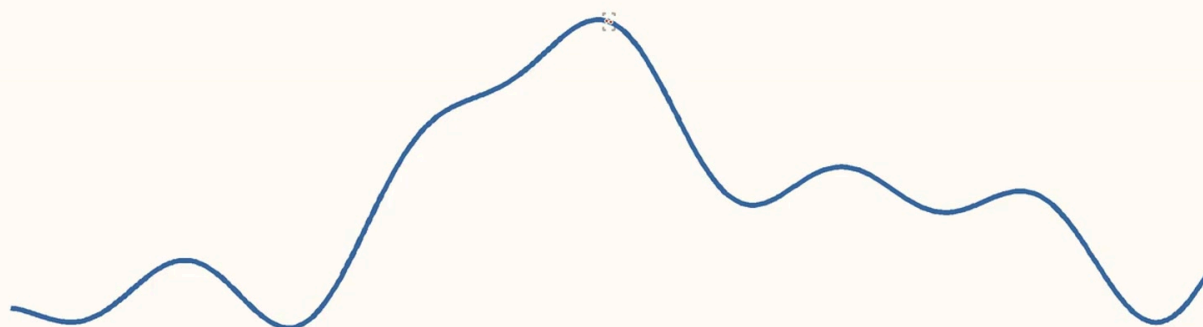
Generator of Translations

We have a function representation. Now we want to find an infinitesimal generator that acts on functions to produce a translation. We can do so by considering what happens to a function's value under "tiny" displacements. In that case, the new value $f(x - a)$ is close to the old value $f(x)$, and correction is given, in the infinitesimal limit, by the slope at x . This is the same idea that any curve becomes flat when zoomed in sufficiently:

A function value is translated by the function's slope

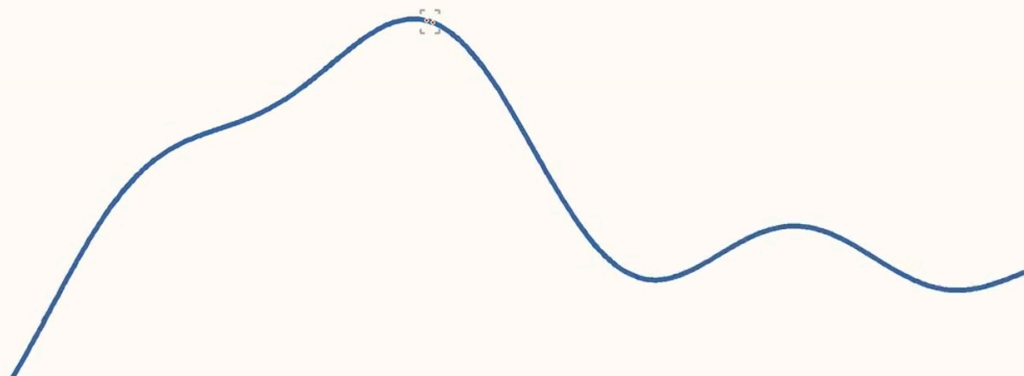
Animation 13 / frame 1 of 4 / 0.30 s

Every smooth curve looks straight up close



Animation 13 / frame 2 of 4 / 4.95 s

Every smooth curve looks straight up close



Every smooth curve looks straight up close



Every smooth curve looks straight up close



That is the visual version of the first-order **Taylor expansion** that relates a function's derivatives to its value under translation:

$$f(x - a) = f(x) - a \frac{df}{dx} + O(a^2).$$

As a approaches 0, the higher order terms vanish and the shift in f 's values is produced by the operator d/dx :

$$-i\hat{K} = -\frac{d}{dx}, \quad \hat{K} = -i\frac{d}{dx}, \quad T_a = e^{-ia\hat{K}}.$$

Primer -- Eigenfunctions

Readers familiar with eigenvectors can [skip this primer](#).

Imagine a rubber sheet. We pull on its corners. What does this do to the x - and y -axes? It rotates them toward each other while stretching them. Now, instead choose x and y to be diagonal axes. Now, when we stretch, the long axis is stretched but not rotated and the short axis is compressed but not rotated. The action of this stretching action on these axes is now simple scalar multiplication of the original vector. Once the basis vectors no longer mix, any other vector, that is, any linear combination of the basis vectors, transforms by having its components scaled independently:

Letting s be the factor by which the long-axis component is stretched, we can see how the transformation of an arbitrary vector $\mathbf{r}$ simplifies using the system's natural basis:

(x, y) basis: components mix

$$\begin{aligned} \mathbf{r} &= \frac{1}{2} \begin{pmatrix} s + s^{-1} & s - s^{-1} \\ s - s^{-1} & s + s^{-1} \end{pmatrix} \mathbf{r}_{\text{in}} \\ &= \frac{1}{2} [(s + s^{-1})x + (s - s^{-1})y] \hat{\mathbf{x}} \\ &\quad + \frac{1}{2} [(s - s^{-1})x + (s + s^{-1})y] \hat{\mathbf{y}} \end{aligned}$$

(u, v) basis: components do not mix

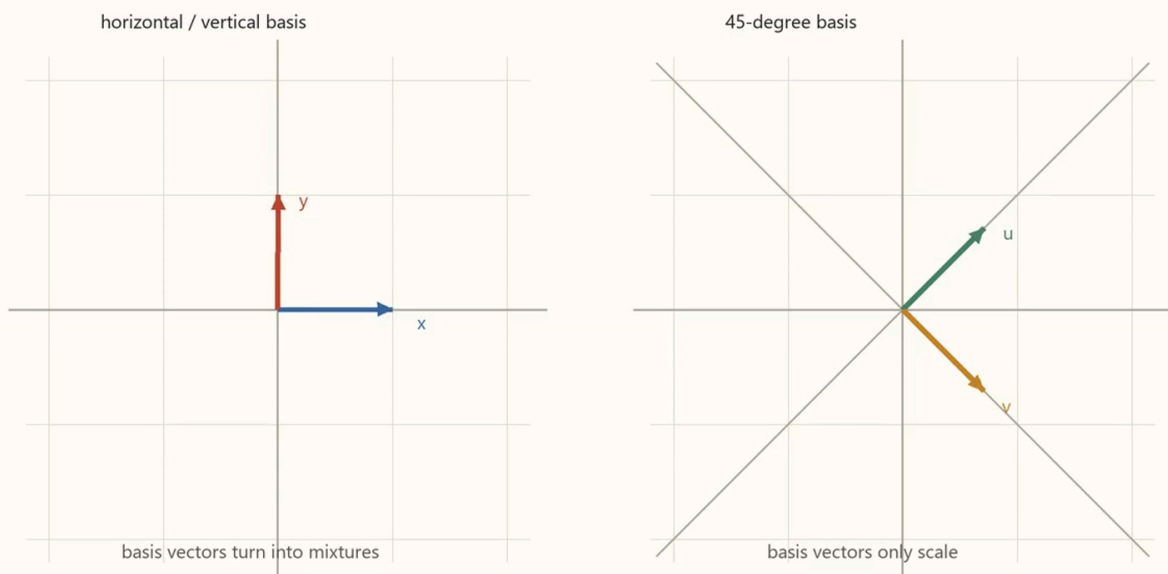
$$\begin{aligned} \mathbf{r} &= \begin{pmatrix} s & 0 \\ 0 & s^{-1} \end{pmatrix} \mathbf{r}_{\text{in}} \\ &= su\hat{\mathbf{u}} + s^{-1}v\hat{\mathbf{v}} \end{aligned}$$

This is readily understood visually:

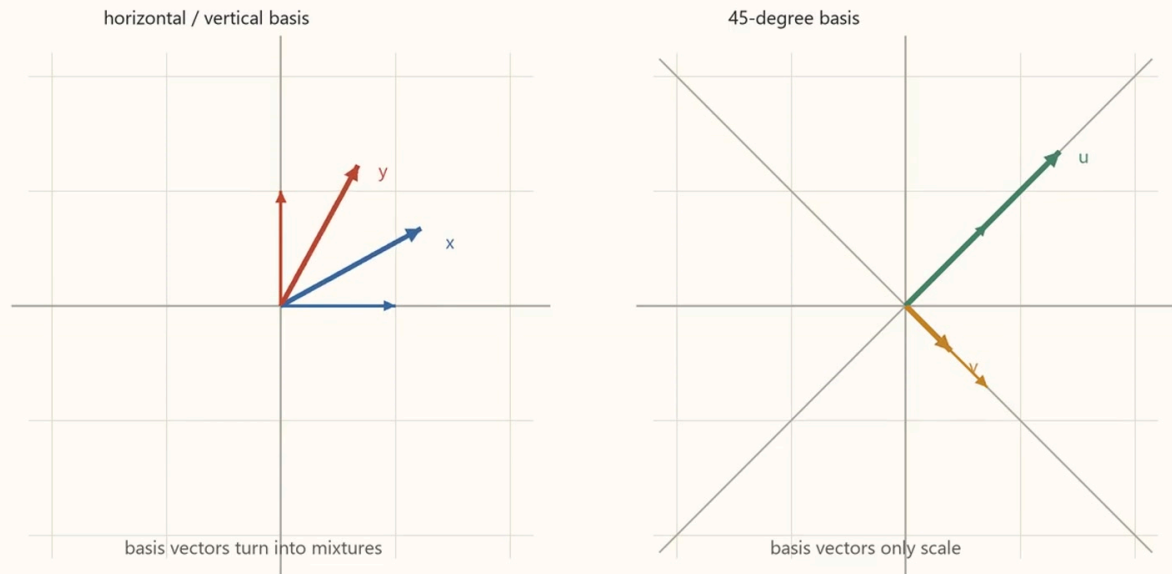
Eigenvectors pictured

Animation 14 / frame 1 of 4 / 0.29 s

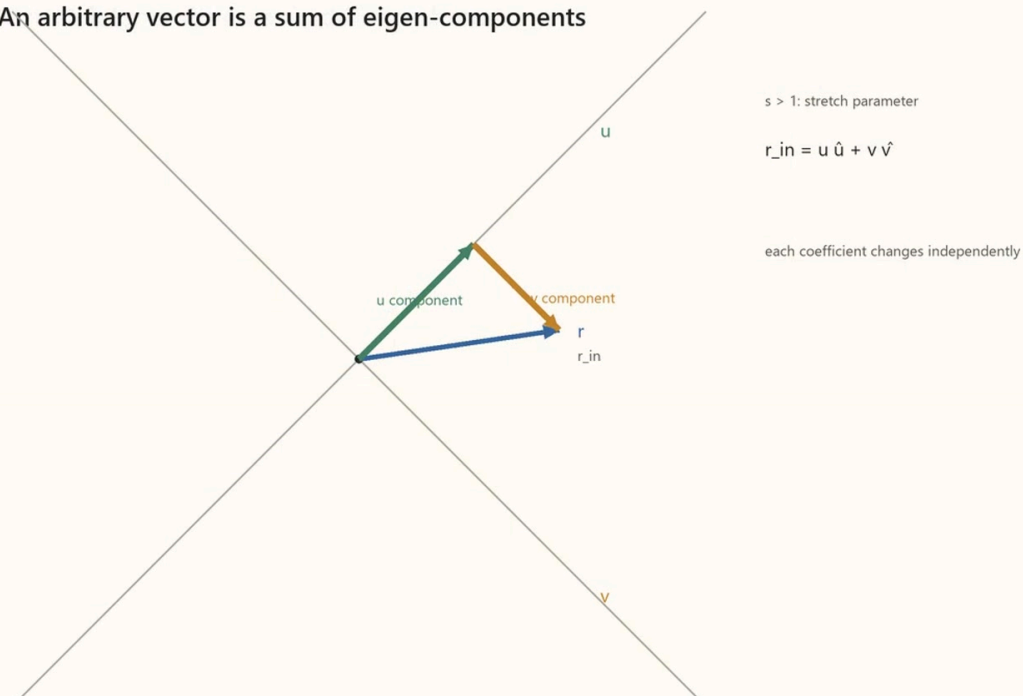
A stretch looks different in different bases

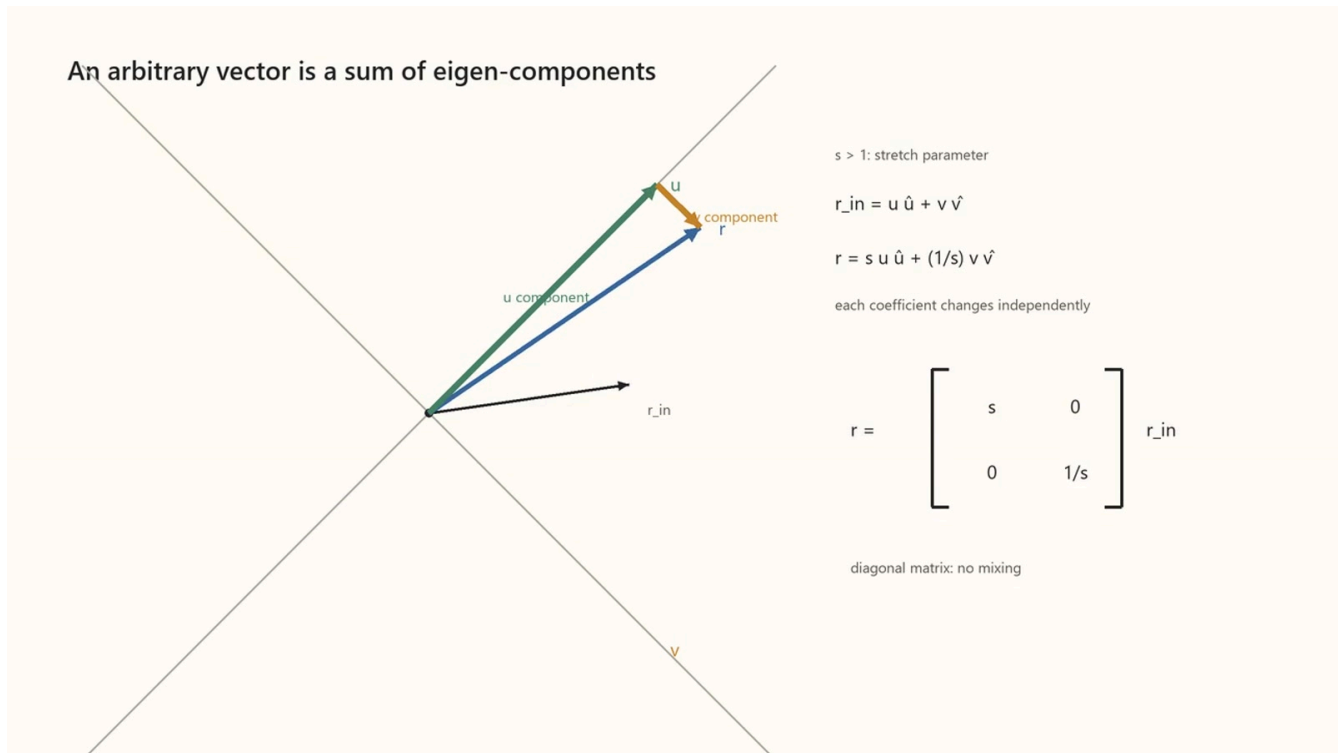


A stretch looks different in different bases



An arbitrary vector is a sum of eigen-components





Now for a bunch of terminology. The "natural" basis vectors are the **eigenvectors**, the values a transformation scales these by are the **eigenvalues** and the basis they form is called the **eigenbasis**. When the "vectors" are functions, we call them **eigenfunctions**. As the transformation matrix is diagonal in the eigenbasis, the procedure for finding an eigenbasis is typically called **diagonalization**.

Eigenfunctions of Translational Symmetry

Since a function is but an infinite-dimensional vector, and the generator d/dx is a way of writing an infinite-dimensional matrix, we can solve a similar eigenvalue equation here, which now takes the form of a differential equation.

The eigenvalue equation is

$$\frac{d}{dx} f(x) = \lambda f(x).$$

This says we are looking for a function whose derivative returns the same function, scaled by a single number. The solution is:

$$f(x) = C e^{\lambda x}.$$

Since:

$$\frac{d}{dx} C e^{\lambda x} = C \lambda e^{\lambda x}$$

the eigenvalue of d/dx is λ .

Translation preserves the shape of a linear combination of eigenfunctions, and with that, its distribution over eigenvalues. When translation is a symmetry of the dynamics, this distribution is conserved. This is the structure behind momentum and energy conservation. As we will see later, momentum is associated with wave number, and energy with frequency. Additionally, in quantum

mechanics, a measurement value must take an eigenvalue of the operator associated with that measurement. When we identify the eigenfunctions and eigenvalues of a symmetry generator, we thereby determine the possible outcomes of the associated measurement. We will have much more to say about this.

Unitarity

We now have a potential way to represent a state in an eigenbasis of translation operators using exponential functions. This basis must satisfy an additional condition. The relationship between two states must be preserved by symmetry actions. That is, states must remain distinguishable. Their "overlap" must be preserved:

$$\langle U\psi_1, U\psi_2 \rangle = \langle \psi_1, \psi_2 \rangle \quad \text{for every symmetry action } U.$$

This property is known as **unitarity**.

It is straightforward but not terribly enlightening to show algebraically that only exponentials with purely imaginary exponents satisfy this requirement. We will look into this more in the next few sections, but for a bit of intuition now, consider that translating real exponentials scales different functions by different amounts, whereas translating complex exponentials, which project to sinusoidal waves, only moves their cyclic pattern, leaving their relative magnitudes intact.

Primer - Complex exponentials and Wave Packets

If you are familiar with complex exponentials, feel free to [skip this section](#).

We asserted above that the complex exponential function describes a plane wave. Let's explain that and generally build some intuition around the complex exponential function. A single parameter complex exponential, $e^{i\theta}$, describes a circle. It is a way to repackage the rotation operator we have already seen:

$$e^{i\theta} \longleftrightarrow e^{\theta J} = R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

i is the generator of rotation in this 1-dimensional complex representation and plays the role J did in the 2-dimensional representation:

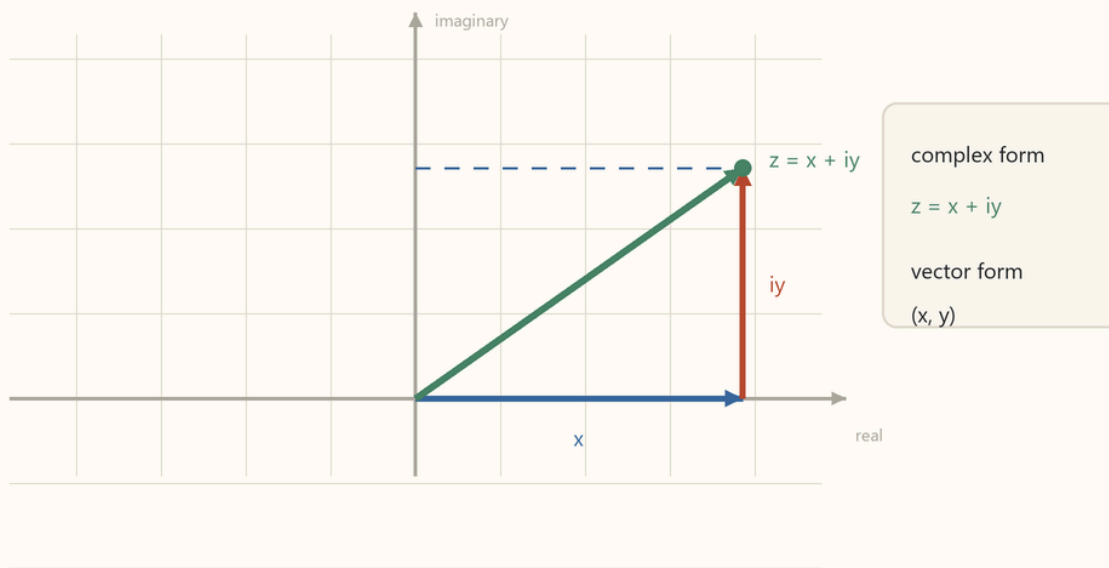
$$i^2 = -1 \longleftrightarrow J^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = -I.$$

How is it possible to replace a **2x2** matrix with a single number i ? The answer is that z has two real-number degrees of freedom. That is, we define a complex number z :

$$z := x + iy,$$

thus a complex number is mapped to a vector in the complex plane.

A complex number is a two-component vector



The complex plane

What action does multiplication by i have at the identity vector 1 ?

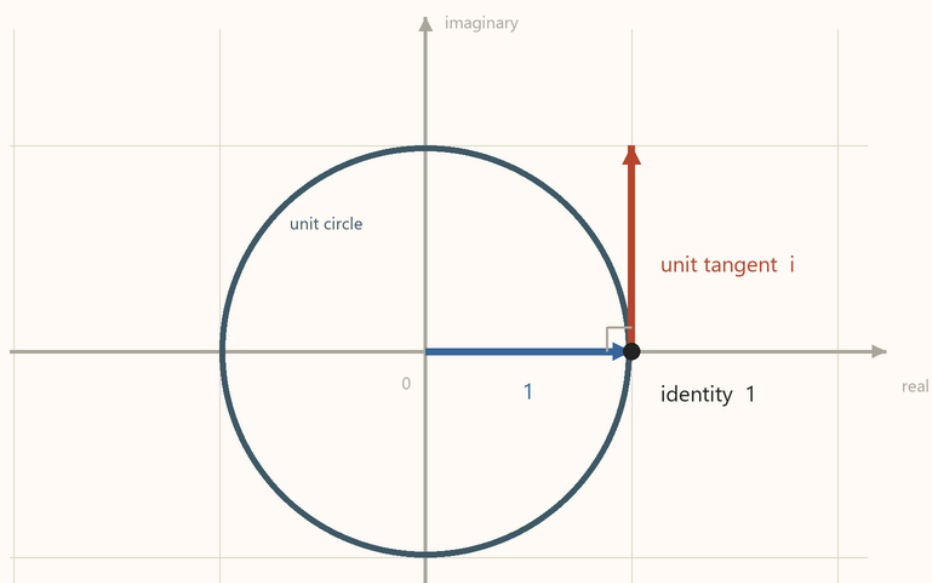
Here:

$$z = 1 = 1 + 0i$$

$$z_{\text{tan}} = zi = 1 \cdot i = i = 0 + 1i$$

This is a vector in the complex plane that is perpendicular to the identity.

The unit tangent at the identity



The action of i

This is exactly the action the $\mathbf{J}$ matrix generator had in 2-dimensions, which we already argued exponentiates to rotation. Let's redo that argument here, as it leads to a gratifying result. Since multiplication by i produces the tangent vector at every point and the tangent vector is the geometric version of the derivative, we have:

$$\frac{dz}{d\theta} = i z(\theta)$$

With $z(0) = 1$, solving this differential equation we then have:

$$z(\theta) = e^{i\theta}$$

At $\theta = \pi$, halfway around the unit circle, we then have:

$$e^{i\pi} = -1$$

This is Euler's identity, satisfyingly relating three of nature's most fundamental constants.

We can map z back to x, y coordinates and obtain Euler's formula:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Engineers often represent real sinusoidal waves with complex exponentials using this formula and reading off the real part as multiplying exponentials is algebraically simpler than multiplying trigonometric functions.

Now let us turn from circular motion to a travelling complex wave. Let the phase vary with position x and time t :

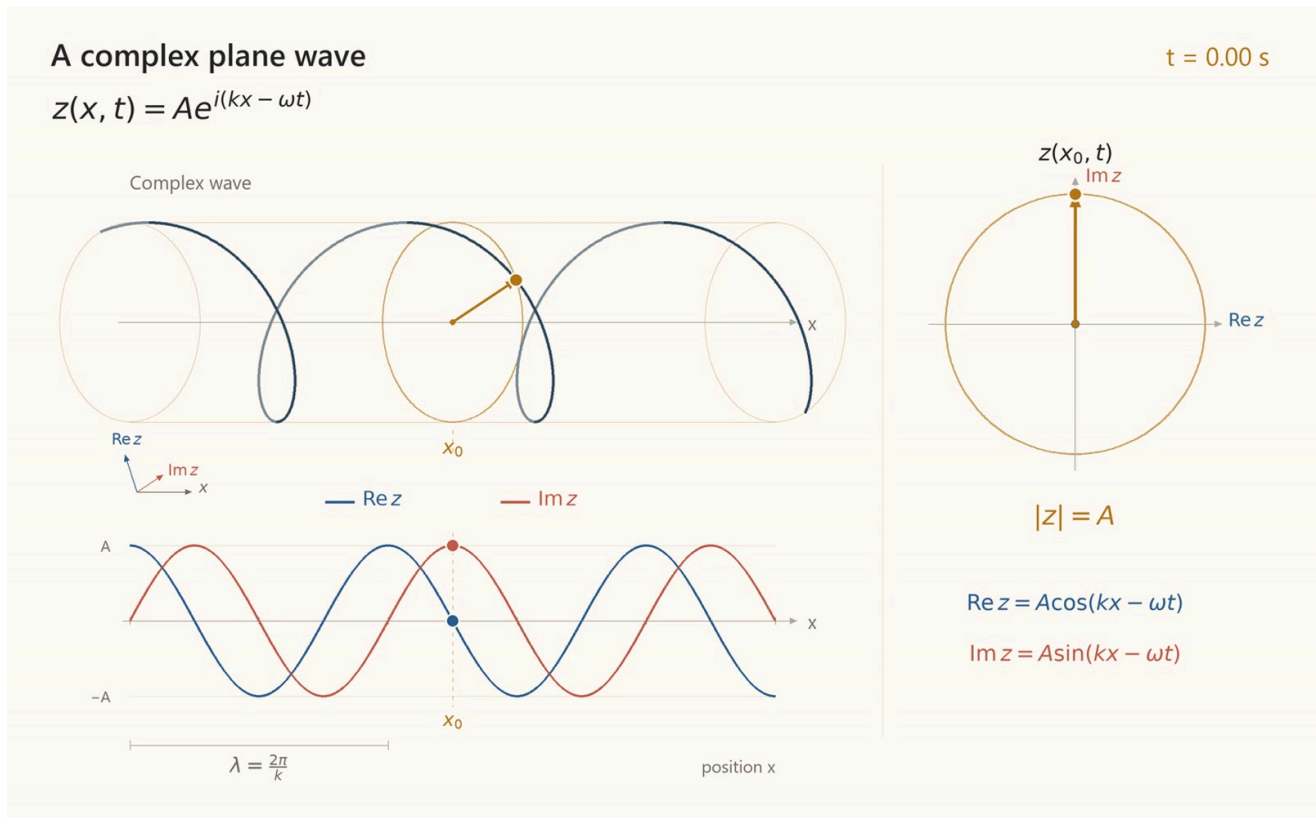
$$\theta(x, t) = kx - \omega t, \quad z(x, t) = Ae^{i(kx - \omega t)}.$$

Here A is the wave's amplitude, λ is its wavelength, $k = 2\pi/\lambda$ is its **wave number**, and ω sets how quickly the phase changes with time.

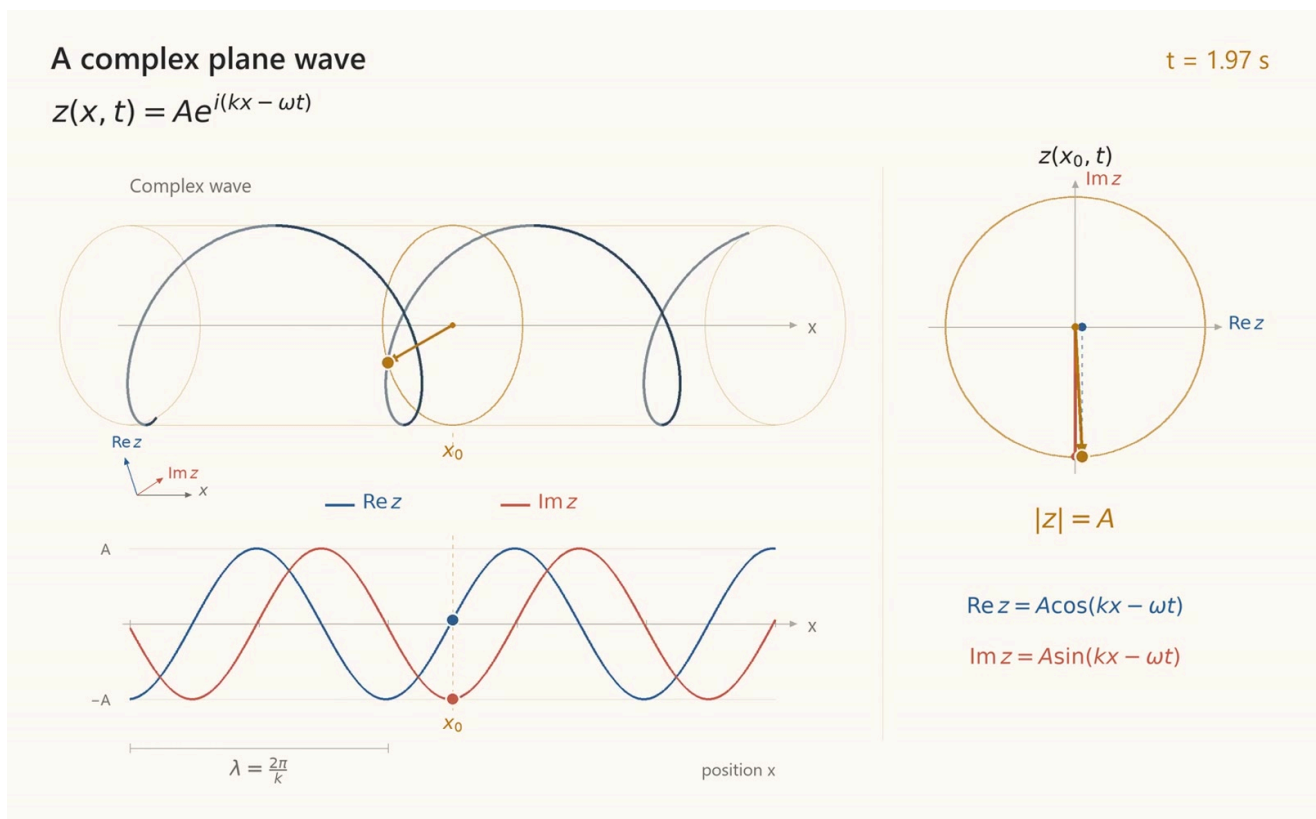
In the animation, position runs along the helix. The other two directions show the real and imaginary parts of the complex value at each position. As time advances, these values rotate and the wave pattern travels along x .

The complex value rotates while its length stays fixed.

Animation 15 / frame 1 of 4 / 0.24 s



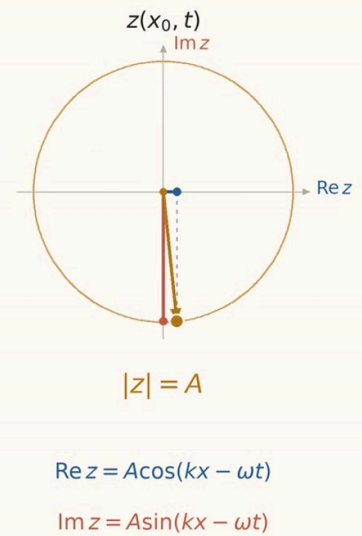
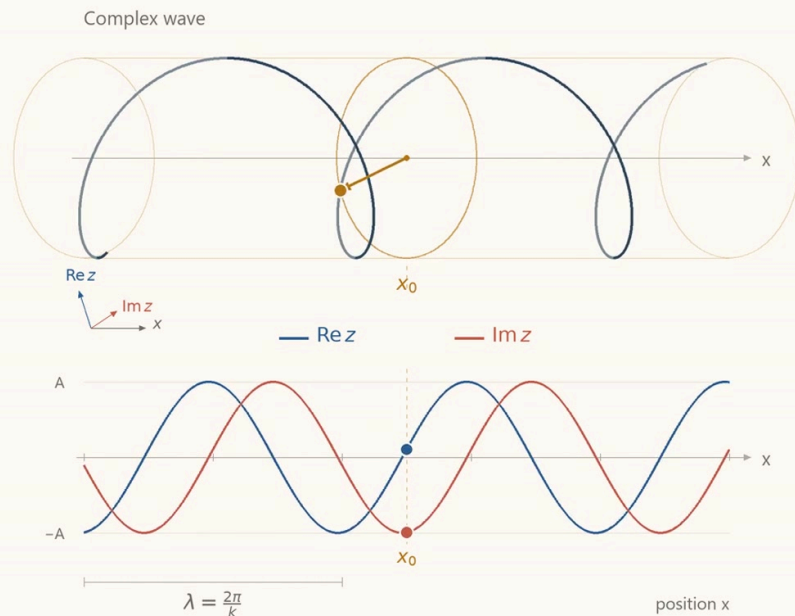
Animation 15 / frame 2 of 4 / 3.96 s



A complex plane wave

$$z(x, t) = Ae^{i(kx - \omega t)}$$

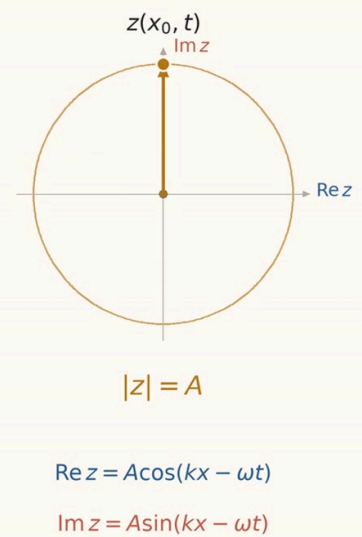
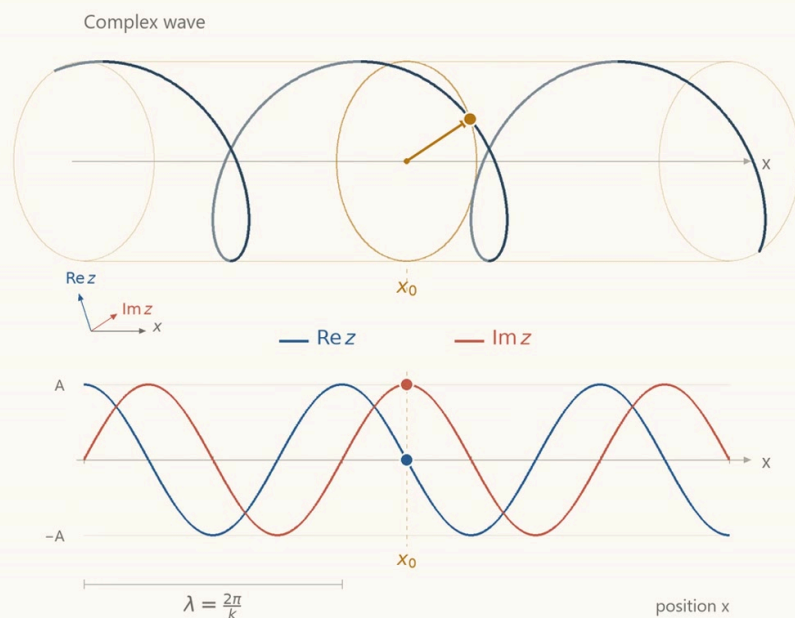
$t = 5.93 \text{ s}$



A complex plane wave

$$z(x, t) = Ae^{i(kx - \omega t)}$$

$t = 8.00 \text{ s}$



The two components are $A \cos \theta$ and $A \sin \theta$. Their values oscillate, while the length of the complex value stays fixed:

$$|z| = \sqrt{A^2 \cos^2 \theta + A^2 \sin^2 \theta} = A.$$

Translating this wave multiplies it by a phase factor:

$$z(x - a, t) = e^{-ika} z(x, t).$$

The eigenvalue e^{-ika} has magnitude one. This is the phase-only scaling allowed by unitarity.

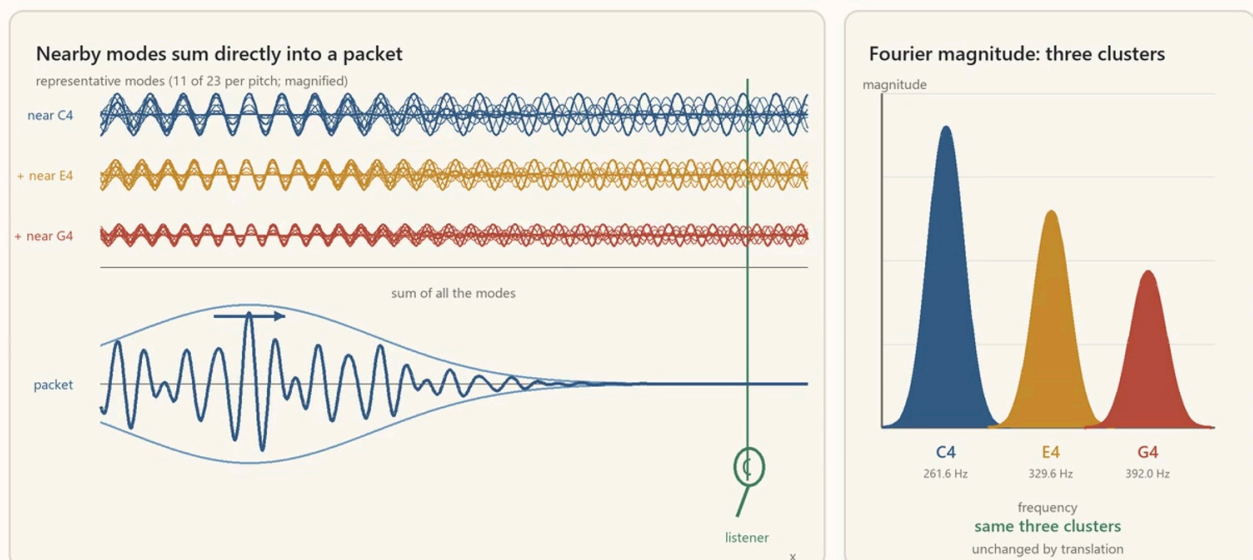
The Fourier Structure of Waves

If you strike a chord on a piano, some complicated function of time describes how the sound pressure reaches your ear. It starts soft, gets louder, softens again. It has discernible main tones, but also a clutter of overtones that comprise the timbre of the piano. While you hear a clear tonal structure, a plot of the sound pressure level over time reaching your ear would completely obscure that structure.

Three-tone chord packet and its Fourier decomposition

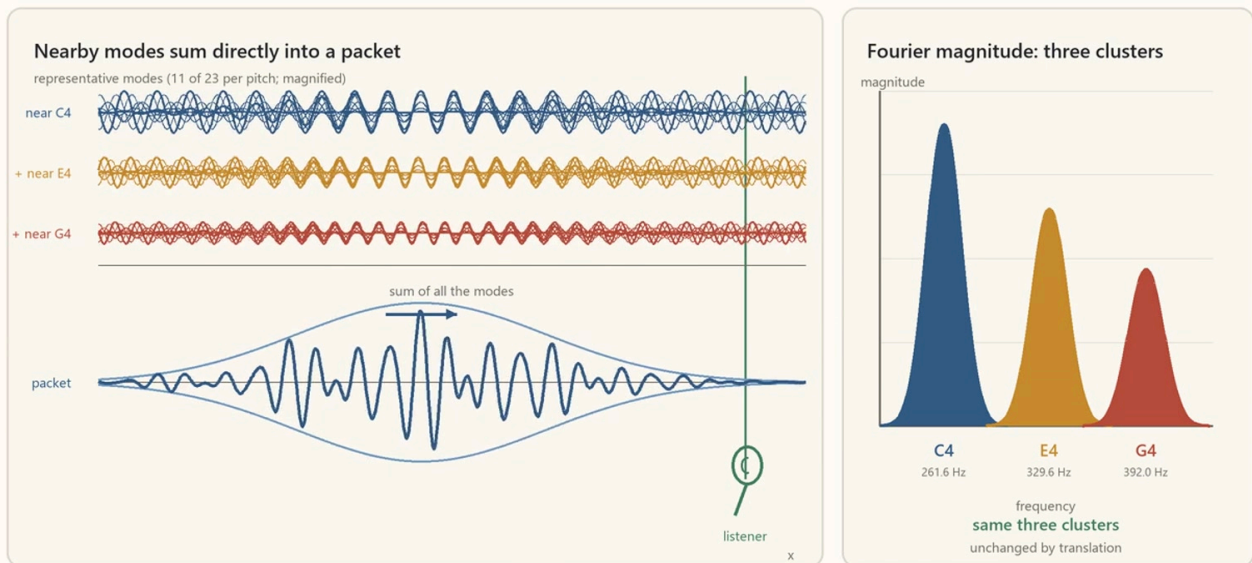
Animation 16 / frame 1 of 4 / 0.16 s

A chord travels; its Fourier composition stays fixed



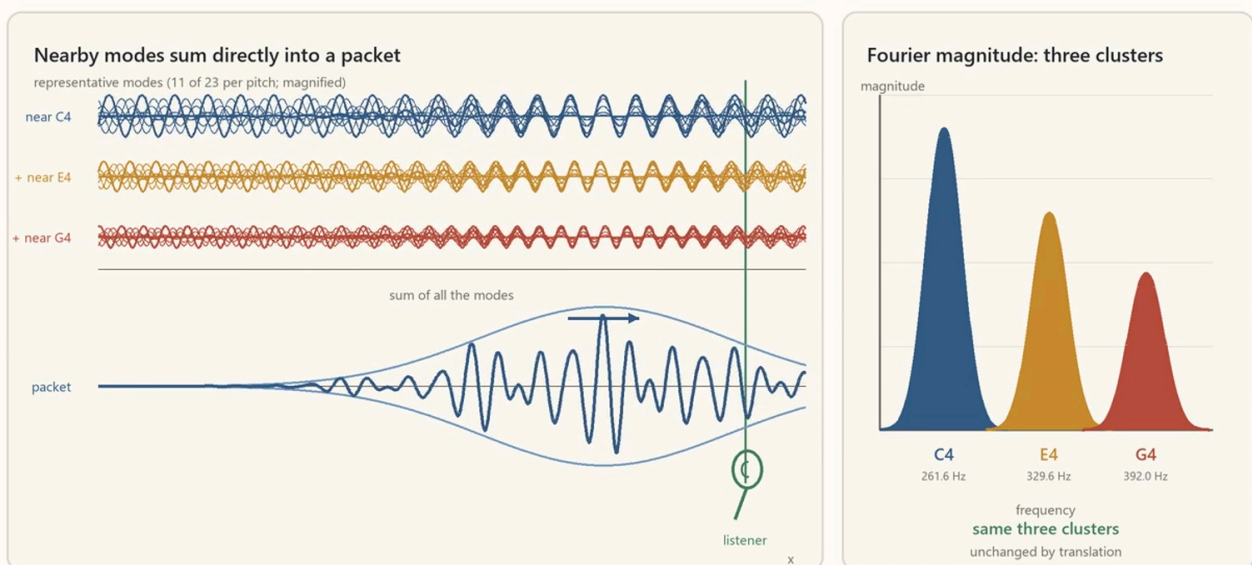
Translation changes the phases of the Fourier components, not their magnitudes.

A chord travels; its Fourier composition stays fixed



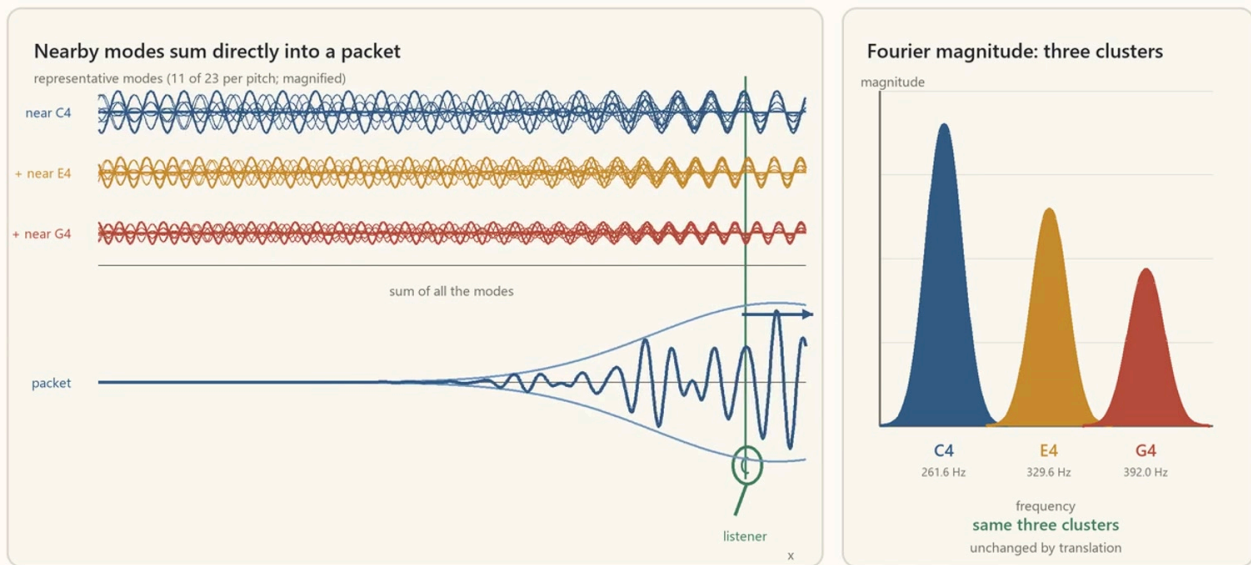
Translation changes the phases of the Fourier components, not their magnitudes.

A chord travels; its Fourier composition stays fixed



Translation changes the phases of the Fourier components, not their magnitudes.

A chord travels; its Fourier composition stays fixed



Translation changes the phases of the Fourier components, not their magnitudes.

But we know something about this random-seeming function reaching your ear. We know it is comprised of mostly 3 pitches, and some overtones. If we plot the sound pressure level as a function of these pitches rather than as a function of time, the structure of our plot clearly reveals what we hear naturally with our ear. What we call a "pitch" is the frequency of a pure **mode**. The pattern of sound pressure over time is the sum, or "composition," of these modes. We can transform back and forth between the wave number and position representations. Typically we label the wave number representation with a tilde:

$$\psi(x) \longleftrightarrow \tilde{\psi}(k)$$

The complex coefficient $\tilde{\psi}(k)$ specifies how strongly each mode contributes and with what phase. We recover the original function by adding these contributions:

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{\psi}(k) e^{ikx} dk.$$

The integral is a continuous sum over wave numbers. The factor $1/\sqrt{2\pi}$ fixes our normalization convention.

Because wave number is conserved, while listeners at different distances from the piano hear the chord at different times, they agree on the tonal and timbral character.

This idea of **decomposing** a wave packet into its "Fourier modes" is remarkably general. Any function whose inner product with itself is finite can be decomposed into a linear combination, or **superposition**, of plane waves.

$$\int_{-\infty}^{\infty} |f(x)|^2 dx < \infty.$$

An exact plane wave extends uniformly to infinity, so its own squared magnitude has an infinite integral. It is an ideal building block. Superpositions of these waves can nevertheless form packets with finite total squared magnitude, which we can normalize as quantum states.

The transformation from an amplitude over translation coordinate to amplitude over wave number coordinate is called a **Fourier transformation**.

Quantum mechanics uses the wave structure we have developed from symmetry. Extending the quantum description to the universe as a whole, some physicists propose a single universal wave function from which observations are, in a loose manner of speaking, sampled. We could apply the ideas of Fourier analysis to shamelessly indulge in mystical cosmology and argue that the structure, the harmony, of the endless complexity that unfolds in time is a single cosmic chord. Perhaps Pythagoras had something like this in mind when he (supposedly) said: "There is geometry in the humming of the strings. There is music in the spacings of the spheres."

The Heisenberg Symmetry Group

We can understand Fourier analysis in terms of the symmetry group that acts on wave functions, the Heisenberg group. This group's actions preserve the overlaps between states, and therefore their distinguishability. They do not in general preserve the behavior of those states as they evolve. In this sense, they are symmetries of state space, whether or not they are also dynamical symmetries of a particular system. This group not only underlies Fourier analysis, but in defining similarity and distinguishability of wave functions, supplies an essential ingredient for a logically viable notion of state.

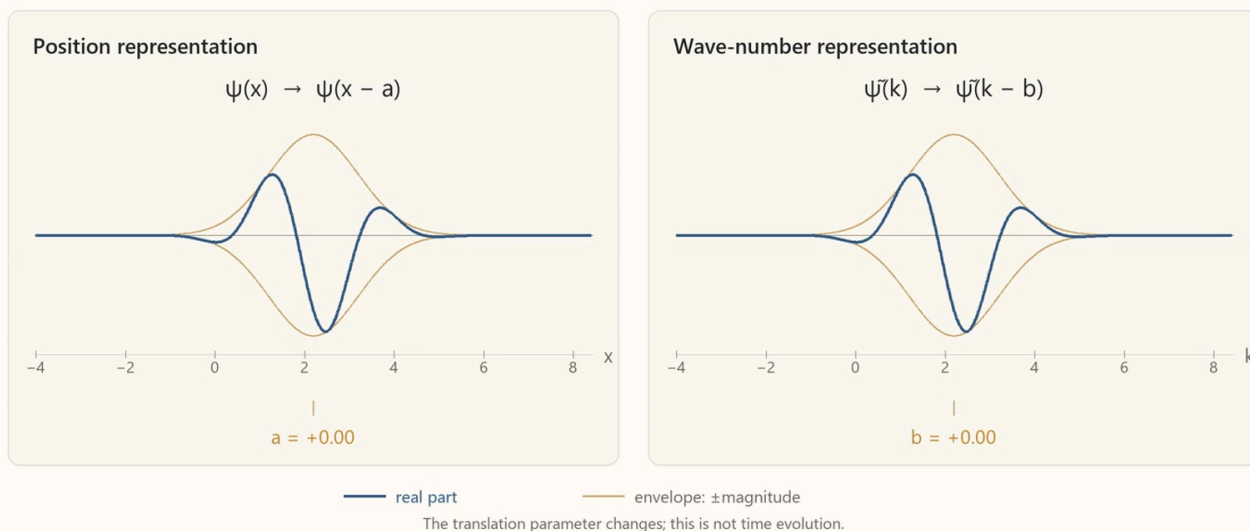
We can translate a wave function either in x -space or in k -space. While shifting the wave number isn't a translation in the familiar physical space we live in, from a mathematical perspective, k -space is the dual, or equivalent up to role reversal, of x -space.

Wave Packet translated in position and momentum space

Animation 17 / frame 1 of 4 / 0.34 s

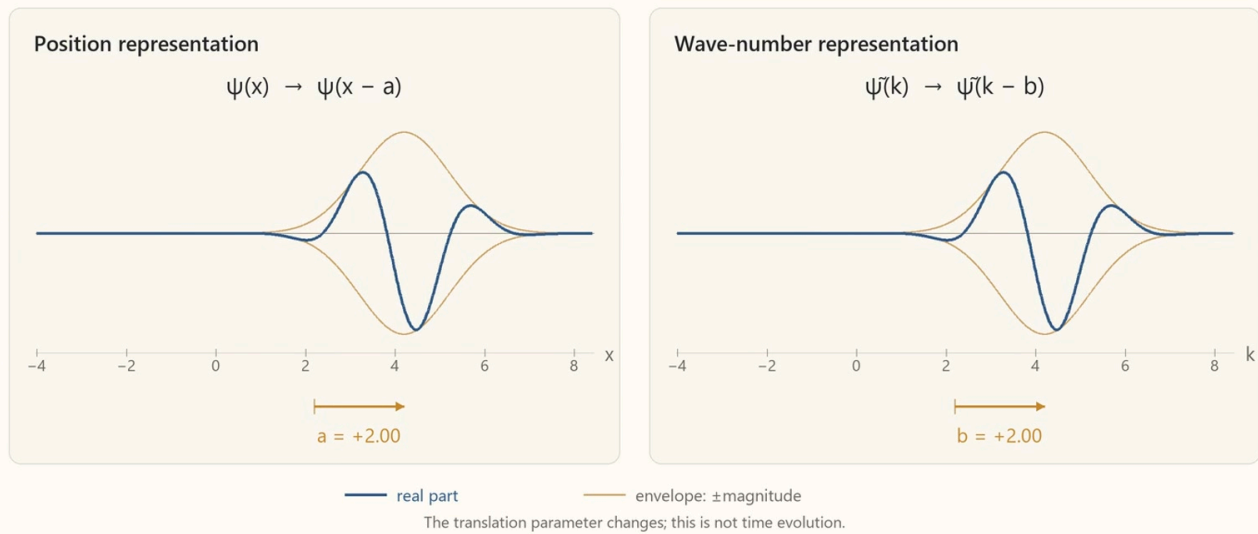
Translation in position and in wave number

Two separate translations of the same starting wave function.



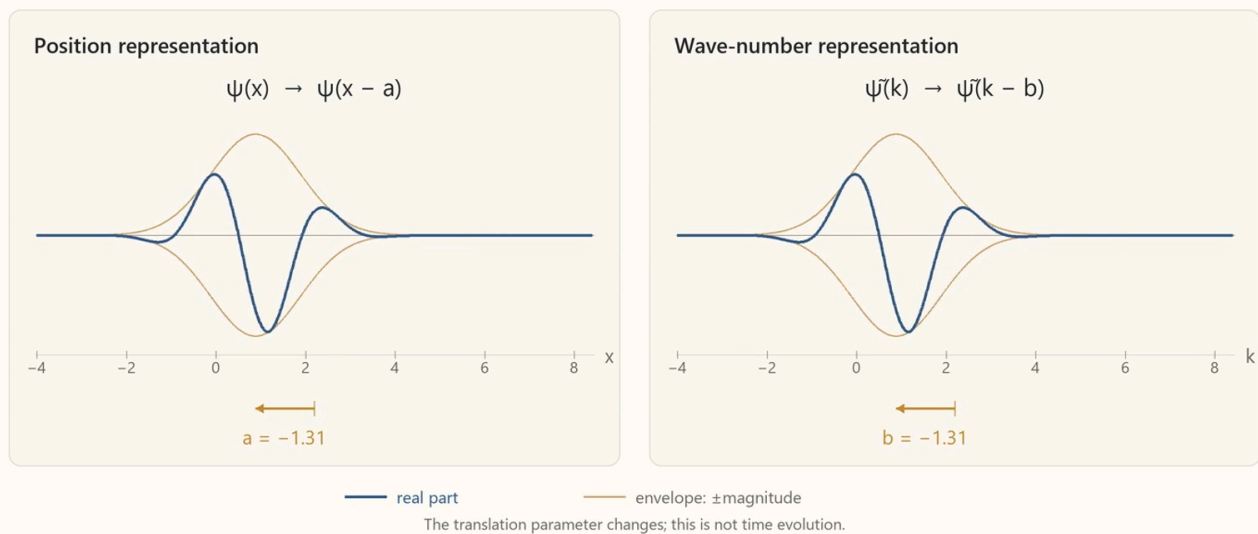
Translation in position and in wave number

Two separate translations of the same starting wave function.



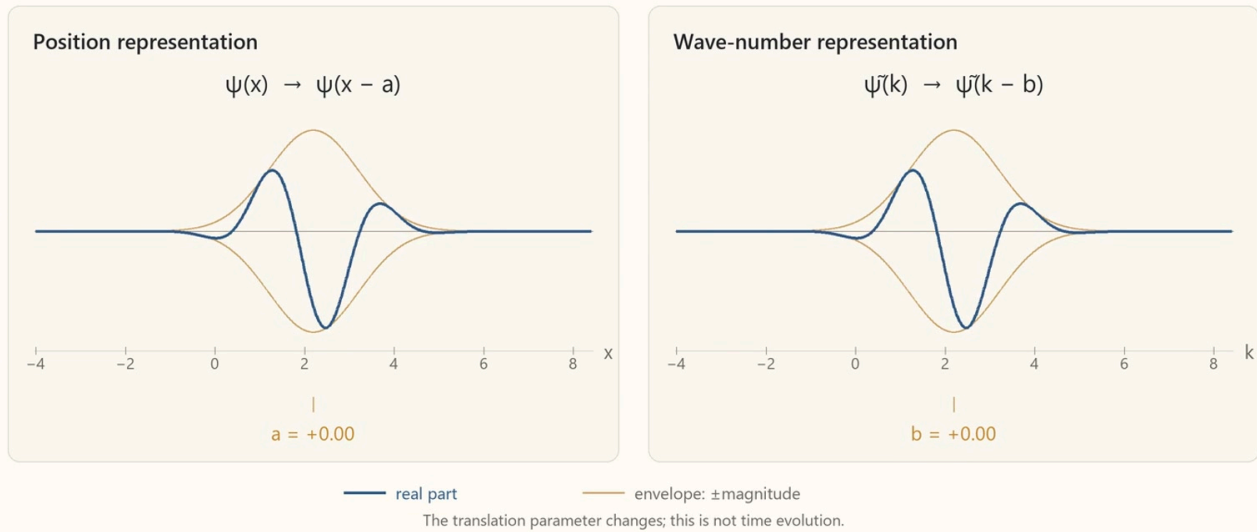
Translation in position and in wave number

Two separate translations of the same starting wave function.



Translation in position and in wave number

Two separate translations of the same starting wave function.



The overlap between two complex wave functions is their **inner product**:

$$\langle \psi_1, \psi_2 \rangle = \int_{-\infty}^{\infty} \psi_1^*(x) \psi_2(x) dx.$$

The star means complex conjugation. Multiplying a complex value by its conjugate gives its squared magnitude, a real number.

The Heisenberg group's transformations preserve this inner product:

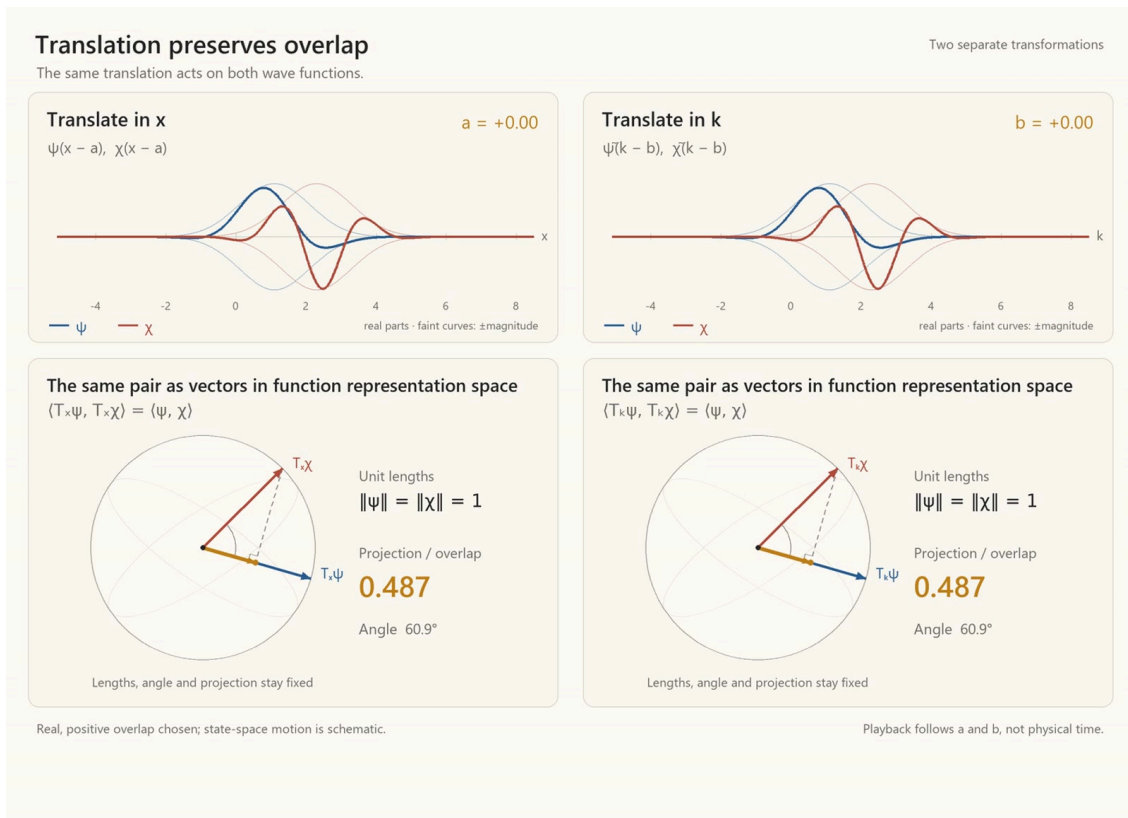
$$\langle U\psi_1, U\psi_2 \rangle = \langle \psi_1, \psi_2 \rangle \quad \text{for every symmetry action } U.$$

This invariance, as we've seen, is called unitarity. It preserves the distinguishability of states under symmetry transformations and makes those transformations reversible. When time evolution is itself unitary, the evolution of the state is deterministic and reversible. Translations in position and wave number preserve the inner product:

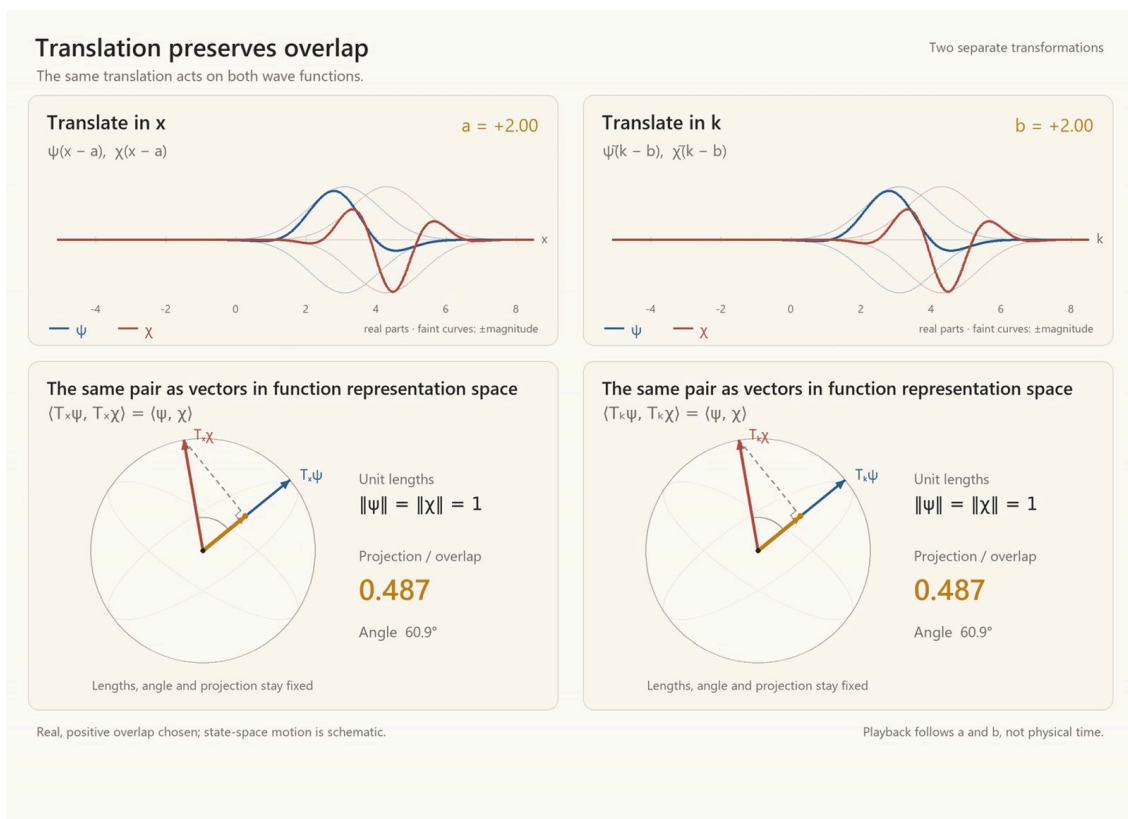
Translation by a in x $\langle T_x(a)\psi, T_x(a)\chi \rangle$ $= \langle \psi, \chi \rangle$	Translation by b in k $\langle T_k(b)\psi, T_k(b)\chi \rangle$ $= \langle \psi, \chi \rangle$
---	---

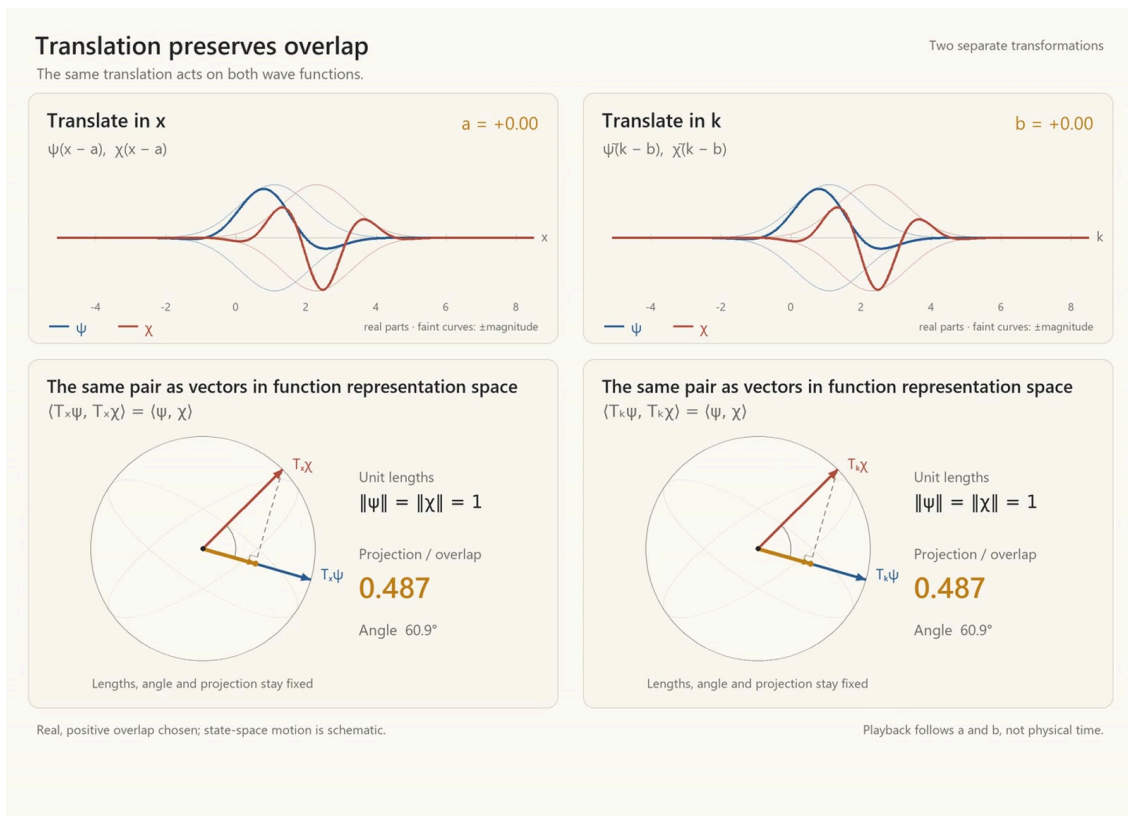
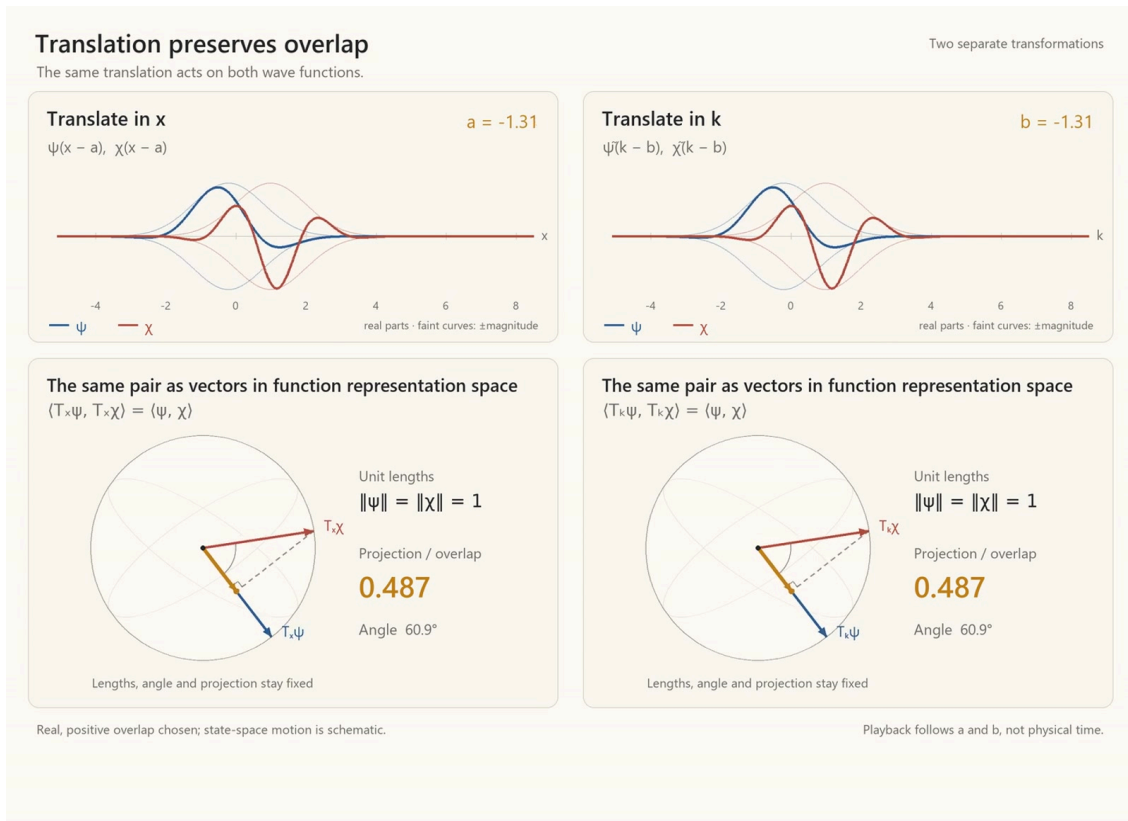
Unitarity of position and wave-number translations

Animation 18 / frame 1 of 4 / 0.34 s



Animation 18 / frame 2 of 4 / 5.61 s



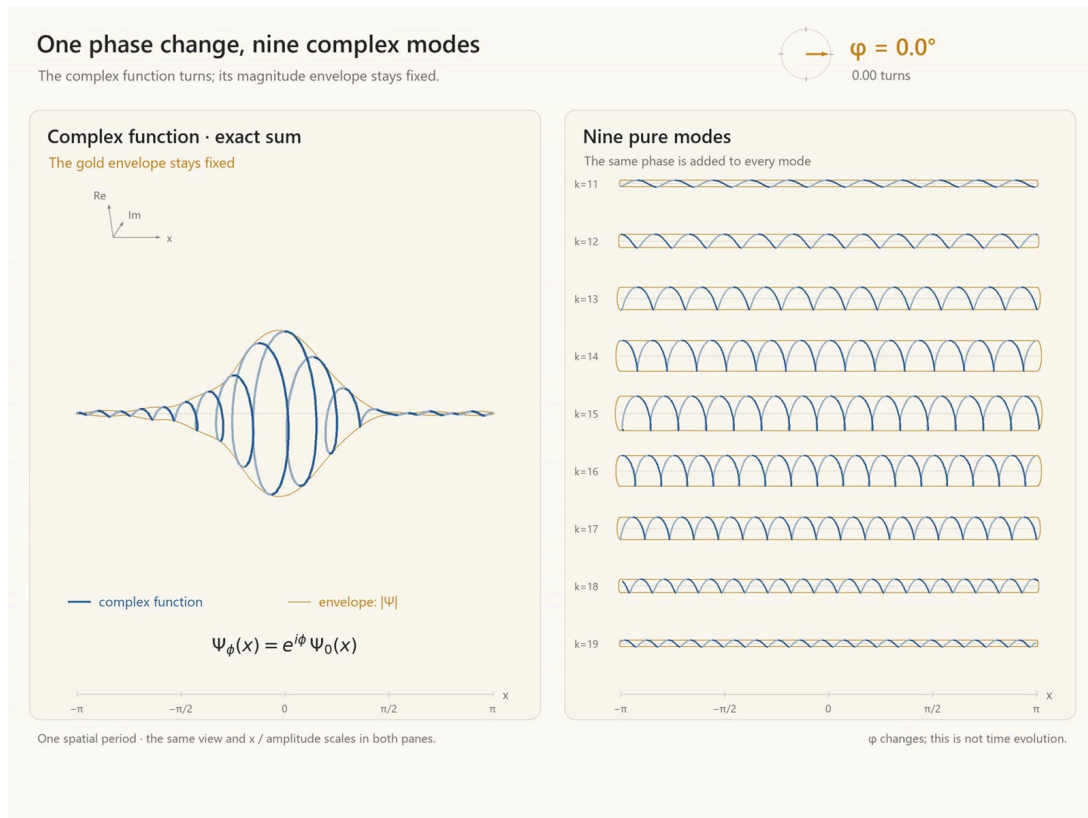


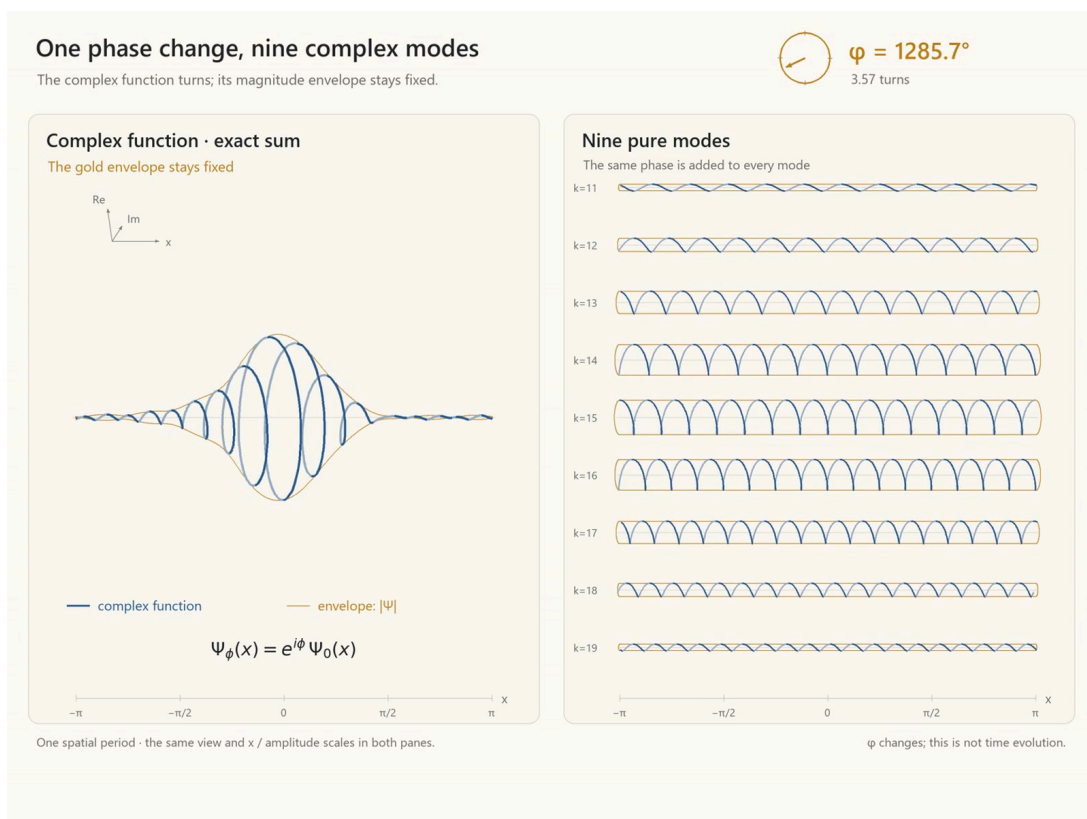
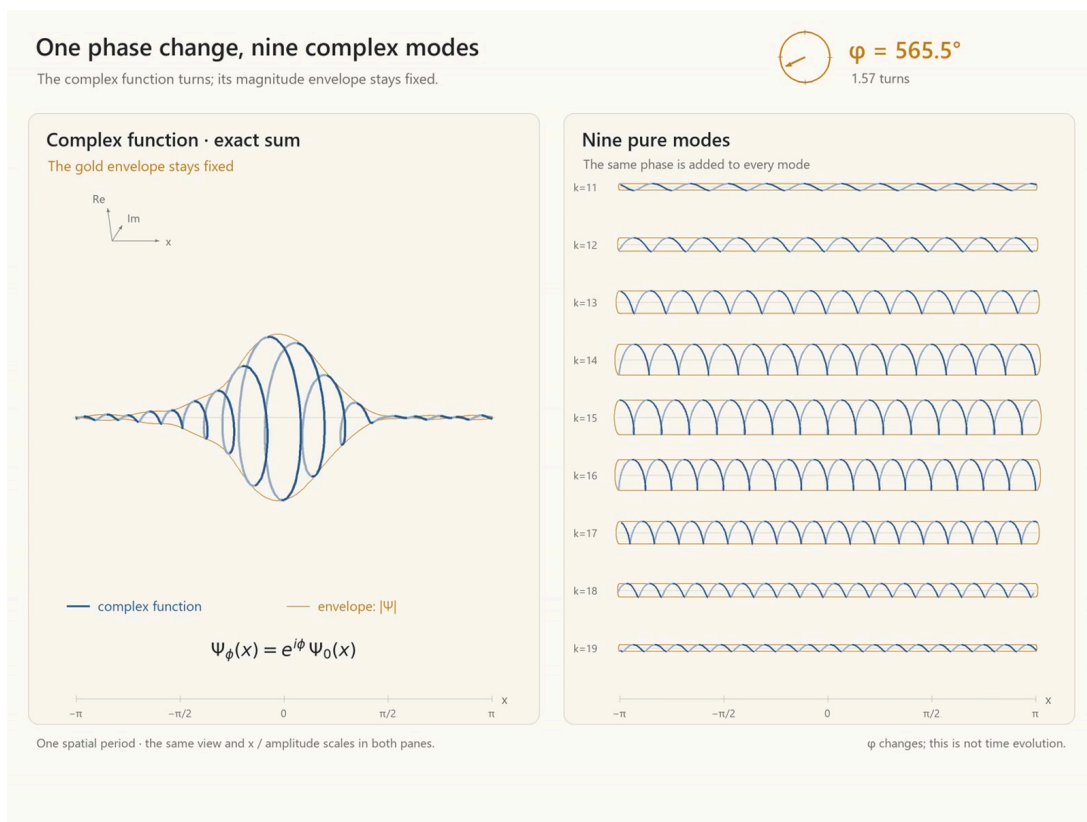
In addition to position and wave number, waves have a third independent way of changing. A wave's **phase**, ϕ , refers to where it is in its cyclic pattern. For example, a phase shift of 2π , or one full "cycle," returns the wave to its exact initial state. We need to be a bit careful here. For a pure mode, shifting its position is indistinguishable from shifting its phase, somewhat in the way the turning of a barbershop

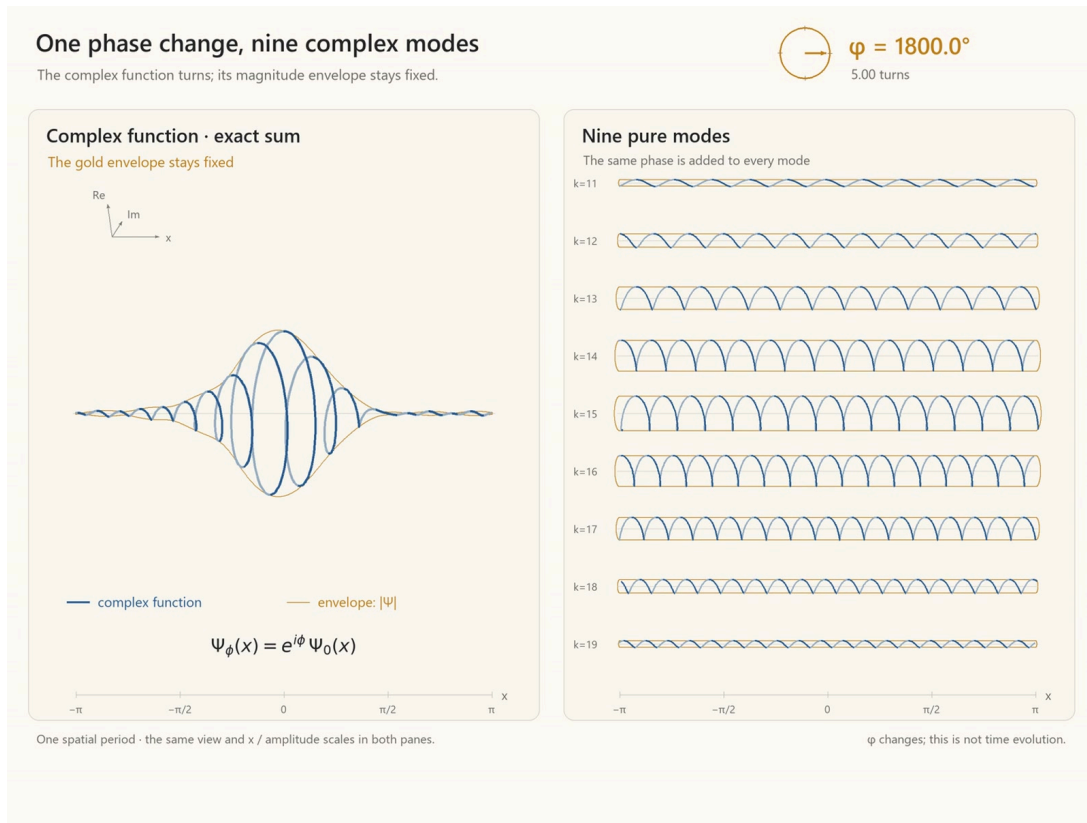
sign appears as though its stripes are moving up and down. We might, then, be tempted to think there is no difference between position and phase shifts. But the single mode is an idealization. In the general case, in which the wave function is a packet composed of modes, position translation shifts the entire function. Phase translation shifts each mode by the same fraction of its cycle, changing the function while leaving its magnitude envelope unchanged.

Visualizing Phase Change

Animation 19 / frame 1 of 4 / 0.36 s







We can also discover and define phase directly from our symmetry group's commutation relations, which gives us a useful algebraic packaging of the group structure. Let's ask the question:

$$[\hat{X}, \hat{K}] = ?$$

If the commutator is non-zero (and linearly independent of $\hat{X}$ and $\hat{K}$), there must be a third kind of transformation in the complete symmetry group. To find the commutator, we follow the usual procedure of translating the function around a loop in $\mathbf{x-k}$ space and asking if the function changes. If it does, $?$ is nonzero, the commutator is the generator of that change, and the action will be identifiably that of a phase shift.

First, write a single mode in the $\mathbf{x}$ and $\mathbf{k}$ representations:

$$\psi_{k_0}(x) = e^{ik_0 x}.$$

$$\tilde{\psi}_{k_0}(k) = \sqrt{2\pi} \delta(k - k_0),$$

where δ is a spike at k_0 that integrates to 1.

Define the $\mathbf{x}$ and $\mathbf{k}$ translation operators in their respective representations:

$$(T_x(a)\psi_{k_0})(x) = \left(e^{-ia\hat{K}}\psi_{k_0}\right)(x) = \psi_{k_0}(x - a).$$

$$(T_k(b)\tilde{\psi}_{k_0})(k) = \tilde{\psi}_{k_0}(k - b).$$

Working in the position basis:

$$(T_k(b)\psi)(x) = \left(e^{ib\hat{X}}\psi\right)(x) = e^{ibx}\psi(x).$$

$\hat{X}$ generates translation in $\mathbf{k}$ -space just as $\hat{K}$ generated translations in $\mathbf{x}$ -space. In $\mathbf{x}$ -space, it multiplies ψ by $\mathbf{x}$ as it weights each component of ψ by its position coordinate.

We can construct a loop by first translating the function by $\mathbf{b}$ in $\mathbf{k}$ and by $\mathbf{a}$ in $\mathbf{x}$, then by $-\mathbf{b}$ in $\mathbf{k}$ and by $-\mathbf{a}$ in $\mathbf{x}$.

$$(T_x(a)T_k(b)\psi_{k_0})(x) = e^{i(k_0+b)(x-a)},$$

$$(T_k(b)T_x(a)\psi_{k_0})(x) = e^{ibx}e^{ik_0(x-a)}.$$

$$(T_x(-a)T_k(-b)T_x(a)T_k(b)\psi_{k_0})(x) = e^{-ib(x+a)}e^{i(k_0+b)x} = e^{-ab[\hat{X}, \hat{K}]\psi_{k_0}(x) = e^{-iab}\psi_{k_0}(x).$$

The two shifts fail to commute by the factor e^{-iab} . Writing $\phi = -ab$, the loop multiplies the function by $e^{i\phi}$. But this is precisely a phase shift, a rotation in the complex plane that “turns” the whole "spiral" of the wave function.

Because this factor is independent of $\mathbf{k}_0$, the same phase shift applies to every mode in a superposition.

Phase translation, then, is a third translation symmetry, whose invariant is the inner product under its unitary action, rounding out the group of $\mathbf{x}$, $\mathbf{k}$, and ϕ coupled through their group structure:

$$\langle T_x(a)\psi, T_x(a)\chi \rangle = \langle \psi, \chi \rangle.$$

$$\langle T_k(b)\psi, T_k(b)\chi \rangle = \langle \psi, \chi \rangle.$$

$$\langle T_\phi(\beta)\psi, T_\phi(\beta)\chi \rangle = \langle \psi, \chi \rangle.$$

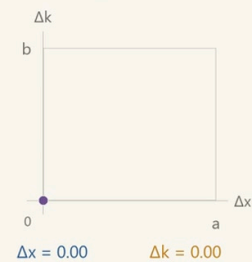
Phase as the $[\hat{X}, \hat{K}]$ commutator

Animation 20 / frame 1 of 6 / 0.32 s

A closed loop leaves a phase

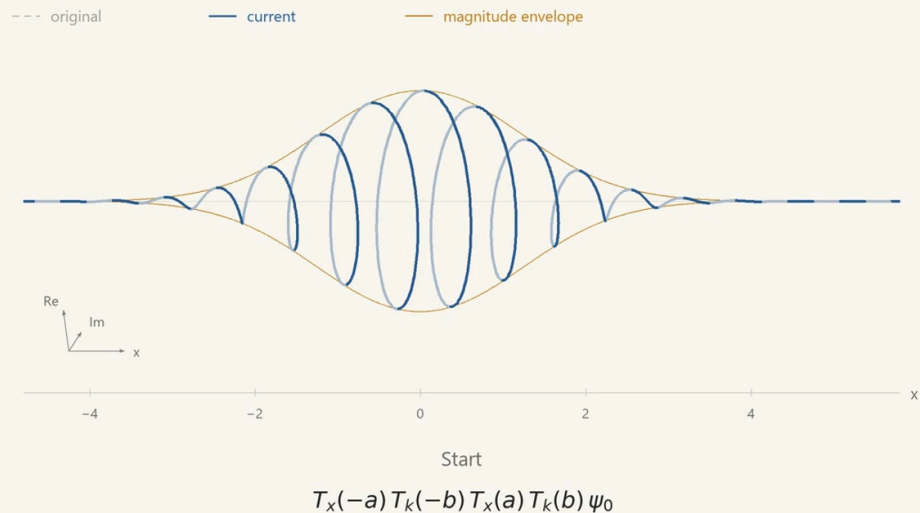
The shifts return to zero. The complex function retains a turn.

The x-k loop



- 1 $T_k(b)$ shift in k
- 2 $T_x(a)$ shift in x
- 3 $T_k(-b)$ return in k
- 4 $T_x(-a)$ return in x

Complex wave function

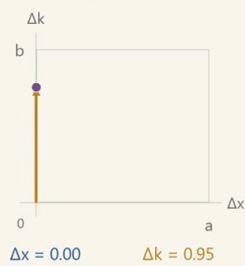


Continuous translations along the four sides; a fixed view of $(x, \text{Re } \psi, \text{Im } \psi)$.

A closed loop leaves a phase

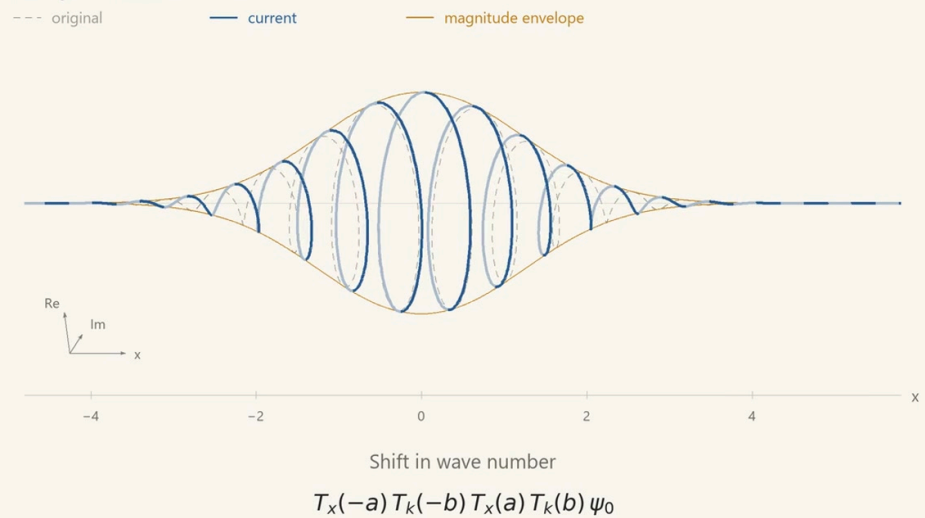
The shifts return to zero. The complex function retains a turn.

The x-k loop



- 1 $T_k(b)$ shift in k
- 2 $T_x(a)$ shift in x
- 3 $T_k(-b)$ return in k
- 4 $T_x(-a)$ return in x

Complex wave function

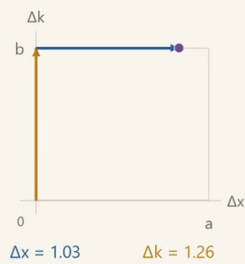


Continuous translations along the four sides; a fixed view of $(x, \text{Re } \psi, \text{Im } \psi)$.

A closed loop leaves a phase

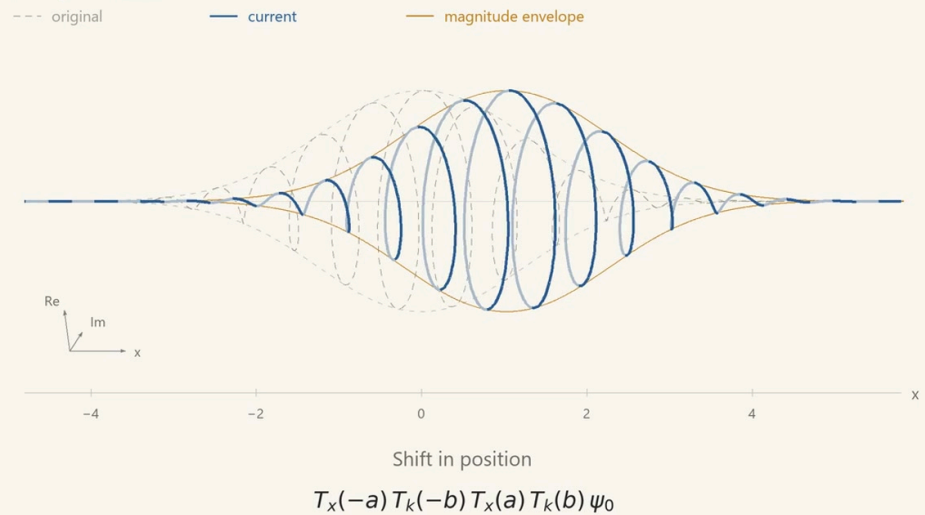
The shifts return to zero. The complex function retains a turn.

The x-k loop



- 1 $T_k(b)$ shift in k
- 2 $T_x(a)$ shift in x
- 3 $T_k(-b)$ return in k
- 4 $T_x(-a)$ return in x

Complex wave function

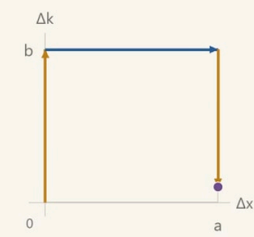


Continuous translations along the four sides; a fixed view of $(x, \text{Re } \psi, \text{Im } \psi)$.

A closed loop leaves a phase

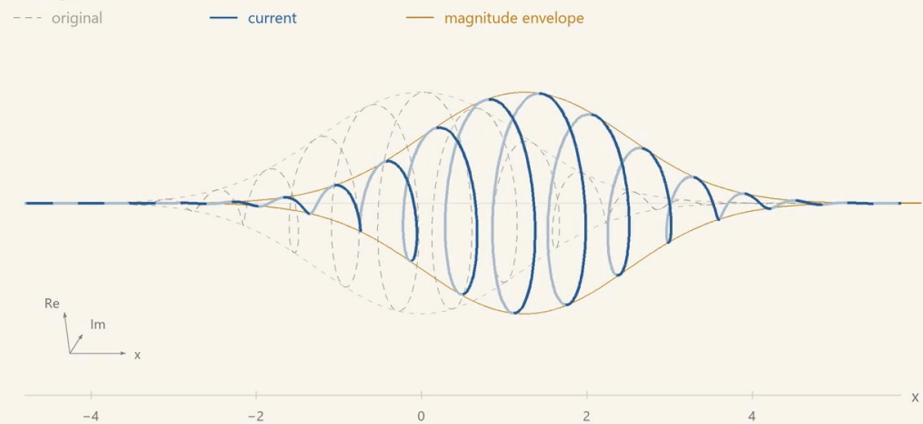
The shifts return to zero. The complex function retains a turn.

The x-k loop



- 1 $T_k(b)$ shift in k
- 2 $T_x(a)$ shift in x
- 3 $T_k(-b)$ return in k
- 4 $T_x(-a)$ return in x

Complex wave function



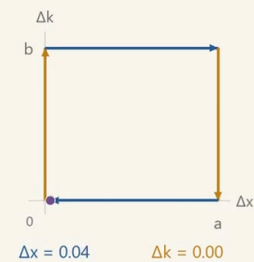
Return in wave number
 $T_x(-a) T_k(-b) T_x(a) T_k(b) \psi_0$

Continuous translations along the four sides; a fixed view of (x, Re ψ , Im ψ).

A closed loop leaves a phase

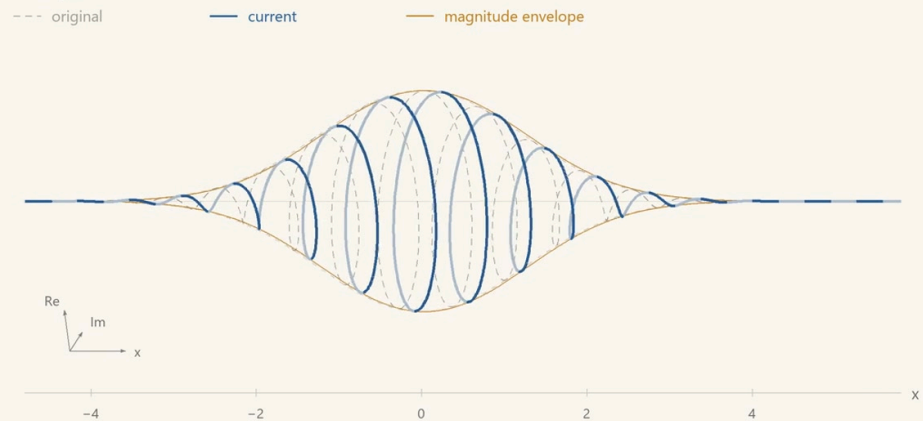
The shifts return to zero. The complex function retains a turn.

The x-k loop



- 1 $T_k(b)$ shift in k
- 2 $T_x(a)$ shift in x
- 3 $T_k(-b)$ return in k
- 4 $T_x(-a)$ return in x

Complex wave function



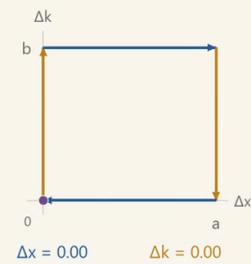
Return in position
 $T_x(-a) T_k(-b) T_x(a) T_k(b) \psi_0$

Continuous translations along the four sides; a fixed view of (x, Re ψ , Im ψ).

A closed loop leaves a phase

The shifts return to zero. The complex function retains a turn.

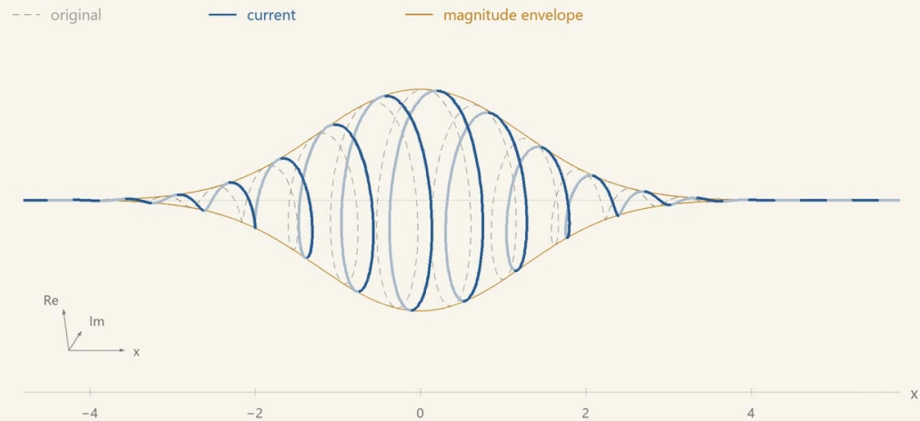
The x - k loop



- 1 $T_k(b)$ shift in k
- 2 $T_x(a)$ shift in x
- 3 $T_k(-b)$ return in k
- 4 $T_x(-a)$ return in x

loop closed

Complex wave function



Same envelope. A quarter-turn of phase remains.

$$\psi_{\text{final}}(x) = e^{-i\pi/2} \psi_0(x)$$

Continuous translations along the four sides; a fixed view of $(x, \text{Re } \psi, \text{Im } \psi)$.

The infinitesimal closed x - k loop is generated by $[\hat{X}, \hat{K}]$. Because the finite loop leaves a phase translation, and phase translations are generated by iI , we have:

$$[\hat{X}, \hat{K}] = iI.$$

We might now ask, is phase a symmetry of physical behavior in the sense that position is? And what is its associated conserved quantity? For waves in everyday life, the answer is “sort of.” A change in phase can be detected by measuring the instantaneous wave value at a given place and time, but at the same time, typical observable properties like pitch and volume (or color and brightness if you prefer light to sound) are in fact the same under phase transformations. (Global phase is the common phase factor that multiplies all components in a packet. Changing relative phases between components can change interference and produce observable effects.) In the realm of quantum mechanics, however, global phase is completely undetectable. In fact, unlike position symmetry which frequently is broken when “external forces” are present, dynamical global phase symmetry is always present. And what, then, is its associated conserved quantity? It is the integral of a wave packet’s squared magnitude, which in quantum mechanics gives total probability. Once normalized to one, it remains one.

Mathematicians love to name groups. Just as we have encountered D_3 and $SO(2)$, our new group has a name, the three-dimensional Heisenberg group, or H_3 . With n spatial dimensions, the group has $2n + 1$ dimensions, where the $+1$ is due to the fact that there is only one phase regardless of the number of spatial dimensions.

Before we close out here, a small amount of house cleaning is needed. First, we have not shown explicitly that changes in ϕ are linearly independent of changes in x and k . They are, as evidenced by the fact that shifts in ϕ can leave the wave function’s position and wave number unchanged. We also did not show that there are *only* 3 generators in H_3 . This is done by showing that the generators close under commutation:

$$[i\hat{X}, -i\hat{K}] = iI, \quad [i\hat{X}, iI] = [-i\hat{K}, iI] = 0.$$

Position / Wave Number Uncertainty

As we know, a single-mode wave function has a single wave number. But what position does it have? There is no way to answer this as the wave is uniform over all position space. The same statement holds in reverse. A wave packet ideally localized at one position is uniform over all k -space. Anywhere in between these extremes, as a wave packet is more localized in one space, it is more spread out in the dual space.

Trading off spread in position for spread in wave number

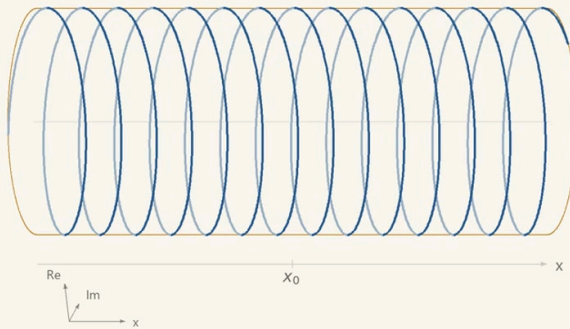
Animation 21 / frame 1 of 4 / 0.24 s

Fourier duality works in both directions

One complex function, two representations

Position representation

$\psi(x)$



One wave number · spread throughout x

$$\psi(x) \propto e^{ik_0(x - x_0)}$$

Wave-number representation

$\bar{\psi}(k)$



Localized at one wave number

$$\bar{\psi}(k) \propto \delta(k - k_0)$$

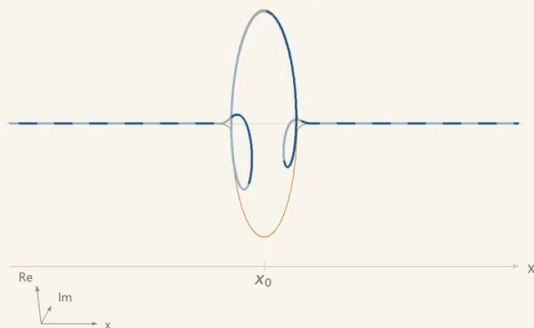
Curve heights rescaled; horizontal scales fixed. Ideal spikes are schematic.

Fourier duality works in both directions

One complex function, two representations

Position representation

$\psi(x)$

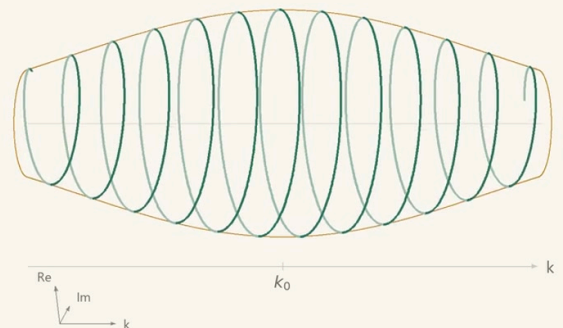


Narrower in position

$$\Delta x = 0.18$$

Wave-number representation

$\tilde{\psi}(k)$



Broader in wave number

$$\Delta k = 2.78$$

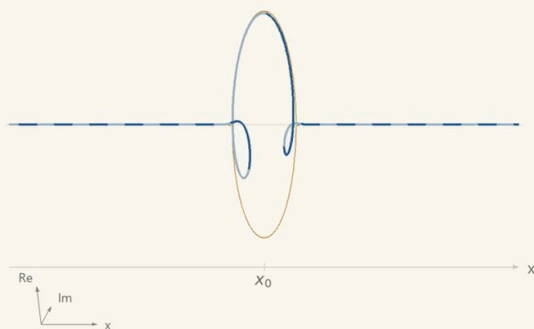
Curve heights rescaled; horizontal scales fixed. Ideal spikes are schematic.

Fourier duality works in both directions

One complex function, two representations

Position representation

$\psi(x)$

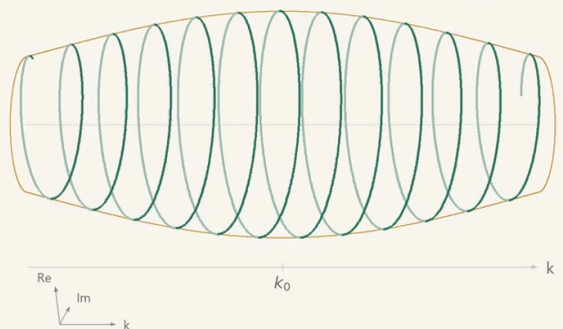


Broader in position

$$\Delta x = 0.15$$

Wave-number representation

$\tilde{\psi}(k)$



Narrower in wave number

$$\Delta k = 3.36$$

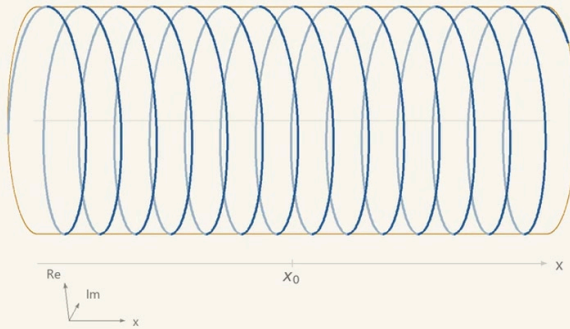
Curve heights rescaled; horizontal scales fixed. Ideal spikes are schematic.

Fourier duality works in both directions

One complex function, two representations

Position representation

$\psi(x)$

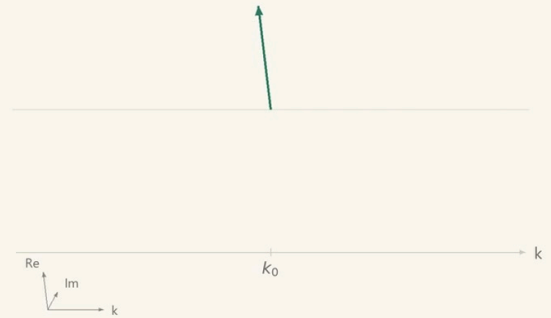


One wave number · spread throughout x

$$\psi(x) \propto e^{ik_0(x-x_0)}$$

Wave-number representation

$\tilde{\psi}(k)$



Localized at one wave number

$$\tilde{\psi}(k) \propto \delta(k-k_0)$$

Curve heights rescaled; horizontal scales fixed. Ideal spikes are schematic.

We can easily see this relationship drawn on paper, but we also hear it in music. The precise pitch of a tuning fork requires long sustain, while the percussive clap of a clave has no clear pitch. This tradeoff is the root of the Heisenberg uncertainty principle, or what is known in popular science as "quantum fuzziness."

With this visual understanding of the tradeoff in the spread of the magnitude envelopes in x and k space, we can define the corresponding statistical standard deviation, or uncertainty, in position and wave number.

For simplicity, set the wave's mean position and wave number at their respective origins. Then:

$$(\sigma_x)^2 = \int_{-\infty}^{\infty} x^2 |\psi(x)|^2 dx,$$

and likewise for σ_k .

σ^2 is a probability-weighted average of squared distances from the distribution's center. This is the textbook definition of variance.

From the commutation relation

$$[\hat{X}, \hat{K}] = iI,$$

one can derive the position/wave-number "uncertainty relation":

$$\Delta x \Delta k \geq \frac{1}{2}.$$

This derivation is rather involved, but we can get a feel for the result by plotting the squared magnitude of a wave function and labelling the standard deviation.

Trade-off in uncertainty of position and wave number

Animation 22 / frame 1 of 4 / 0.24 s

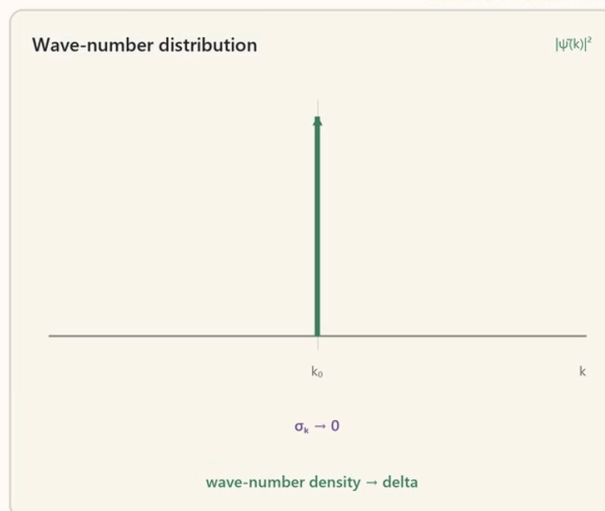
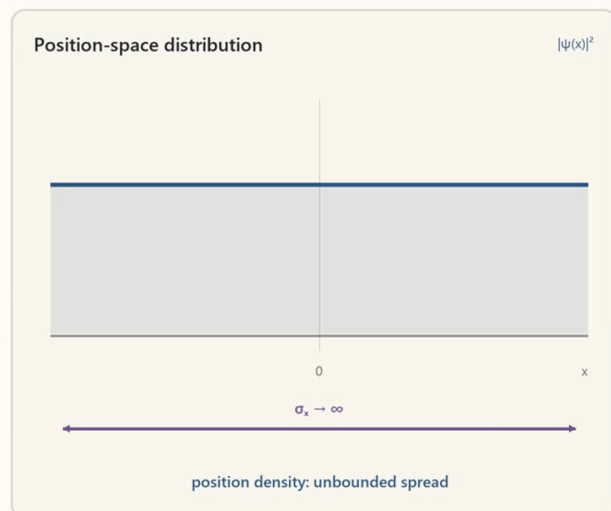
Uncertainty lives in the squared magnitudes

Exact wave-number limit

position density spreads without bound; wave-number density localizes

shape vertically rescaled - not physical time

ideal limit of the Gaussian family



Animation 22 / frame 2 of 4 / 3.96 s

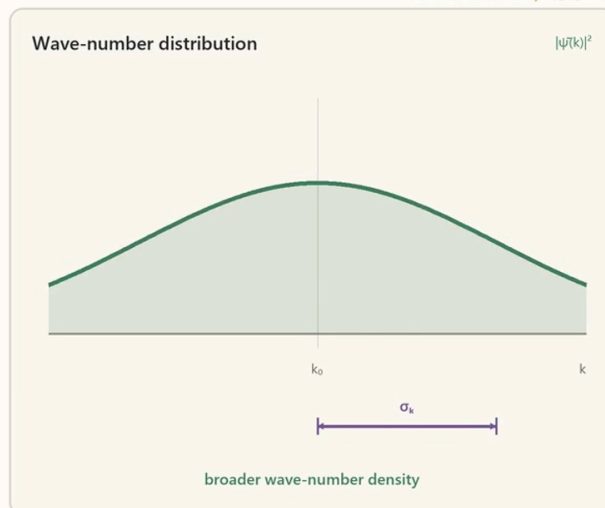
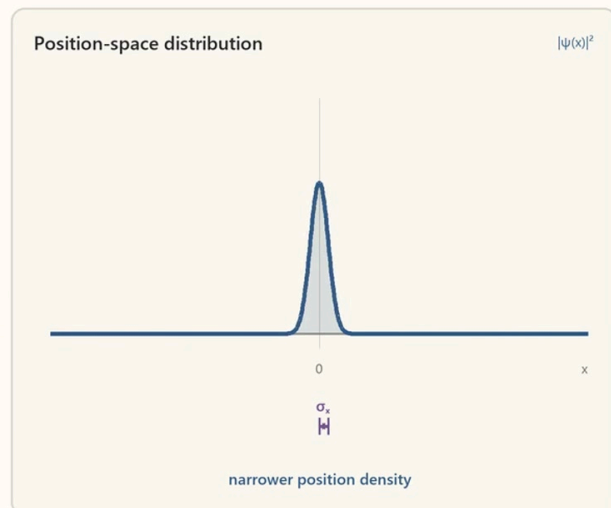
Uncertainty lives in the squared magnitudes

Trade statistical width between representations

as σ_x decreases, σ_k increases

shape vertically rescaled - not physical time

finite Gaussian family: $\sigma_x \sigma_k = 1/2$

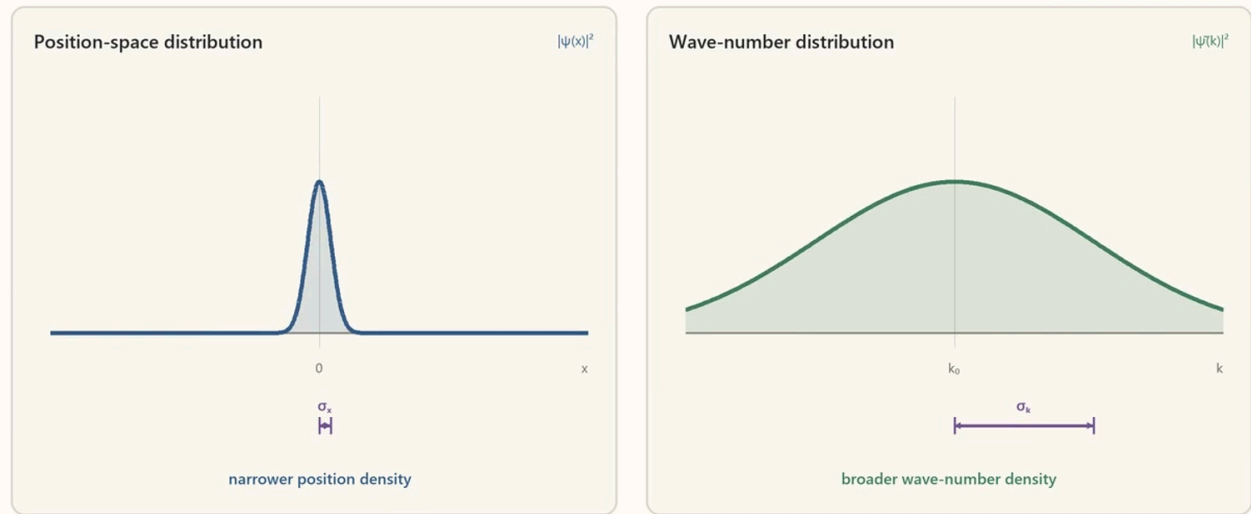


Uncertainty lives in the squared magnitudes

Return from the exact-position limit
the reciprocal standard deviations approach balance

shape vertically rescaled - not physical time

finite Gaussian family: $\sigma_x \sigma_k = 1/2$

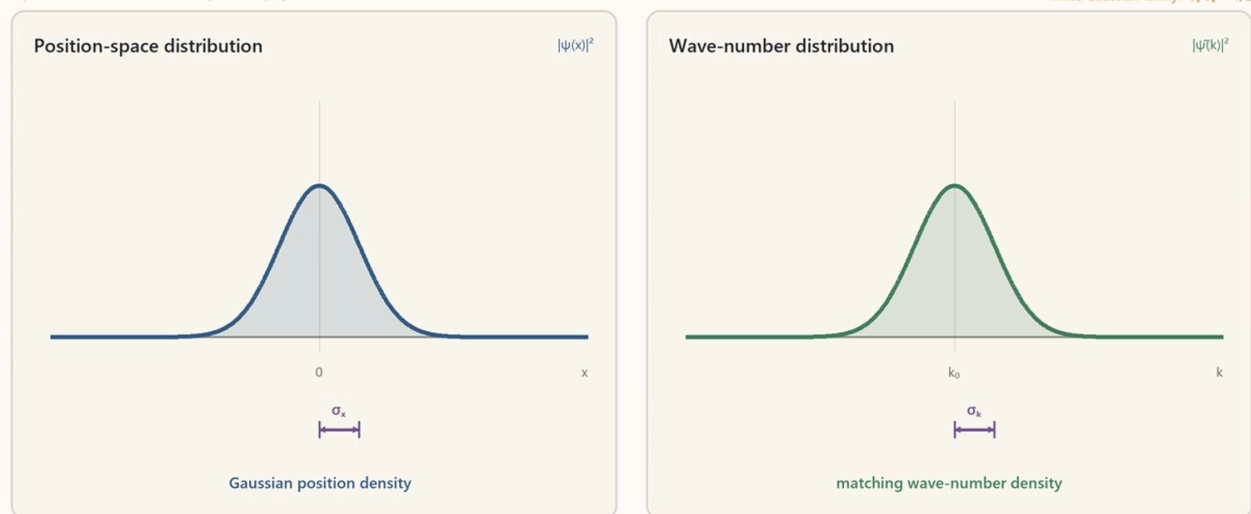


Uncertainty lives in the squared magnitudes

Balanced Gaussian distributions
equal standard deviations on reciprocal display scales

shape vertically rescaled - not physical time

finite Gaussian family: $\sigma_x \sigma_k = 1/2$



Wave Propagation and Interference

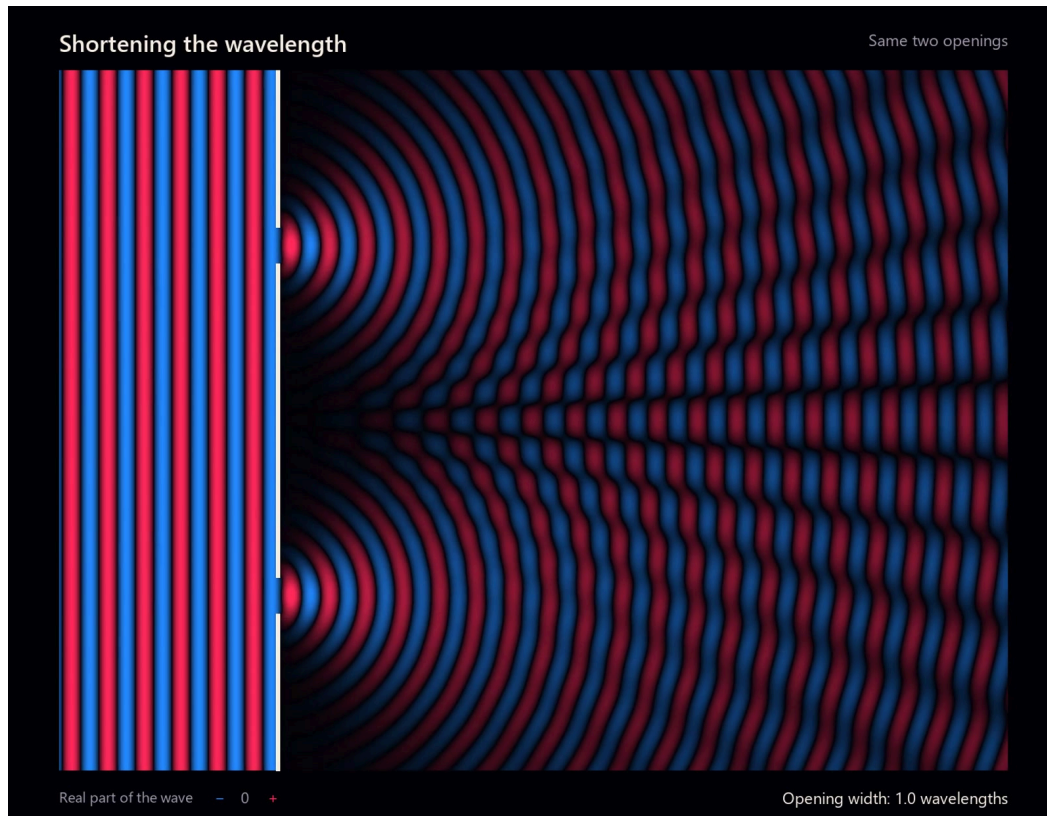
Everyone who has taken a high-school physics class knows that given a particular setup, the laws of motion tell us the path an object follows. For example, under constant acceleration:

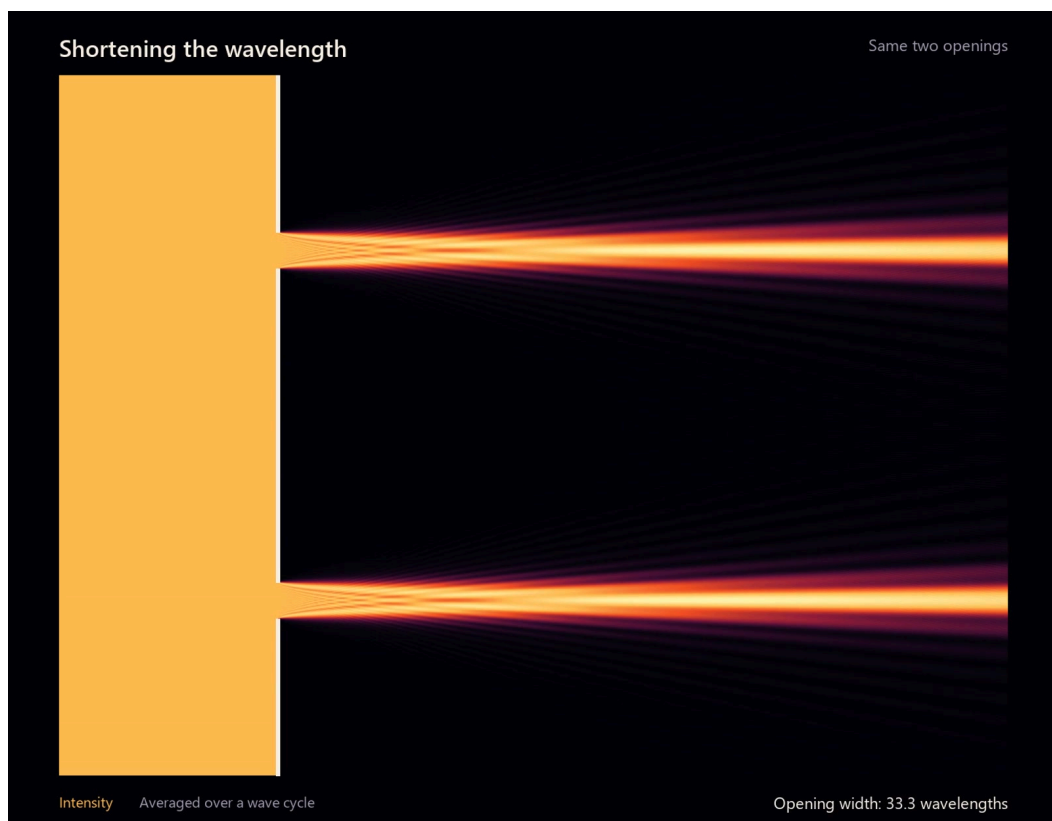
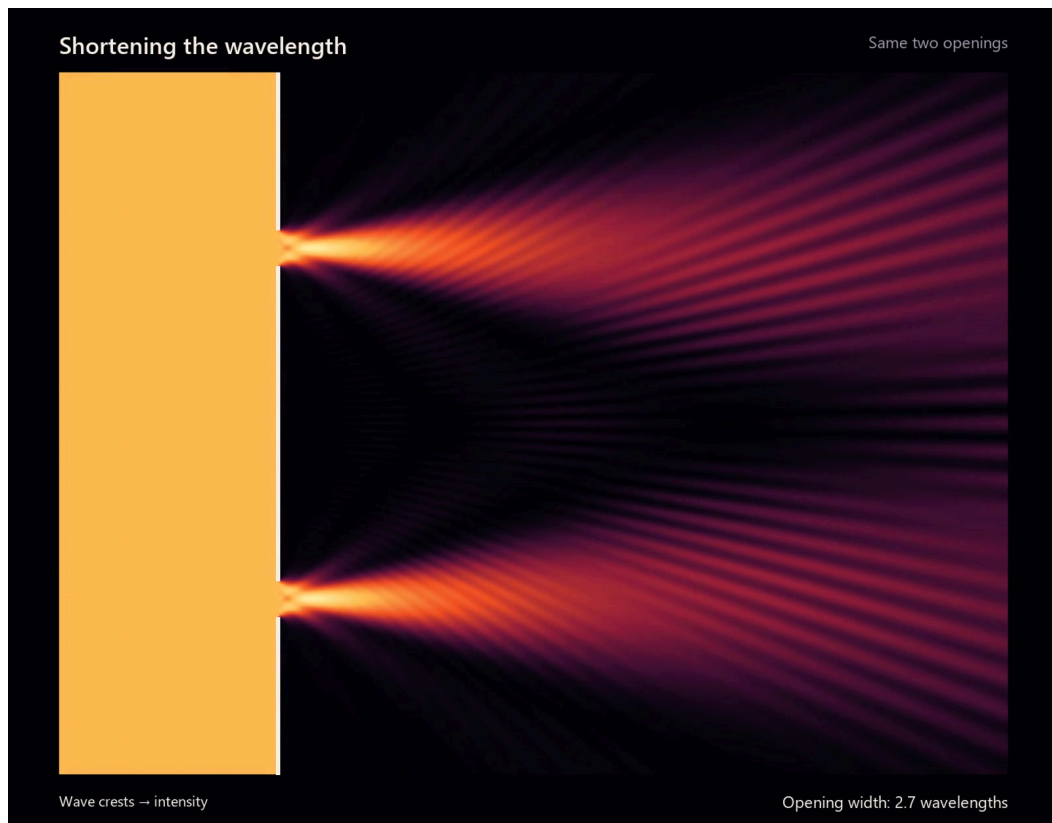
$$x = x_0 + v_0 t + \frac{1}{2} a t^2$$

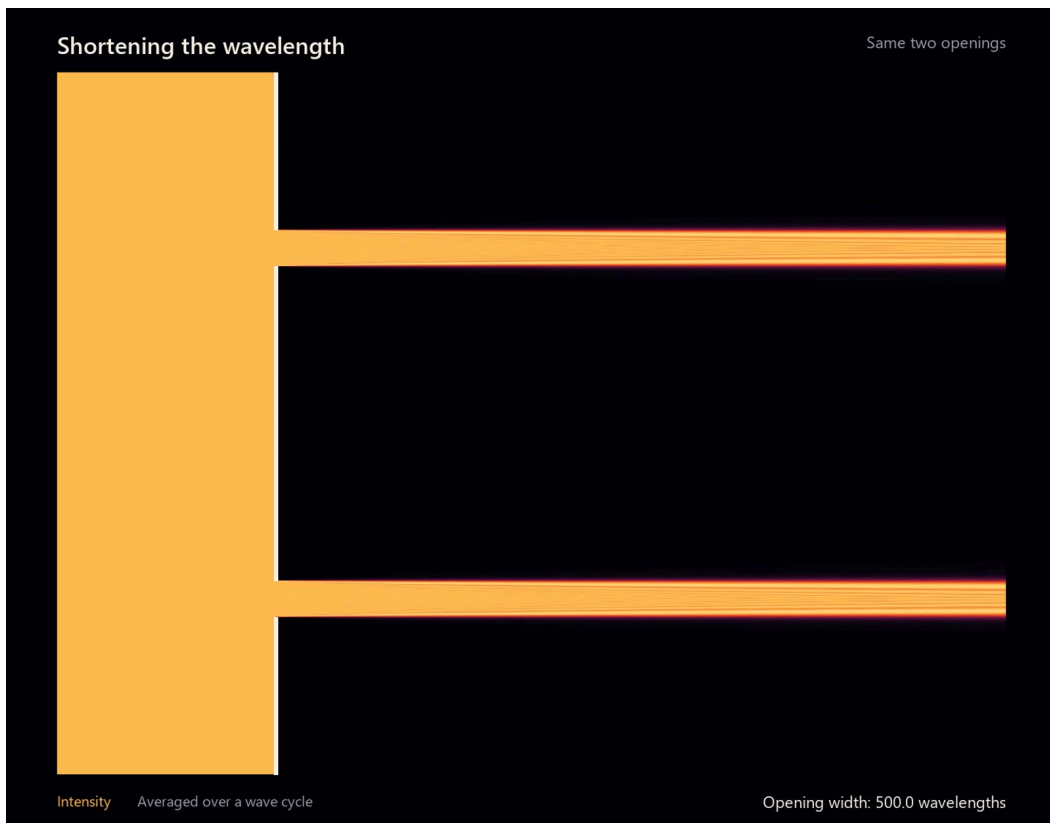
Such "physically valid" paths in the macroscopic world have a fascinating quality that can be leveraged to find the laws of motion that predict them. They are such that some quantity associated with possible paths, which is called **action**, is extremized at the valid path. In the next sections, we will explore the relationship between objects following paths and wave propagation. Just as an object following a definite path extremizes action, a wave, in the regime where it behaves like a ray, follows a path that extremizes accumulated phase. We will establish how contributions from many possible paths combine to produce this behavior, giving us a bridge between wave propagation and action extremization. We will then see how this stationary-path limit connects the quantum wave description we have alluded to with the well-defined paths that action extremization predicts for everyday macroscopic objects.

Interference and Rays

Animation 23 / frame 1 of 4 / 0.40 s







Thus far we have described the symmetry group of a wave with a single translation direction $\mathbf{x}$ and wave number, $\mathbf{k}$. If the wave is to propagate, we also require that the wave represent time translation. We also need to require that time translation commute with spatial translation, for otherwise, it would change the mode composition over time, and spatial translation would no longer be a symmetry of nature. A single-mode travelling wave is then given by:

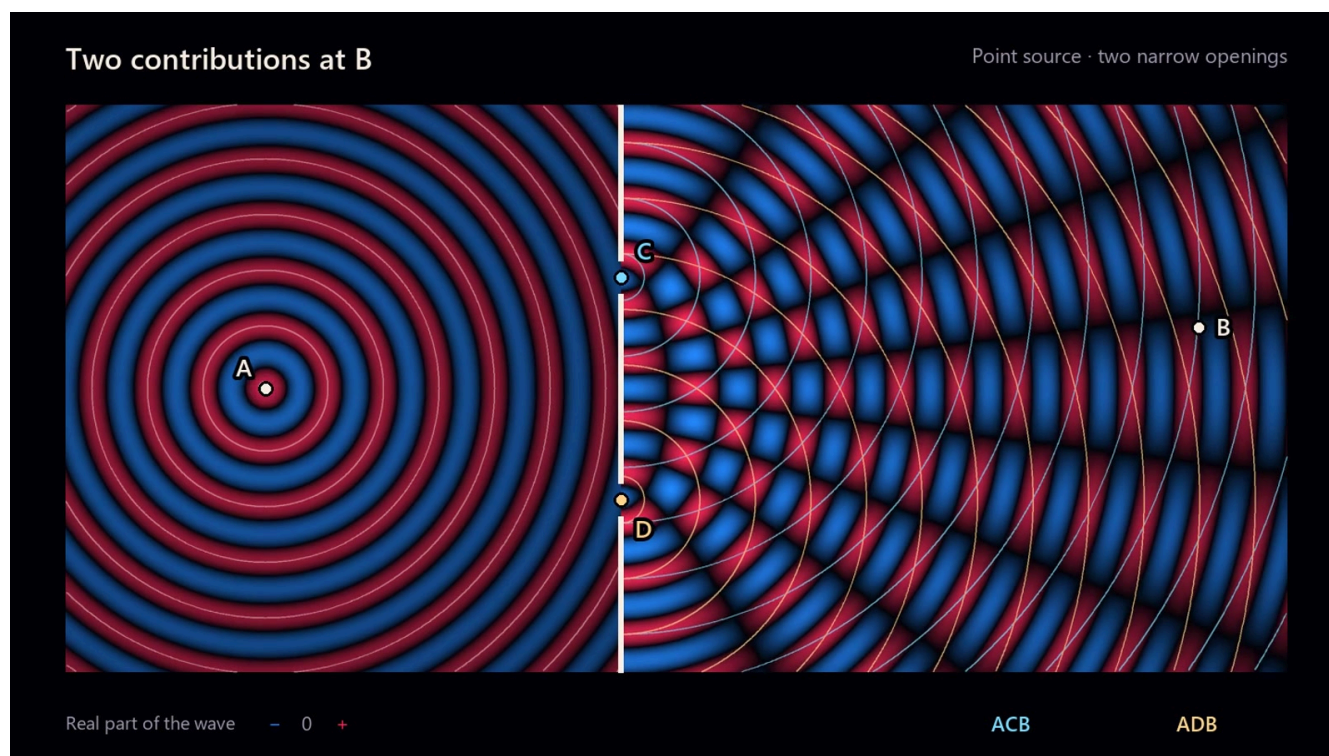
$$M_0 e^{i[kx - \omega t]}.$$

Because time is special, ω is called (angular) **frequency**, not wave number, but from a mathematical perspective, it is just another wave number.

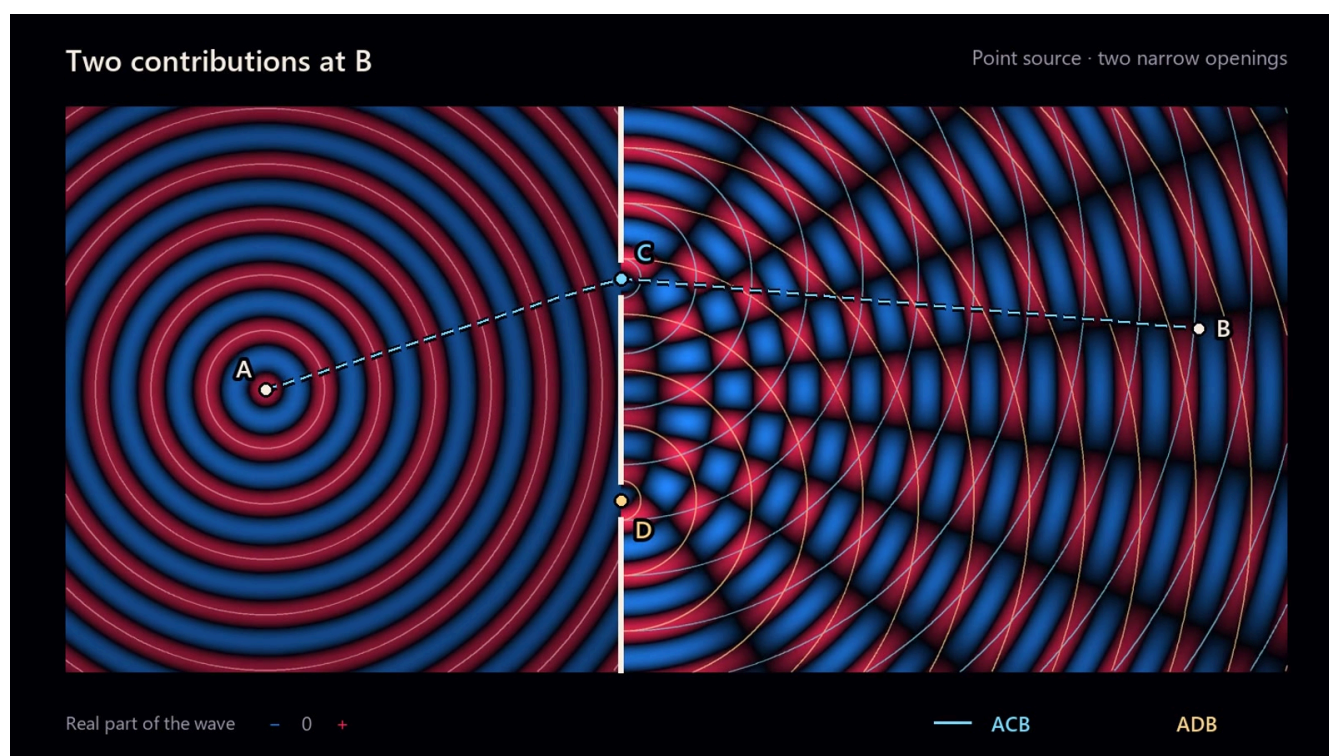
Let us now ask the question: how do we find the amplitude at some point $\mathbf{B}$ from some initial state of a wave? To do this, we can decompose the contributions into those from individual paths, starting with a very simple model. First, let's construct a point source of single-mode spherical waves emanating from $\mathbf{A}$. Then let's add a barrier with two slits through which the wave can pass, $\mathbf{C}$ and $\mathbf{D}$. This setup allows us to calculate the amplitude at $\mathbf{B}$ by combining only the amplitudes associated with the two paths $\mathbf{ACB}$ and $\mathbf{ADB}$.

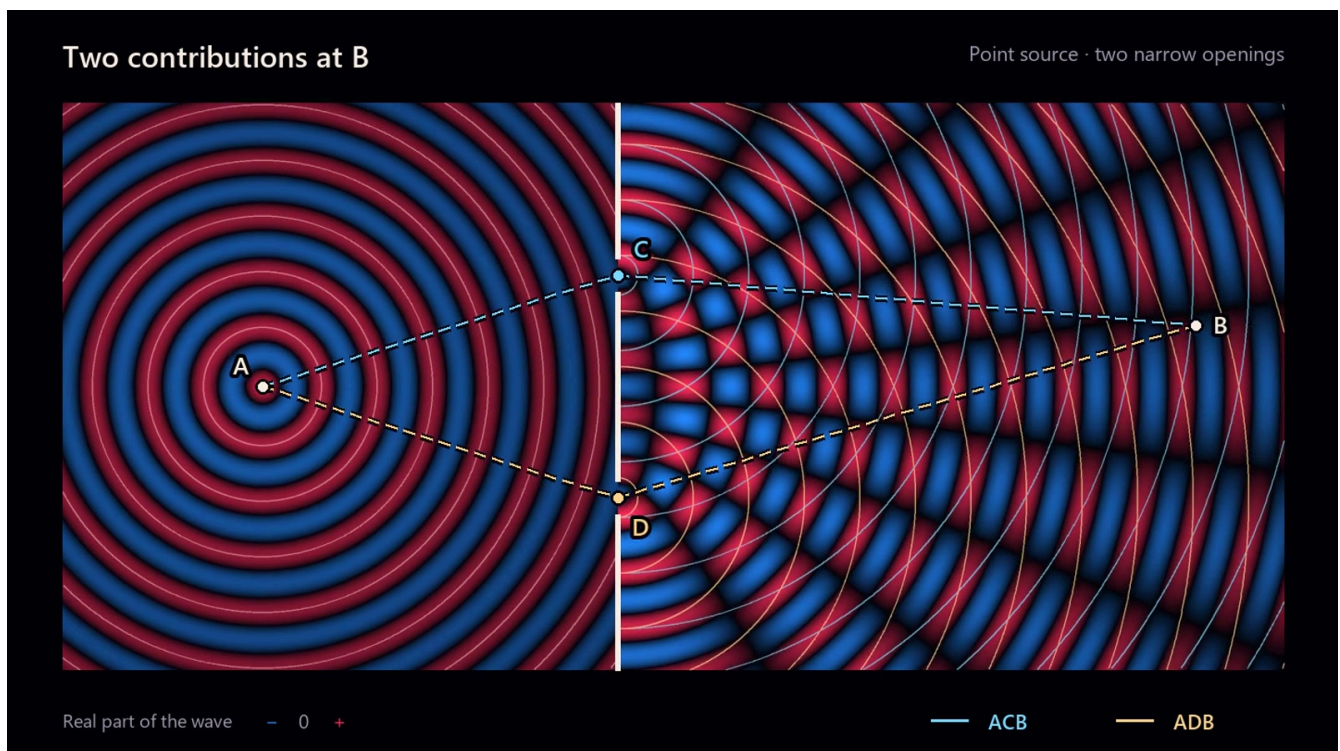
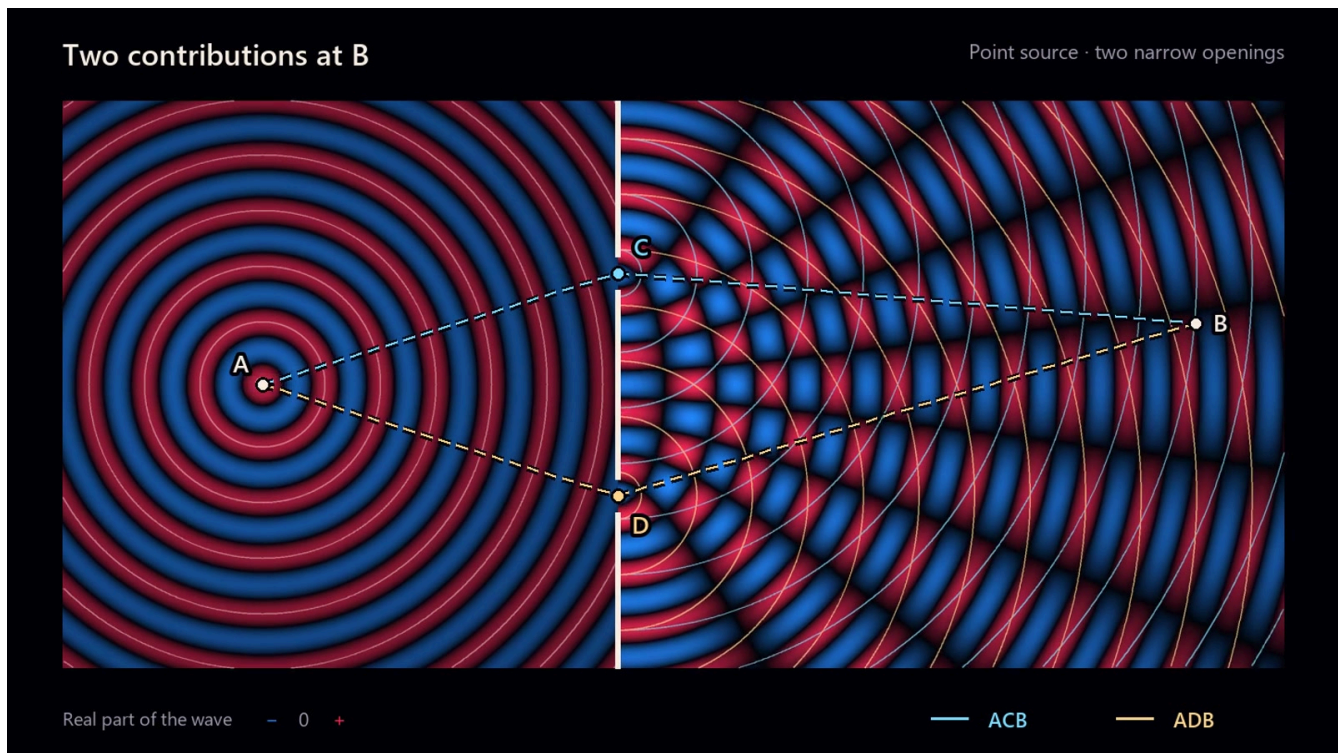
Two Path Interference

Animation 24 / frame 1 of 4 / 0.20 s



Animation 24 / frame 2 of 4 / 3.30 s





First, let's figure out how any one straight segment of a plane wave contributes to the amplitude at its endpoint. From:

$$e^{i[kx - \omega t]}.$$

we can identify that:

$$\Delta\phi = k \Delta\ell - \omega \Delta t,$$

where $\Delta\ell$ is distance along the ray. In our setup, the segments combine into two candidate paths from A to B :

$$A \rightarrow C \rightarrow B \quad \text{and} \quad A \rightarrow D \rightarrow B.$$

We can now ask how each path contributes to the amplitude at B . The magnitude simply falls off as $1/r$ in accordance with spherical geometry. Also, since all contributions arrive at B at the same observation time, their time-dependent phase is the same. The remaining things to calculate are their path-dependent spatial phases. Let ℓ_{AC} be the length of segment AC , and likewise for the other segments. The phase advances are:

$$\begin{aligned} \phi_{AC} &= k\ell_{AC}, & \phi_{CB} &= k\ell_{CB}, \\ \phi_{AD} &= k\ell_{AD}, & \phi_{DB} &= k\ell_{DB}. \end{aligned}$$

To transform the wave function along the path, we compose, or multiply, the phase actions. Because multiplying phase factors adds the angles in their exponents, we can simply add the phase angle contributions to calculate the total phase advance along ACB and ADB , respectively. Letting ϕ_0 include the original phase at A and the temporal phase advance common to both paths, we have:

$$\begin{aligned} \Phi_{ACB} &= \phi_0 + \phi_{AC} + \phi_{CB} = \phi_0 + k(\ell_{AC} + \ell_{CB}), \\ \Phi_{ADB} &= \phi_0 + \phi_{AD} + \phi_{DB} = \phi_0 + k(\ell_{AD} + \ell_{DB}). \end{aligned}$$

If we let M_{ACB} and M_{ADB} denote the magnitudes of the two path contributions, we then have the total contributions of each path at B :

$$\begin{aligned} \Psi_{ACB}(B) &= M_{ACB}e^{i(\phi_0 + \phi_{AC} + \phi_{CB})}, \\ \Psi_{ADB}(B) &= M_{ADB}e^{i(\phi_0 + \phi_{AD} + \phi_{DB})}. \end{aligned}$$

What then is the total amplitude at B ? It is just the sum of the two contributions arriving there. This is the principle of superposition which manifests visually as interference:

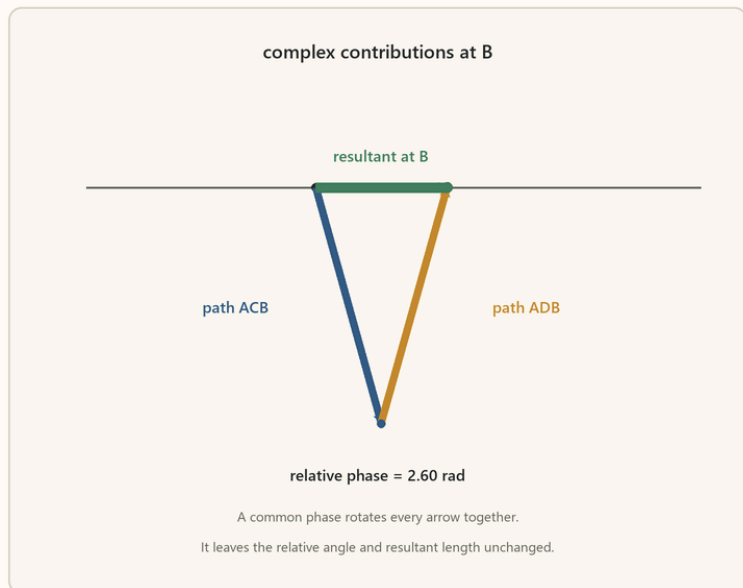
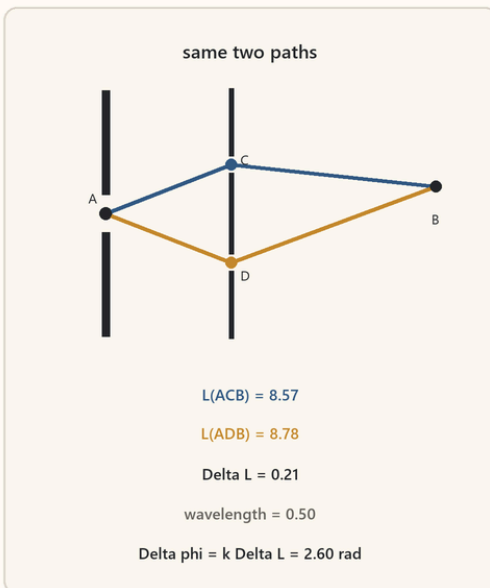
$$\Psi_B = M_{ACB}e^{i(\phi_0 + \phi_{AC} + \phi_{CB})} + M_{ADB}e^{i(\phi_0 + \phi_{AD} + \phi_{DB})}.$$

Compose phase actions along a route. Add amplitudes across candidate routes.

We can plot the contribution from each path ACB and ADB in the complex plane, showing their sum by drawing them tip-to-tail. The vector from the beginning of the first arrow to the end of the second is the amplitude at B . Its length is the magnitude at B and its angle is the phase at B .

The two path contributions add tip to tail

Equal magnitudes; the path difference determines their relative phase.

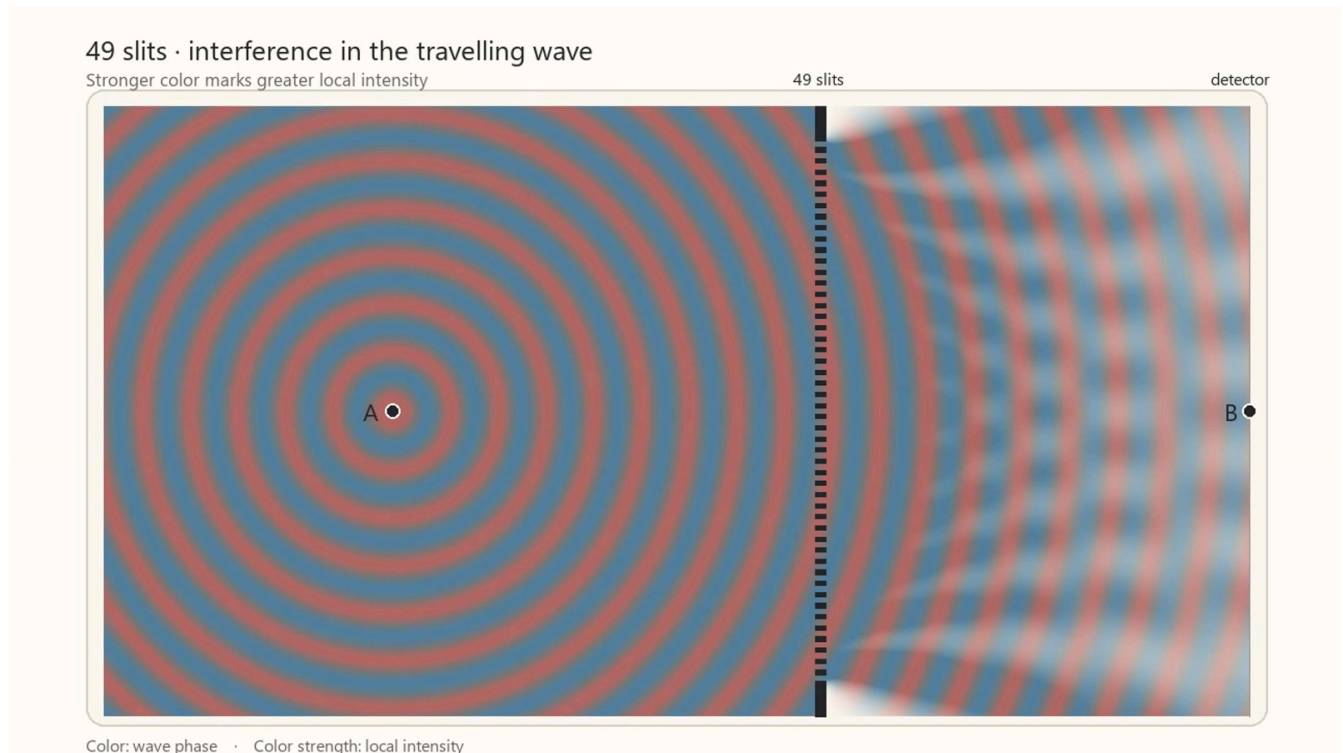


*Amplitude at **B** is the sum of contributions from **ACB** and **ADB***

We can repeat the same procedure with many more paths. As the path deviates more from a straight, minimum length path, it has a greater first-order change in phase. (This is the common result from calculus that near a function's minimum, there is no change to the value of the function in the first order of the argument.) When the candidate paths are far from the stationary value, their phases vary greatly, effectively cancelling out their contributions to the total sum. On the other hand, the phases of the paths near the stationary path align and dominate the sum. The green line in the tip-to-tail pane of the animation shows the sum of each of these contributions, giving the amplitude at **B**. The resulting intensity on the projection screen is the square of this magnitude.

Seeing stationarity emerge in closely spaced paths

Animation 25 / frame 1 of 4 / 0.45 s



Animation 25 / frame 2 of 4 / 7.42 s

Many paths · one consistent wave amplitude

Complete propagation phases and weights · the same sum drives the detector

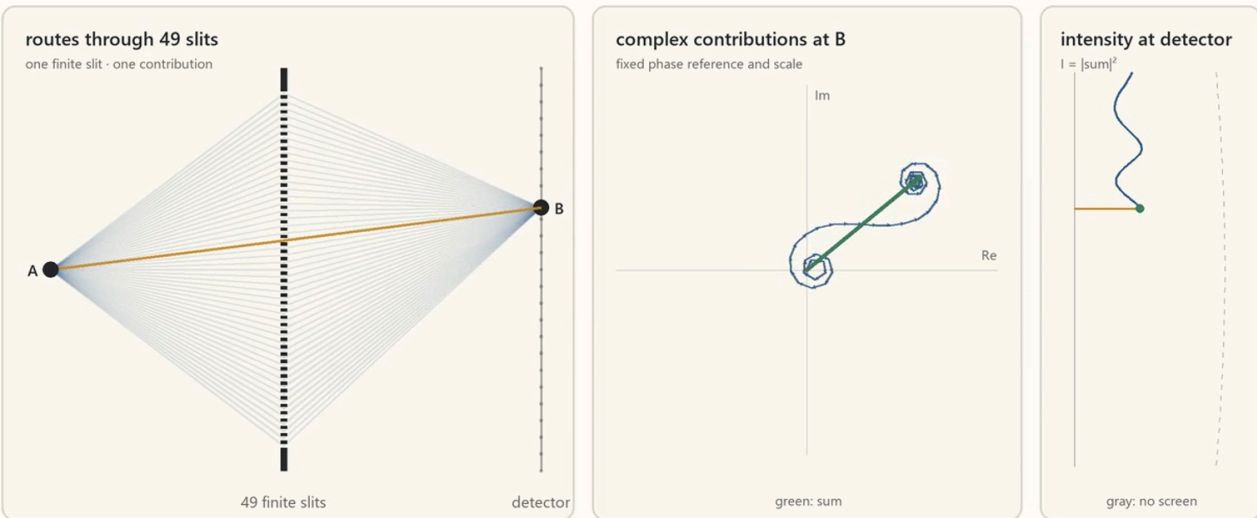


Phase zero: the complete sum at central B. The reference stays fixed as B moves.

2D scalar diffraction · 49 finite slits with opaque gaps · lengths carry amplitude

Many paths · one consistent wave amplitude

Complete propagation phases and weights · the same sum drives the detector

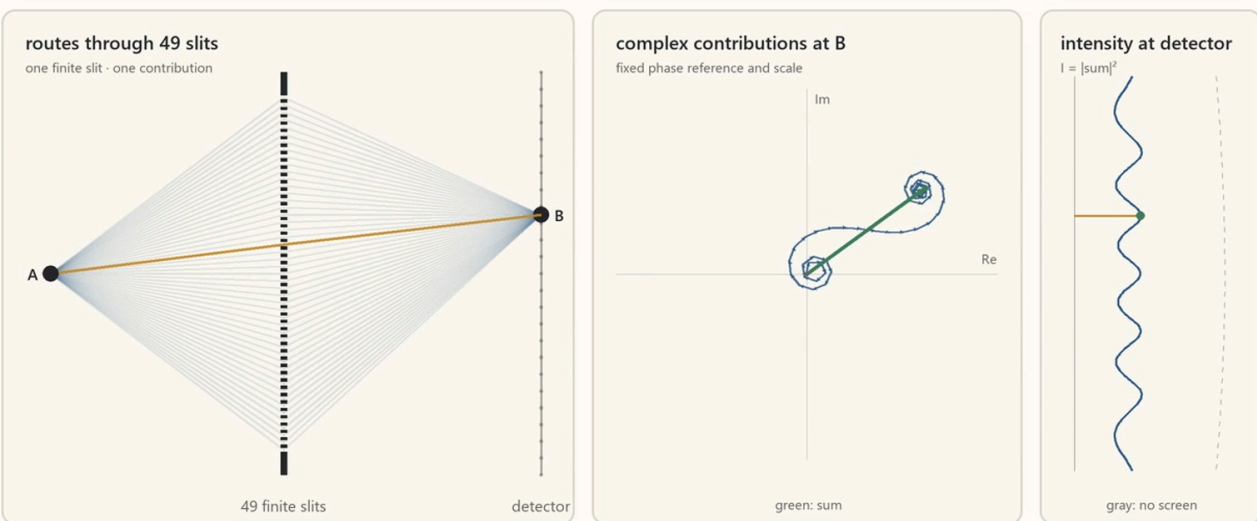


Phase zero: the complete sum at central B. The reference stays fixed as B moves.

2D scalar diffraction · 49 finite slits with opaque gaps · lengths carry amplitude

Many paths · one consistent wave amplitude

Complete propagation phases and weights · the same sum drives the detector



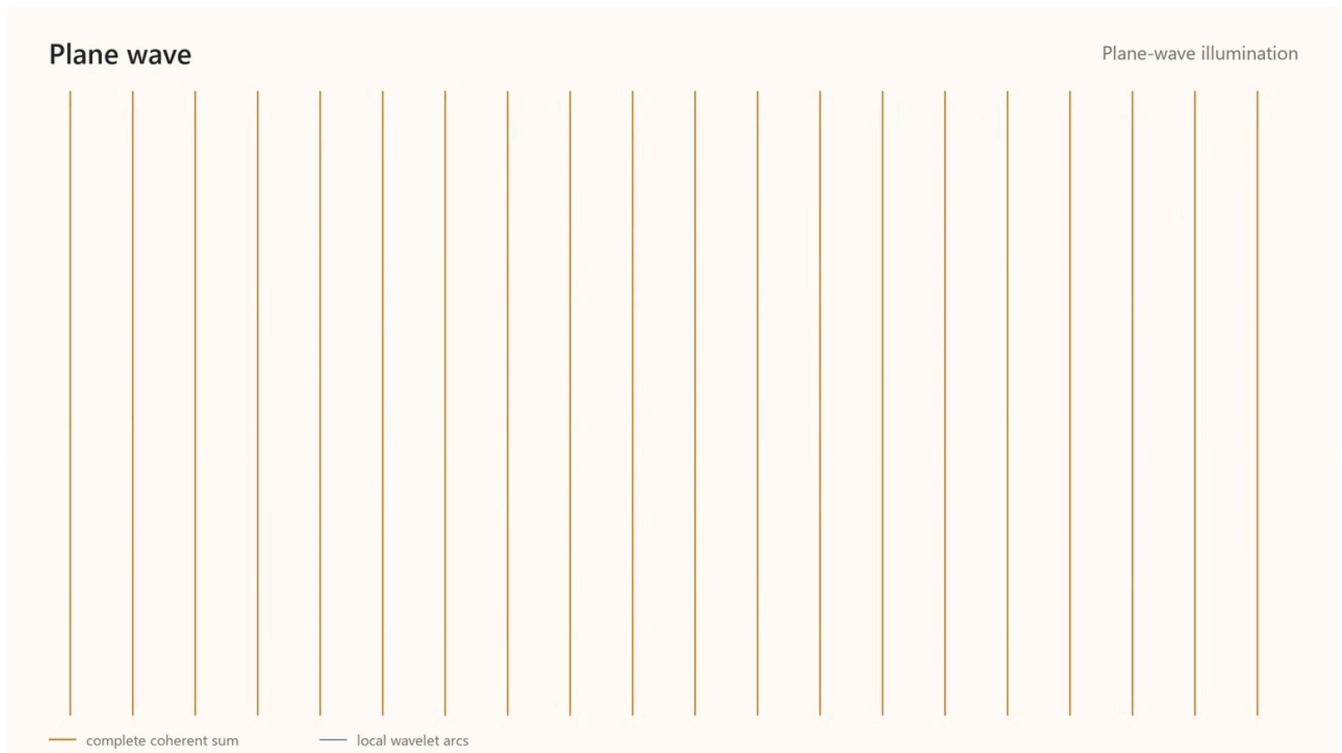
Phase zero: the complete sum at central B. The reference stays fixed as B moves.

2D scalar diffraction · 49 finite slits with opaque gaps · lengths carry amplitude

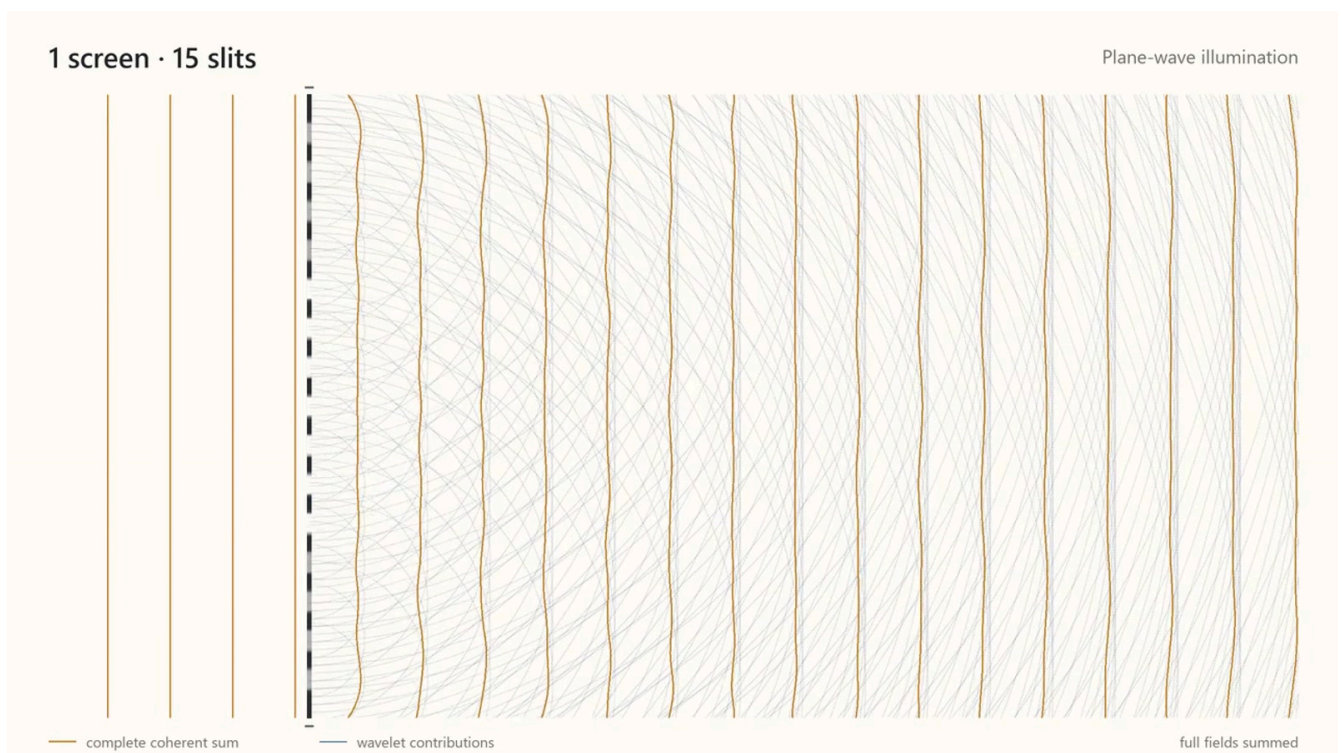
We can extend this procedure to its limit and include infinitely many screens with infinitely many slits, and recover unobstructed propagation. In the following animation, we start with a plane wave and recover that same wave. This construction, which provides a bridge to Feynman's path integral formulation of quantum mechanics, has its roots in Huygens' wavelets from the late 1600s, extended by Fresnel to include interference in the early 1800s.

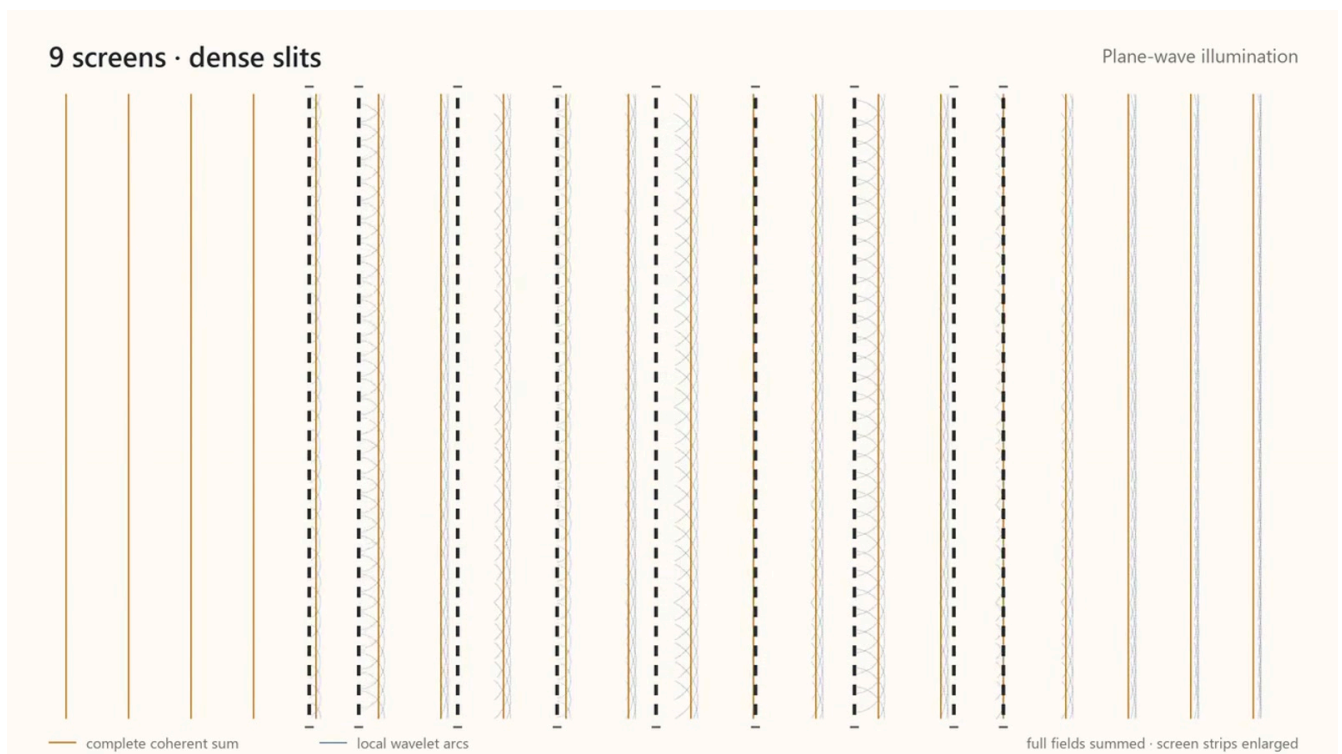
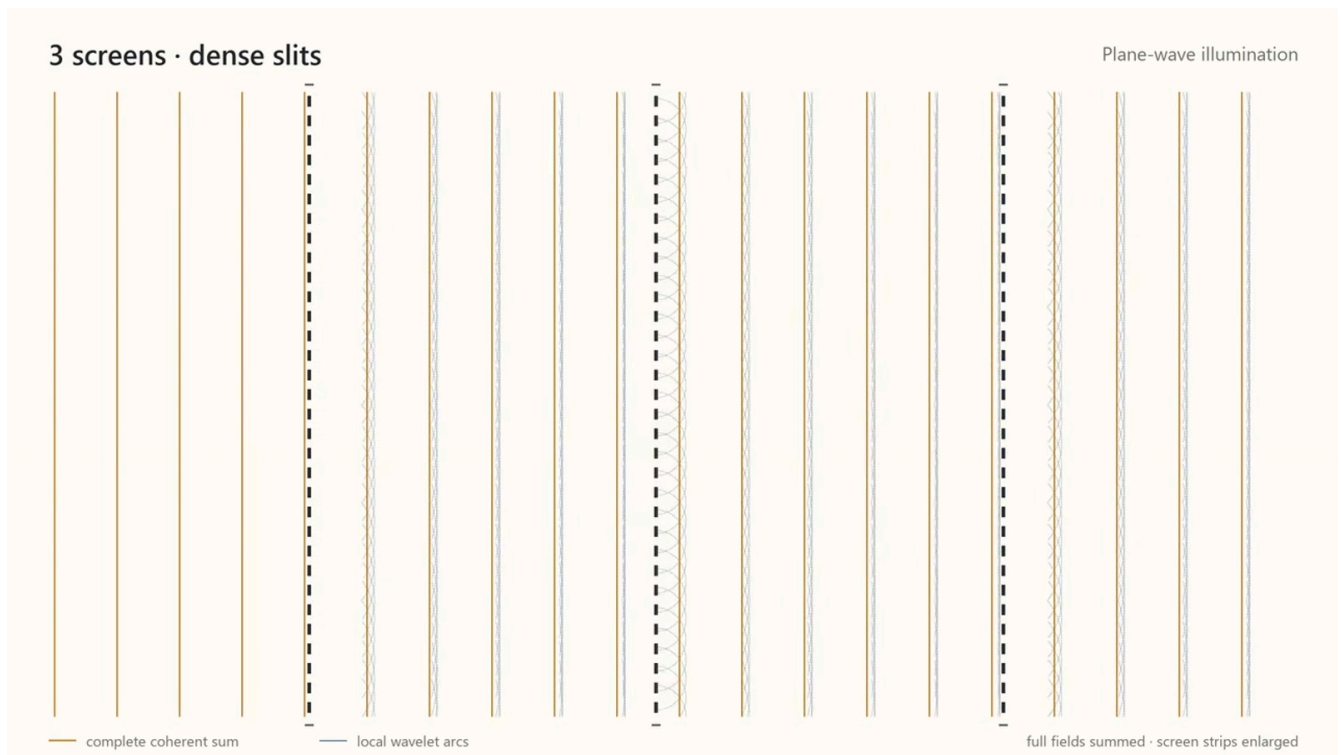
Huygens–Fresnel principle

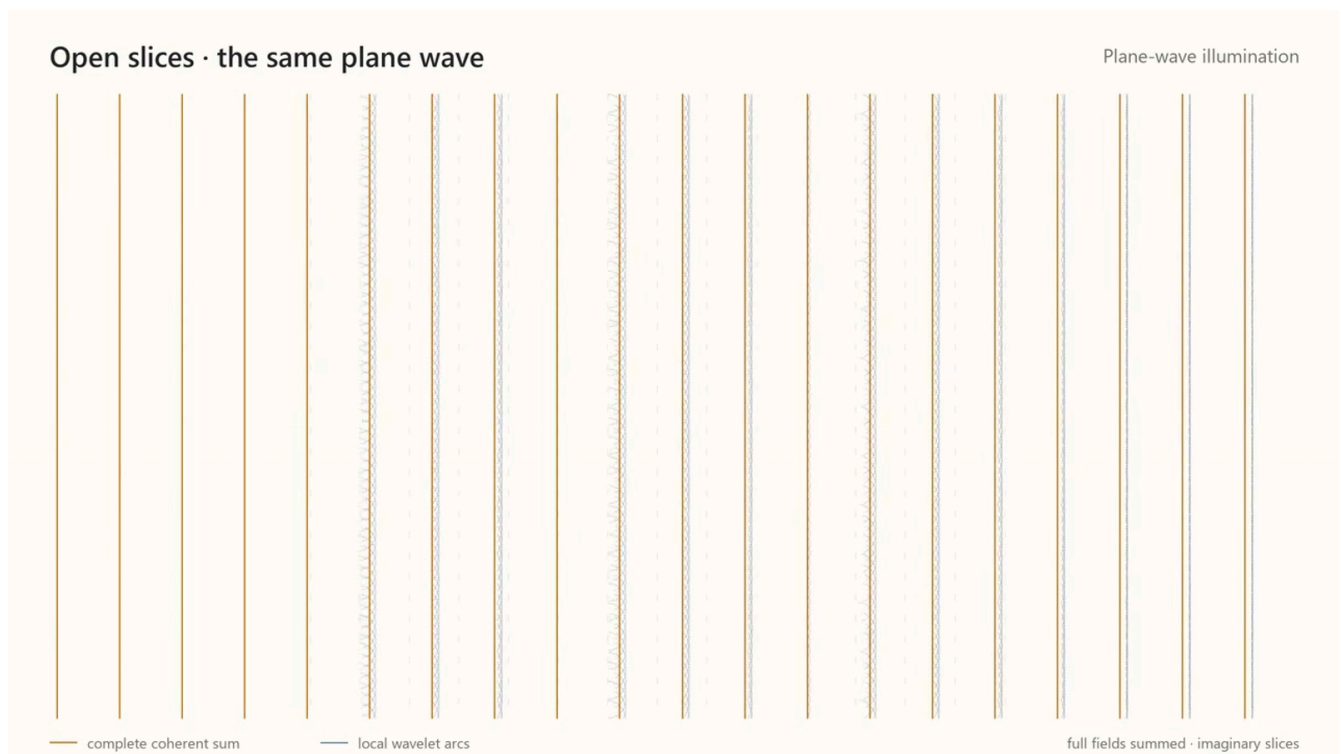
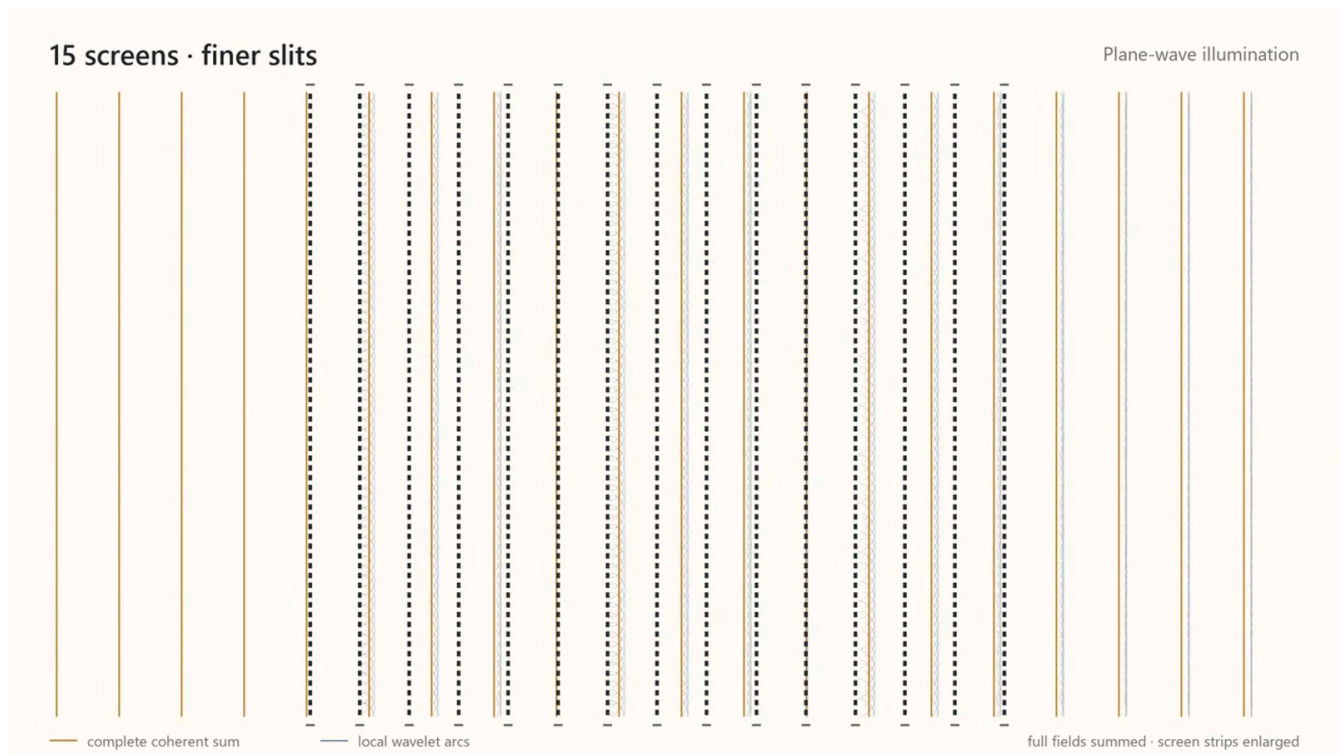
Animation 26 / frame 1 of 6 / 0.48 s



Animation 26 / frame 2 of 6 / 4.80 s







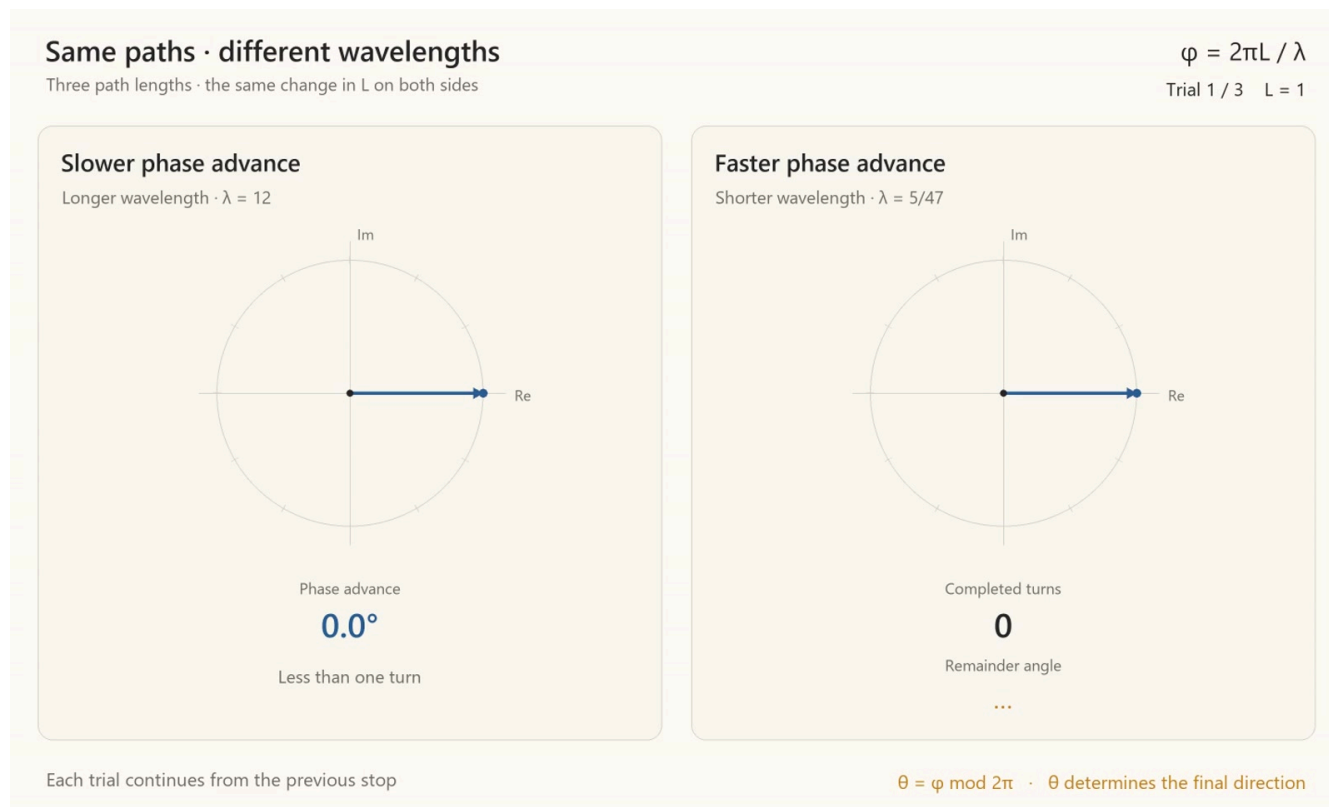
Let us now ask one more question. What happens to our tip-to-tail diagram of the path contributions when we vary the wavelength relative to the slit? As the wavelength becomes small, even a slight change in path length can produce a large phase change:

$$\phi = \frac{2\pi L}{\lambda} = 2\pi n + \theta, \quad 0 \leq \theta < 2\pi$$

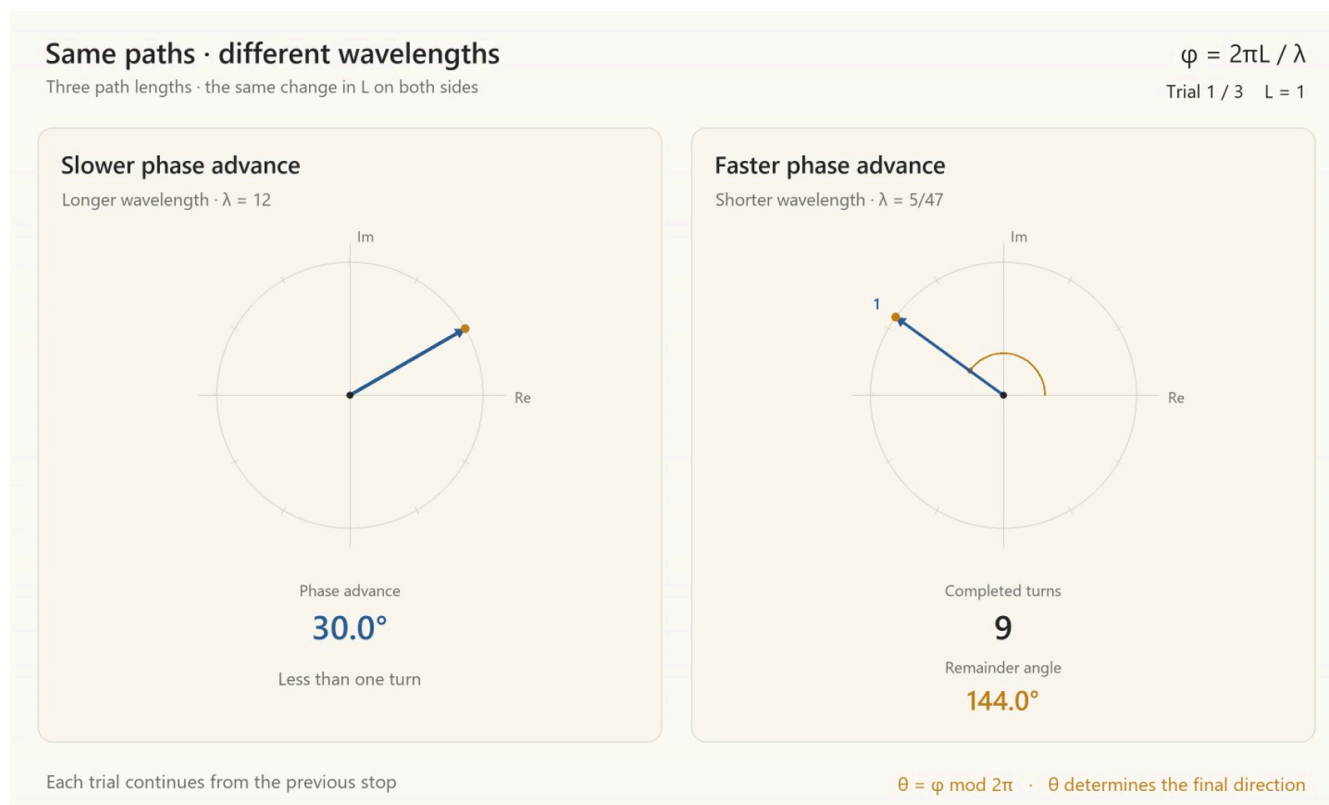
Away from a stationary path, the phase winds through many cycles over a small range of paths, leaving an effectively random phase remainder so that contributions away from stationary paths largely cancel, while those near stationary paths dominate the sum.

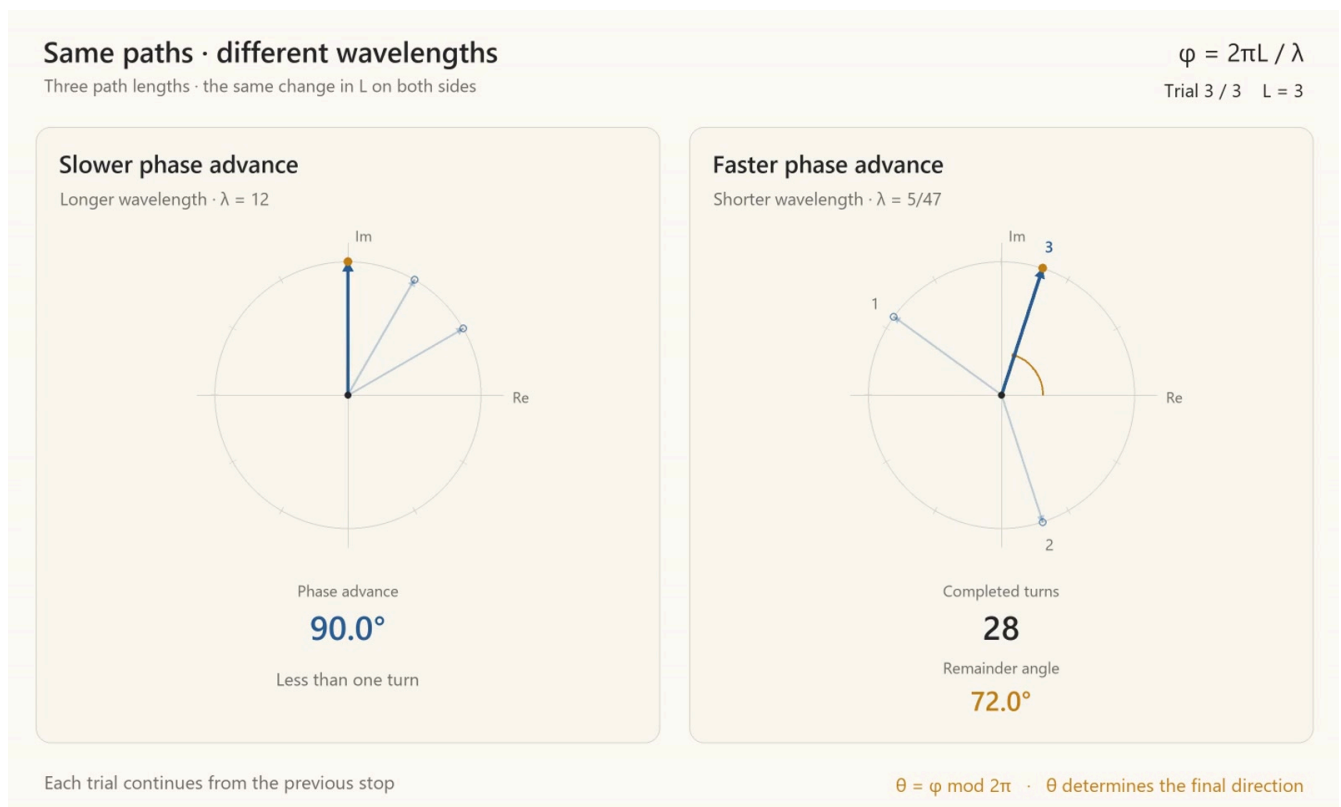
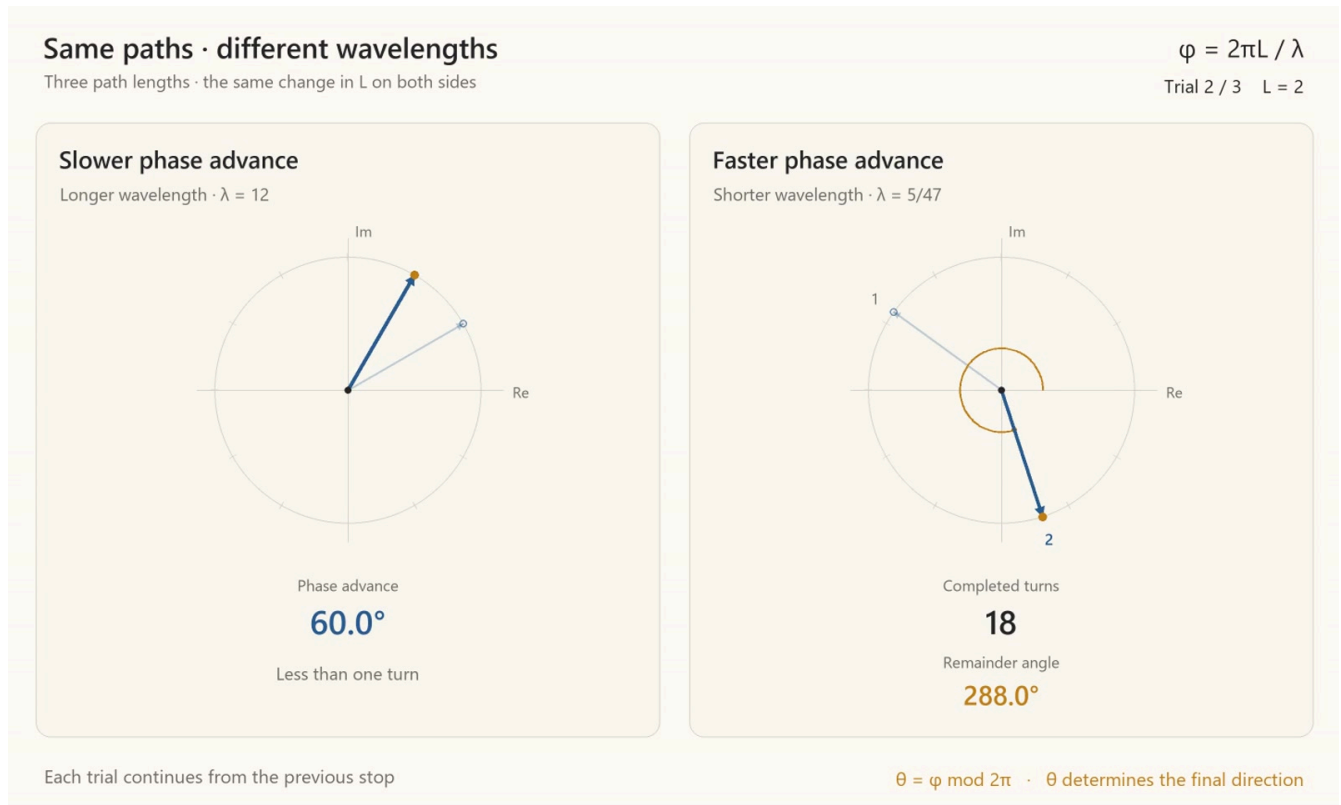
Three trials accumulate a quarter turn at the longer wavelength while the shorter wavelength produces many rotations

Animation 27 / frame 1 of 4 / 0.34 s



Animation 27 / frame 2 of 4 / 5.61 s



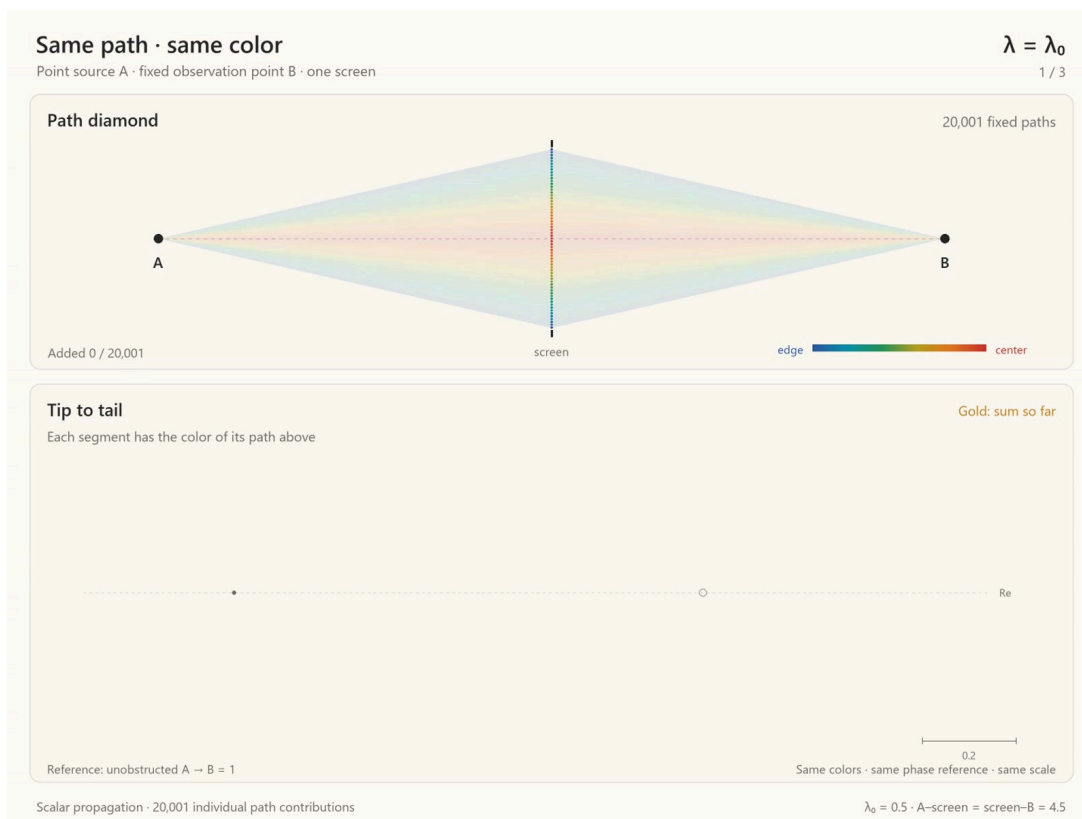


In this limit, waves passing through slits behave as rays. A narrow beam through one opening reaches a small region of the screen, much as a thrown ball does.

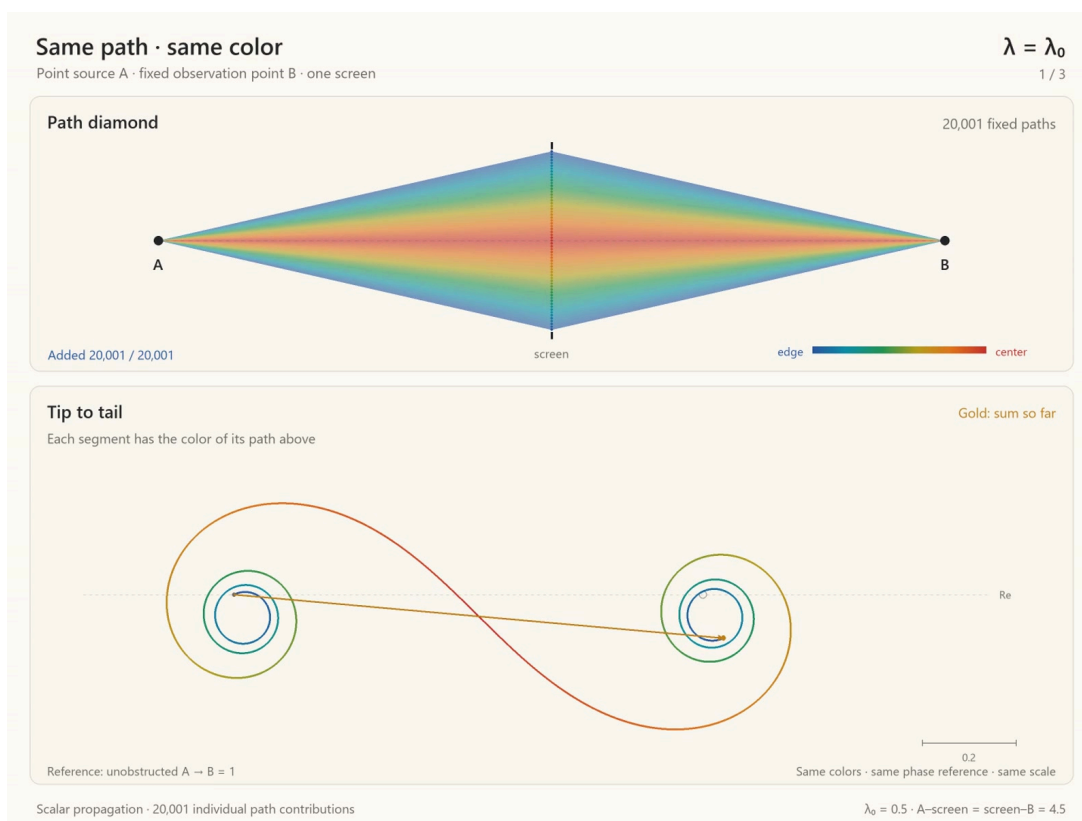
Each path shares its color with its contribution to the tip-to-tail sum. As the wavelength shrinks, an increasingly narrow range of paths around the straight path carries the sum forward.

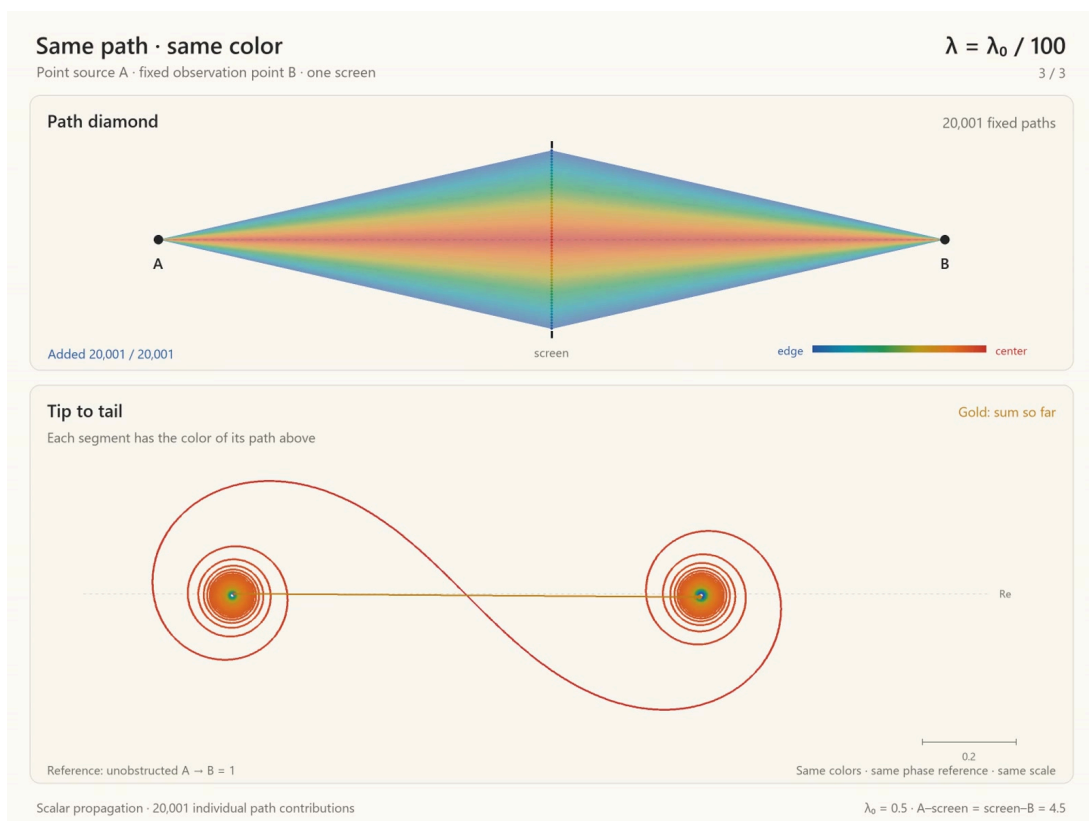
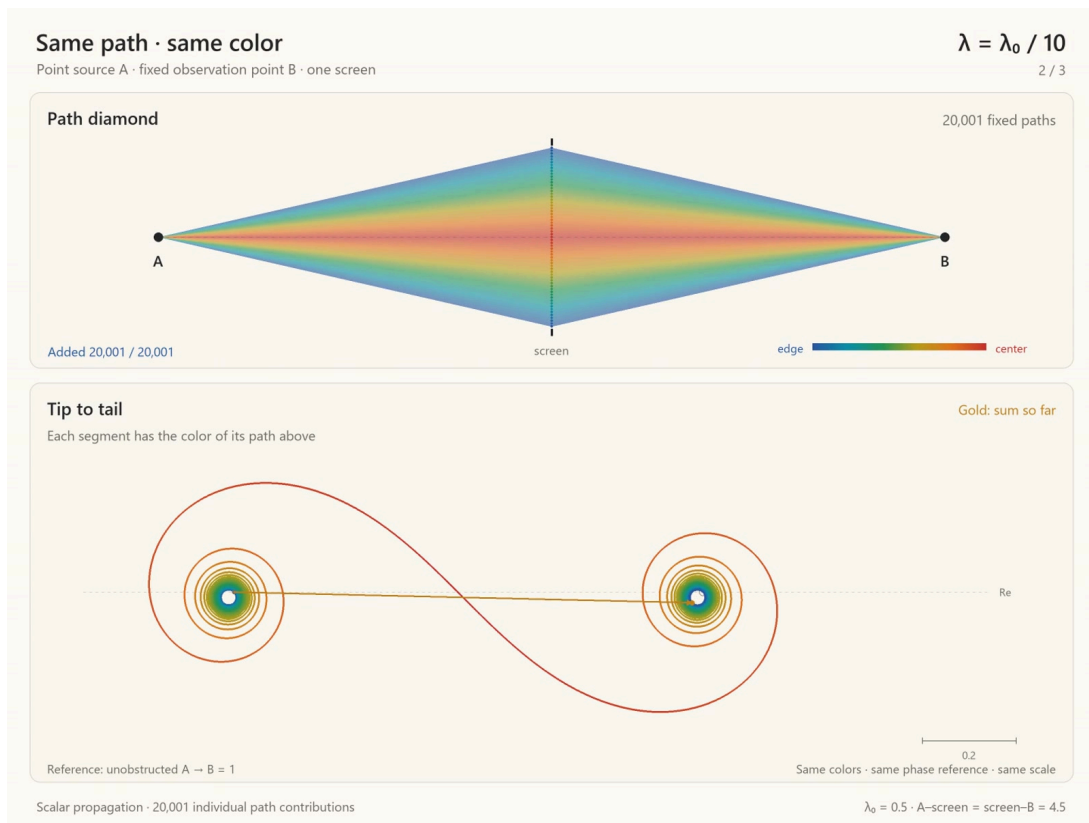
Matching colors connect candidate paths to their contributions in the tip-to-tail sum as wavelength decreases

Animation 28 / frame 1 of 4 / 0.72 s



Animation 28 / frame 2 of 4 / 11.88 s





From Wave Mechanics to Quantum Mechanics

Quantum mechanics gives wave intensity a new physical interpretation. For a normalized wave function, the squared magnitude gives the probability density for a position measurement:

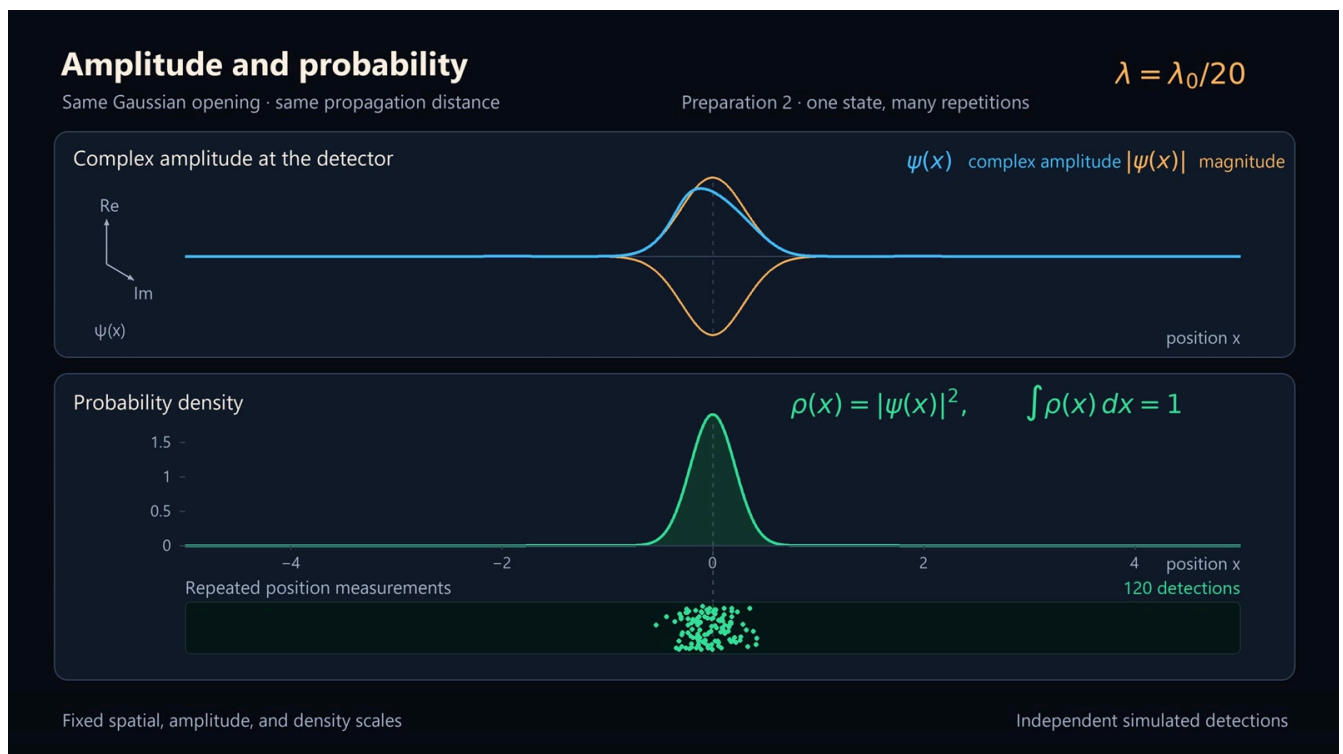
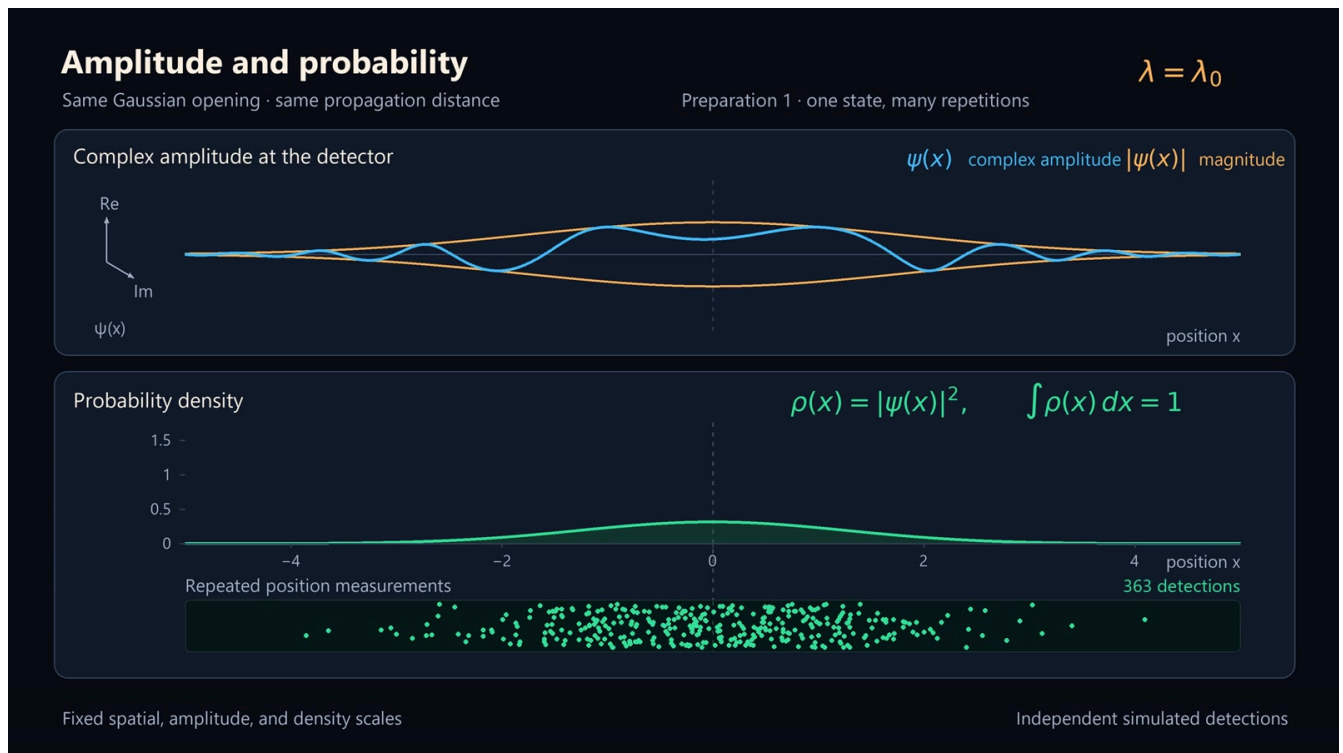
$$\rho(x) = |\psi(x)|^2, \quad \int_{-\infty}^{\infty} \rho(x) dx = 1.$$

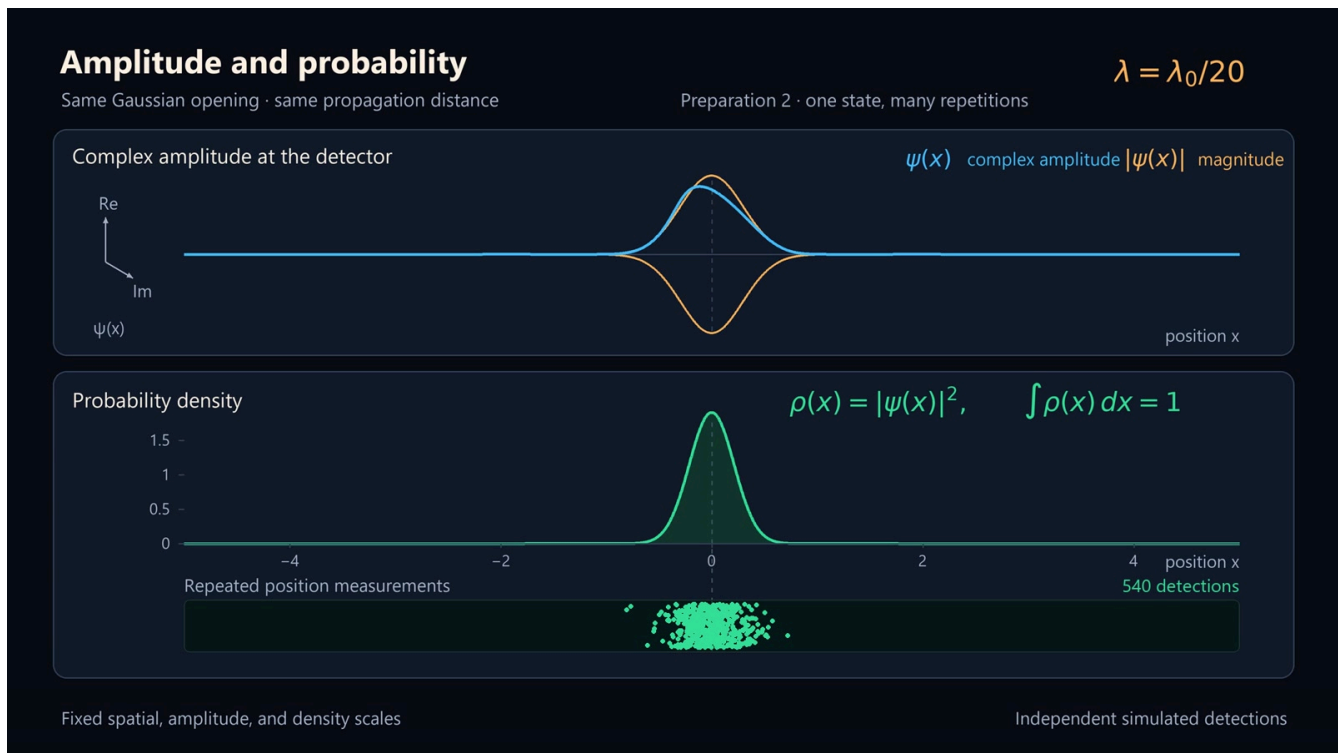
We could write the same relationship in the wave-number representation. In either case, the interpretation is as follows. A state is a superposition of the eigenstates in the chosen basis. An ideal measurement leaves the state in a specific eigenstate of the measured operator. The value of the measurement is the eigenvalue of that state. The squared magnitudes of the coefficients multiplying those eigenstates determine the outcome probabilities. They must sum to one for discrete outcomes. For continuous quantities such as position and wave number, they give probability densities that integrate to one. In this view, we may say, loosely, and only if we are so inclined, that the thing that "is" is a wave packet, and the wave-number distribution we measure is given by the squared magnitudes of its Fourier components.

From intensity to position measurement probability density

Animation 29 / frame 1 of 4 / 0.56 s







To connect our wave description to mechanics, we need to relate phase to action.

Action, the quantity extremized by a physically valid path, can be constructed from the structure of **spacetime**, as articulated in the theory of special relativity, which will be the topic of our next chapter. Crudely speaking, because the quantity to be extremized must be agreed upon by all observers, it is natural that it should be an invariant of symmetry actions on spacetime. This leads to the result that the action is, in free motion, for massive bodies, proportional to an invariant built from translations — the time elapsed along a path as measured in a body's rest frame — times a dual invariant built from time and space translation generators. The former quantity is called **proper time** while the latter is the body's **mass**.

$$S = -m\tau \quad (c = 1)$$

At the same time, we can calculate how phase advances along a path in spacetime. Introducing κ , the invariant built from wave function spacetime-translation generators, a single-mode plane wave along its inertial **worldline**, or path in a position vs time plot, is:

$$e^{-i\kappa\tau}$$

We then have:

$$\text{constant} = \frac{S}{\phi} = \frac{m}{\kappa}$$

For a single free massive particle, multiplying the action by a constant leaves its stationary paths unchanged. A collision couples the variations of two particle types.

Let's consider an idealized brief collision with free motion before and after. Neglecting the interaction region's contribution to first-order changes in action and phase, each segment contributes $-m_i\tau_i$ to the action and, in the limit where waves follow definite paths, $-\kappa_i\tau_i$ to its propagation phase. The

wave and mechanical descriptions must predict the same collision. We can hold the initial and final positions and times fixed and slightly shift the candidate collision point. At the stationary-phase point, the two particles' first-order phase changes cancel.

$$\delta\phi_{\text{total}} = \delta\phi_1 + \delta\phi_2 = 0.$$

With $a_i = m_i/\kappa_i$, this gives

$$\begin{aligned}\delta\phi_1 &= \epsilon, & \delta\phi_2 &= -\epsilon, \\ \delta S &= a_1\delta\phi_1 + a_2\delta\phi_2 = (a_1 - a_2)\epsilon.\end{aligned}$$

If the collision changes a particle's speed or direction, we can shift the candidate collision point so that its phase changes to first order. Choose such a shift, so $\epsilon \neq 0$. For the action to be stationary at the same point, $a_1 = a_2$. The shared conversion factor is $\hbar$.

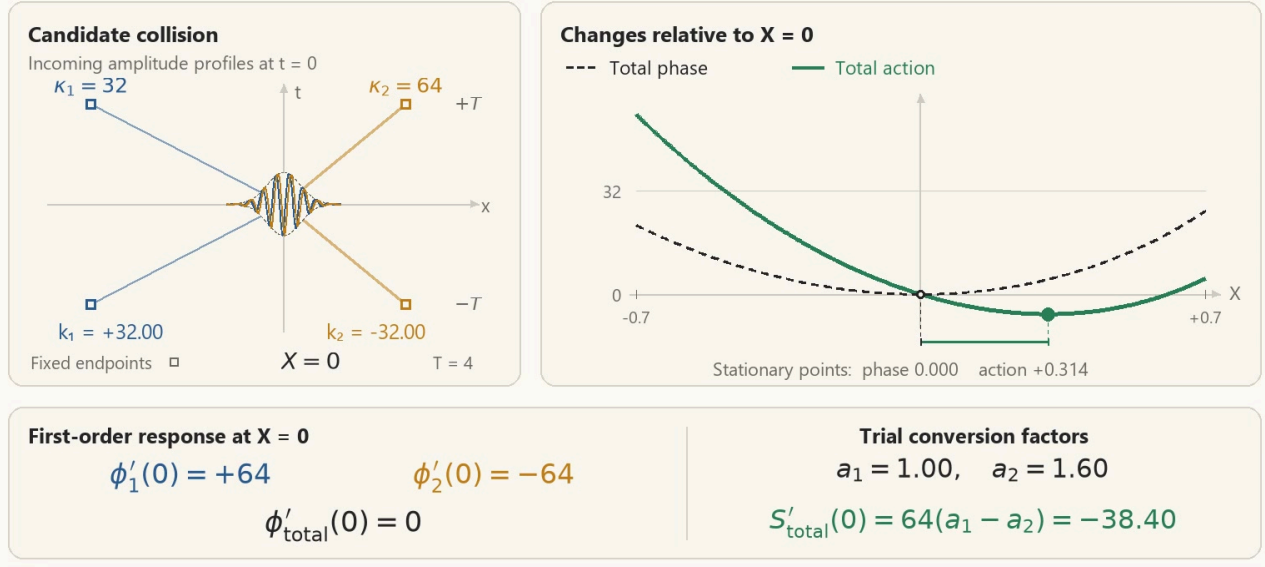
A shared conversion factor makes phase and action select the same collision

Animation 30 / frame 1 of 6 / 0.60 s

A shared scale for phase and action

$$S_i = a_i\phi_i, \quad a_i = \frac{m_i}{\kappa_i}$$

Unequal factors · phase and action predict different collision points



Vary X , not time · fixed packet envelopes · negligible interaction contribution · illustrative units

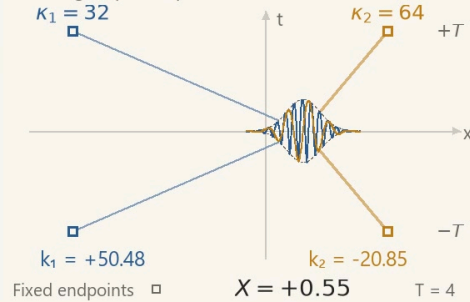
A shared scale for phase and action

$$S_i = a_i \phi_i, \quad a_i = \frac{m_i}{\kappa_i}$$

Unequal factors · phase and action predict different collision points

Candidate collision

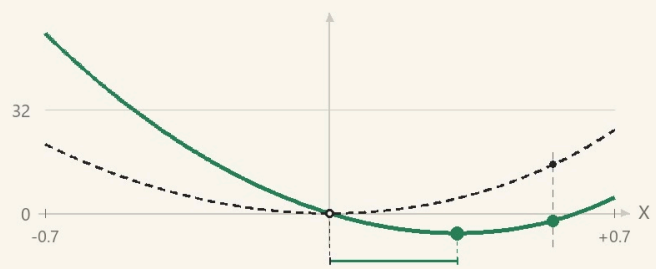
Incoming amplitude profiles at $t = 0$



Changes relative to $X = 0$

--- Total phase

— Total action



First-order response at $X = 0$

$$\phi'_1(0) = +64$$

$$\phi'_2(0) = -64$$

$$\phi'_{\text{total}}(0) = 0$$

Trial conversion factors

$$a_1 = 1.00, \quad a_2 = 1.60$$

$$S'_{\text{total}}(0) = 64(a_1 - a_2) = -38.40$$

Vary X , not time · fixed packet envelopes · negligible interaction contribution · illustrative units

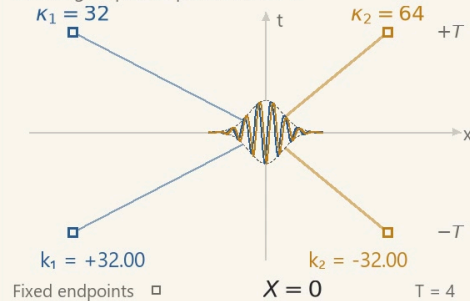
A shared scale for phase and action

$$S_i = a_i \phi_i, \quad a_i = \frac{m_i}{\kappa_i}$$

Test two conversion factors · hold the collision point at $X = 0$

Candidate collision

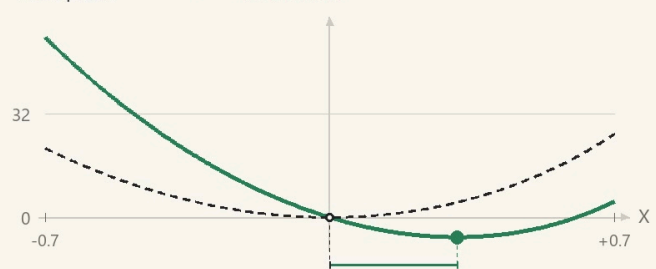
Incoming amplitude profiles at $t = 0$



Changes relative to $X = 0$

--- Total phase

— Total action



First-order response at $X = 0$

$$\phi'_1(0) = +64$$

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$$\phi'_{\text{total}}(0) = 0$$

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$$S'_{\text{total}}(0) = 64(a_1 - a_2) = -38.40$$

Vary X , not time · fixed packet envelopes · negligible interaction contribution · illustrative units

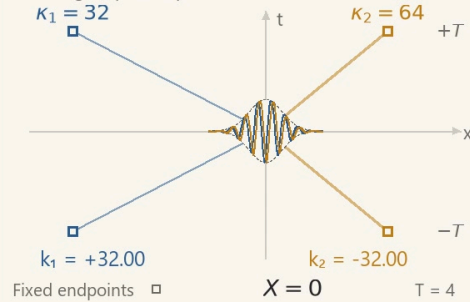
A shared scale for phase and action

$$S_i = a_i \phi_i, \quad a_i = \frac{m_i}{K_i}$$

Equal factors · repeat the same collision-point sweep

Candidate collision

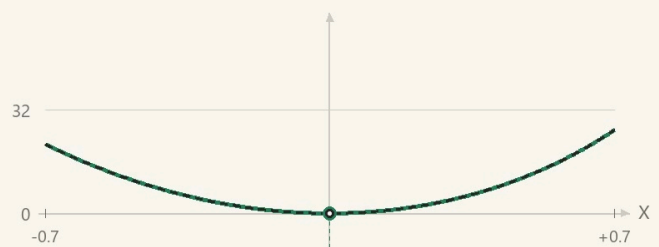
Incoming amplitude profiles at $t = 0$



Changes relative to $X = 0$

--- Total phase

— Total action



Stationary points: phase 0.000 action +0.000

First-order response at $X = 0$

$$\phi'_1(0) = +64$$

$$\phi'_2(0) = -64$$

$$\phi'_{\text{total}}(0) = 0$$

Shared conversion factor

$$a_1 = 1.00, \quad a_2 = 1.00$$

$$S'_{\text{total}}(0) = 64(a_1 - a_2) = 0$$

Vary X , not time · fixed packet envelopes · negligible interaction contribution · illustrative units

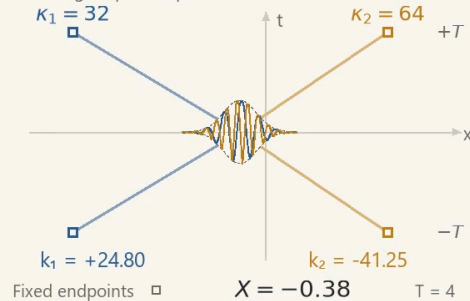
A shared scale for phase and action

$$S_i = a_i \phi_i, \quad a_i = \frac{m_i}{K_i}$$

Equal factors · repeat the same collision-point sweep

Candidate collision

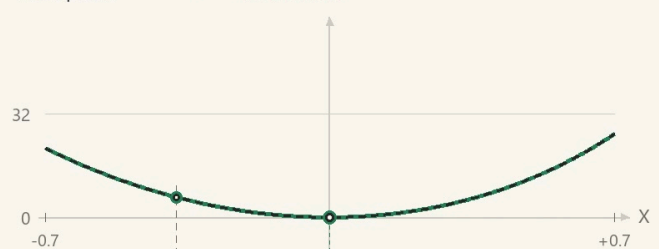
Incoming amplitude profiles at $t = 0$



Changes relative to $X = 0$

--- Total phase

— Total action



Stationary points: phase 0.000 action +0.000

First-order response at $X = 0$

$$\phi'_1(0) = +64$$

$$\phi'_2(0) = -64$$

$$\phi'_{\text{total}}(0) = 0$$

Shared conversion factor

$$a_1 = 1.00, \quad a_2 = 1.00$$

$$S'_{\text{total}}(0) = 64(a_1 - a_2) = 0$$

Vary X , not time · fixed packet envelopes · negligible interaction contribution · illustrative units

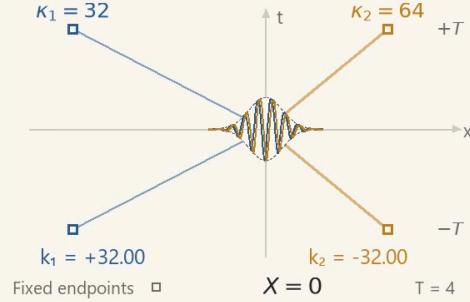
A shared scale for phase and action

$$S_i = a_i \phi_i, \quad a_i = \frac{m_i}{\kappa_i}$$

Equal factors · phase and action predict the same collision point

Candidate collision

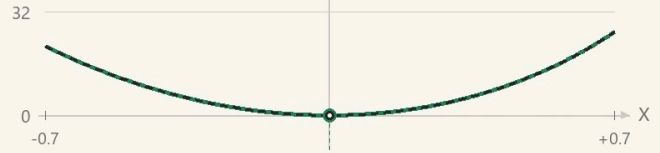
Incoming amplitude profiles at $t = 0$



Changes relative to $X = 0$

--- Total phase

— Total action



First-order response at $X = 0$

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$$\phi'_2(0) = -64$$

$$\phi'_{\text{total}}(0) = 0$$

Shared conversion factor

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$$S'_{\text{total}}(0) = 64(a_1 - a_2) = 0$$

Vary X , not time · fixed packet envelopes · negligible interaction contribution · illustrative units

We can measure its numerical value in the lab by comparing mass, as measured through collisions, to wavelength, as measured through interference. We then find:

$$\frac{S}{\phi} := \hbar \approx 1.055 \times 10^{-34} \text{ J s.}$$

The smaller $\hbar$ is, the more phase cycles a given action difference produces. For candidate paths, we use the wave mode appropriate to each segment and add the accumulated phases, following the procedure described earlier. For two candidate paths:

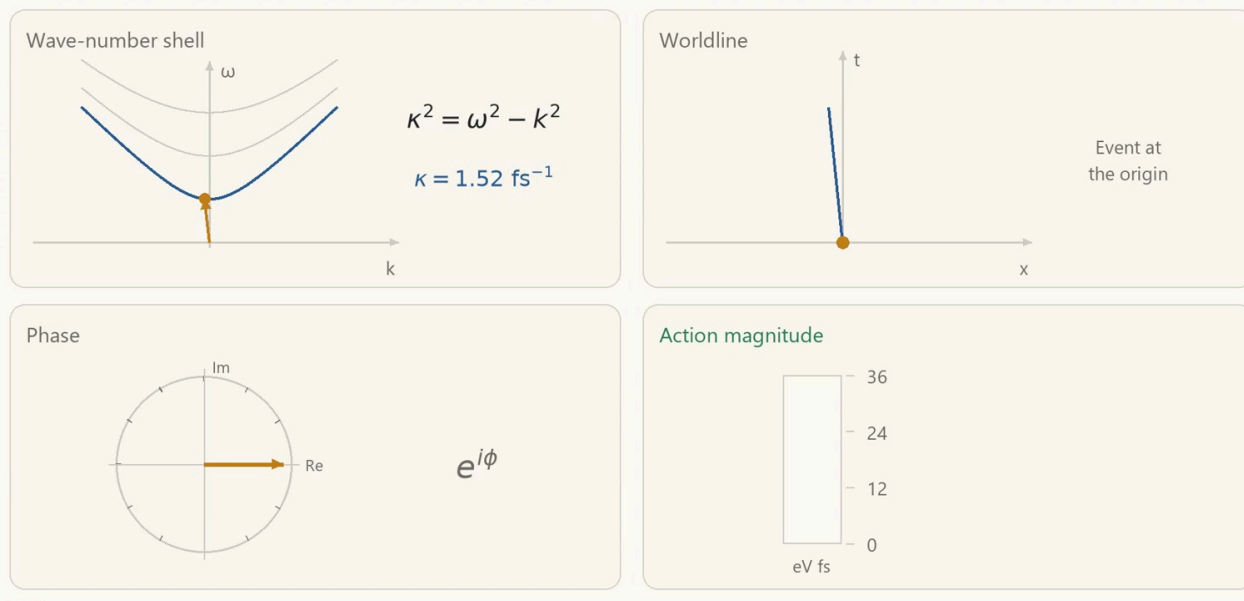
$$\Delta\phi = \frac{\Delta S}{\hbar}.$$

Three wave-number shells share the same proper-time interval while phase cycles and action accumulates at corresponding rates

Animation 31 / frame 1 of 6 / 0.48 s

$\hbar$ relates phase to action

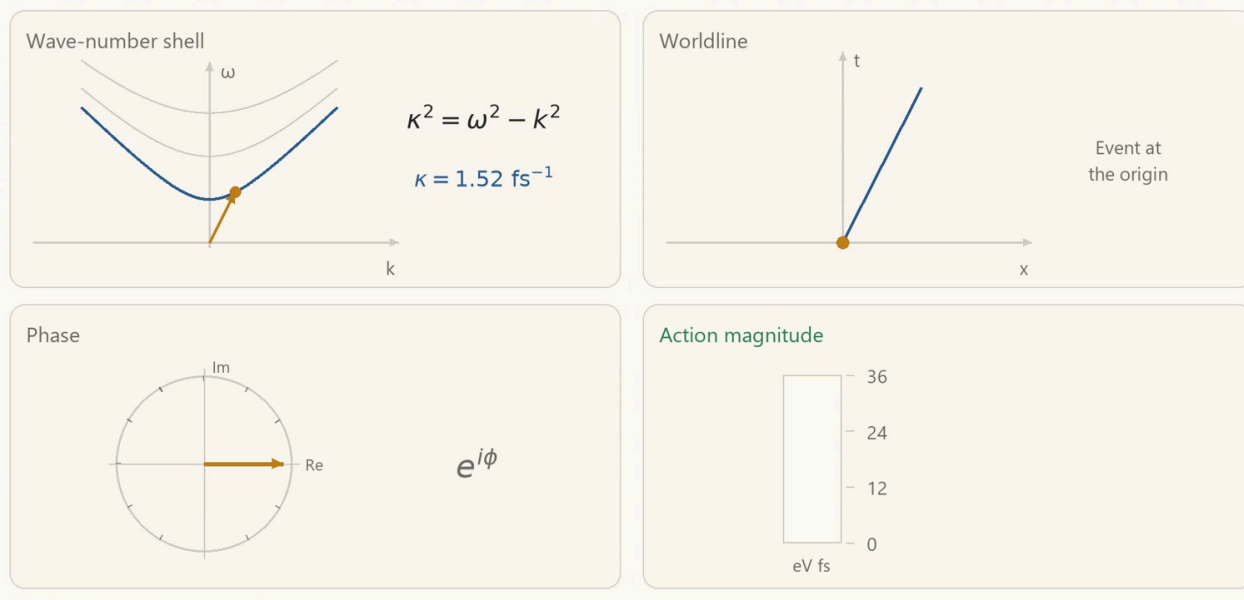
Change the frame. Keep the event fixed.



Animation 31 / frame 2 of 6 / 4.80 s

$\hbar$ relates phase to action

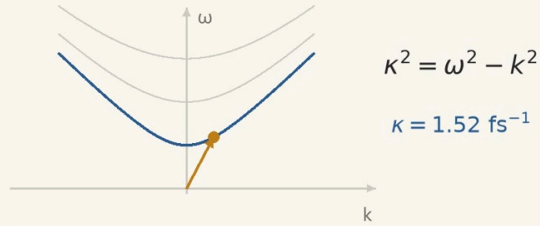
Change the frame. Keep the event fixed.



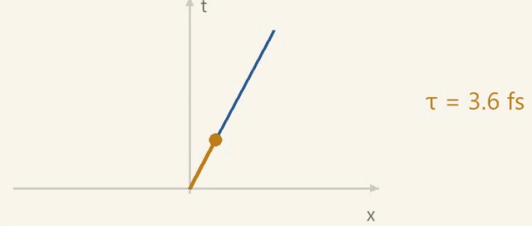
$\hbar$ relates phase to action

Shell 1 of 3 · The same proper-time interval.

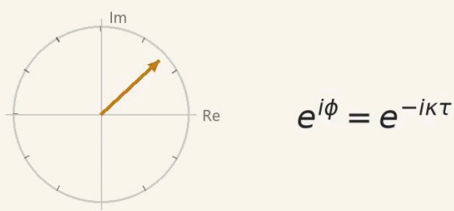
Wave-number shell



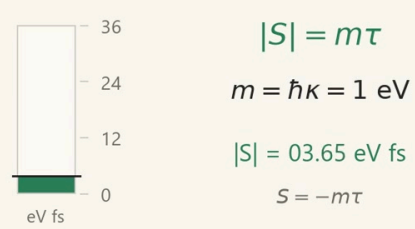
Worldline



Phase



Action magnitude

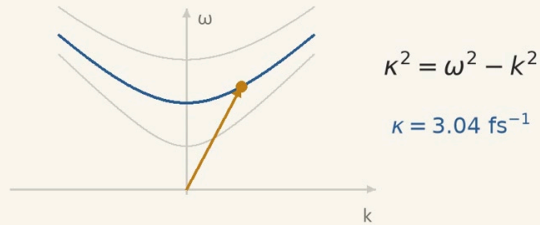


Free motion · illustrative masses · $c = 1$

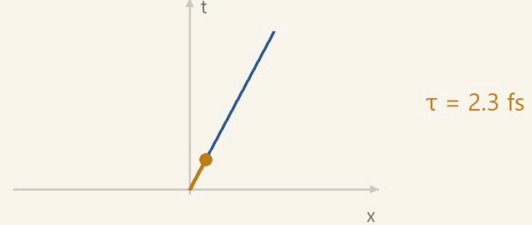
$\hbar$ relates phase to action

Shell 2 of 3 · The same proper-time interval.

Wave-number shell



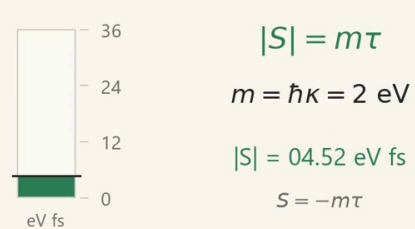
Worldline



Phase



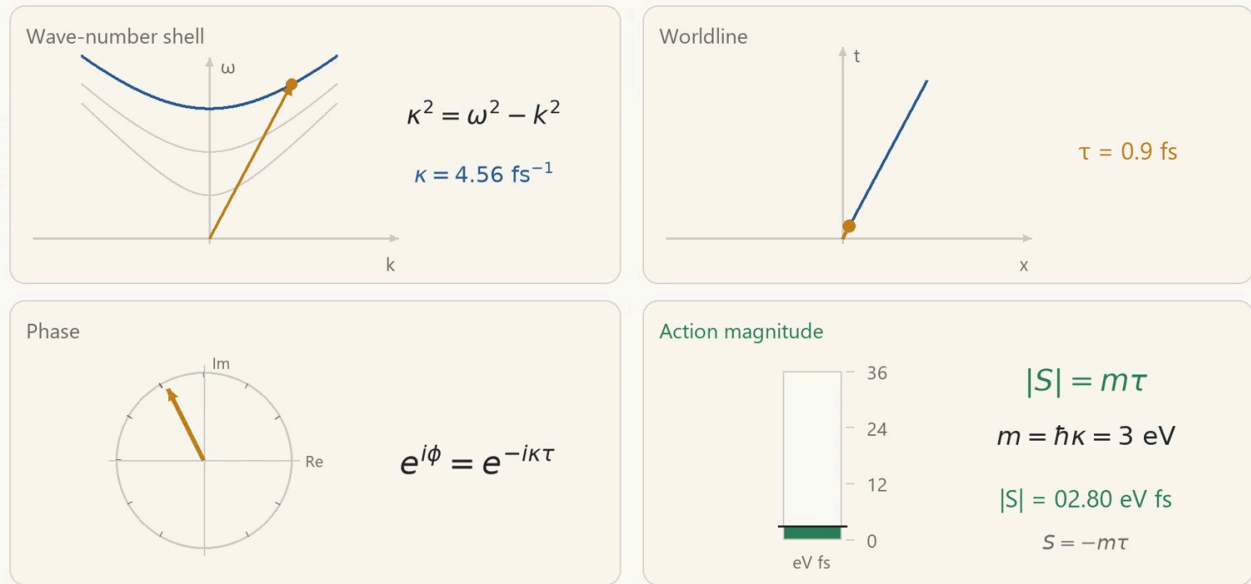
Action magnitude



Free motion · illustrative masses · $c = 1$

$\hbar$ relates phase to action

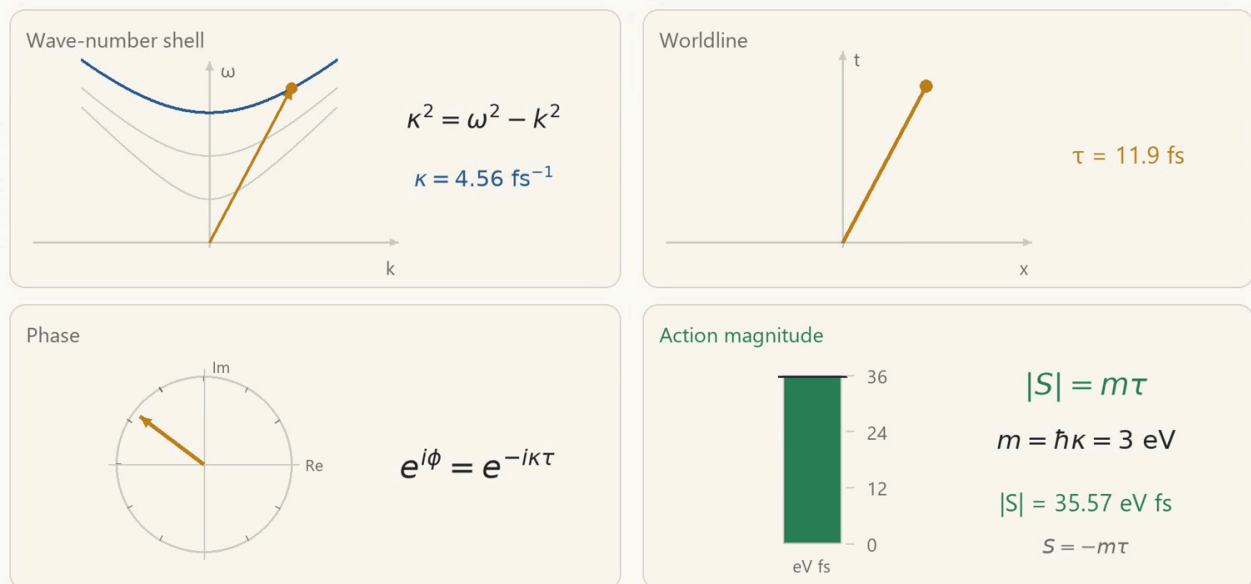
Shell 3 of 3 · The same proper-time interval.



Free motion · illustrative masses · $c = 1$

$\hbar$ relates phase to action

Shell 3 of 3 · The same proper-time interval.



Free motion · illustrative masses · $c = 1$

But this is exactly our condition for the path sum to be dominated by stationary paths. For everyday bodies, $\hbar$ is tiny compared with action differences so that their motion is effectively deterministic. Are we saying that the laws of motion we learn in high school physics are an approximation? Yes, a very good approximation.

We can now express our H_3 commutation relation in units of action:

$$[\hat{X}, \hat{K}] = iI \xrightarrow{\times \hbar} [\hat{X}, \hbar \hat{K}] = i\hbar I$$

We then have:

$$\hat{P} := \hbar \hat{K}$$

where $\hat{P}$ generates position translations with the same scaling that relates phase to action. Its eigenvalue is **momentum**, p , giving us the quantum **canonical commutation relation**:

$$[\hat{X}, \hat{P}] = i\hbar I$$

Recall that x and p are the eigenvalues of the $\hat{X}$ and $\hat{P}$ operators, respectively, acting on the wave function. We then have:

$$\Delta x \Delta k \geq \frac{1}{2} \xrightarrow{\Delta p = \hbar \Delta k} \Delta x \Delta p \geq \frac{\hbar}{2}$$

This is the Heisenberg uncertainty relation, which states that a quantum state cannot have perfectly sharp values of both position and momentum, which becomes relevant at subatomic scales.

The canonical commutation relation, along with the definitions of $\hat{X}$ and $\hat{P}$, is also useful as a starting point from which to derive quantum theory's general law of motion.